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Luís Felipe Sobreira Amaral

Contributions to the continuity
problem of Lyapunov exponents
for linear cocycles over
hyperbolic base maps

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LUÍS FELIPE SOBREIRA AMARAL

Tese defendida e aprovada pela banca examinadora constituída por:

Profª. Karina Daniela Marín
Orientadora - UFMG

Prof. Jamerson Douglas dos Santos Bezerra
UFC

Prof. Javier Alexis Correa Mayobre
UFMG

Prof. Lucas Backes
UFRGS

Profª. Sylvie Marie Oliffson Kamphorst Leal da Silva
UFMG

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To my wife and parents

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“I wish it need not have happened in my time,” said Frodo.
“So do I,” said Gandalf, “and so do all who live to see such times. But that is not for them to decide. All we have to decide is what to do with the time that is given us.”

The Lord of the Rings, J.R.R. Tolkien

The Eternal enters time, the Whole lies hidden in the part.

Fides et Ratio, St. John Paul II

Resumo¹

Vamos fixar um sistema dinâmico composto por um homeomorfismo hiperbólico $f: M \rightarrow M$ definido em um espaço métrico compacto munido de uma probabilidade μ que seja f -ergódica e que possua estrutura de produto local e suporte total. Dada uma aplicação contínua $B: M \rightarrow \text{GL}(2, \mathbb{R})$ (isto é, cada elemento da imagem é uma transformação linear $\mathbb{R}^2 \rightarrow \mathbb{R}^2$ invertível), podemos construir seu cociclo linear $F_B: M \times \mathbb{R}^2 \rightarrow M \times \mathbb{R}^2$ como sendo $F_B(x, v) = (f(x), B(x) \cdot v)$. Uma ferramenta muito útil para caracterizar o sistema dinâmico gerado pelo cociclo linear sobre o fibrado $M \times \mathbb{R}^2$ são os chamados *expoentes de Lyapunov*: $\lambda_+(B)$ e $\lambda_-(B)$, constantes em μ -quase todo ponto (e, portanto, dependendo apenas da aplicação B que gera o cociclo linear). Ainda não sabemos responder qual o maior conjunto possível de aplicações contínuas $M \rightarrow \text{GL}(2, \mathbb{R})$ no qual os expoentes de Lyapunov variam continuamente. Para este fim, verifica-se que é suficiente considerar aplicações $M \rightarrow \text{SL}(2, \mathbb{R})$ (ou seja, cujos elementos da imagem sejam transformações lineares $\mathbb{R}^2 \rightarrow \mathbb{R}^2$ que preservam áreas). Essas aplicações são importantes devido ao fato de que seus expoentes de Lyapunov são números opostos. O pesquisador Marcelo Viana afirma, em uma de suas conjecturas, que os expoentes de Lyapunov serão contínuos no espaço das aplicações contínuas $B: M \rightarrow \text{SL}(2, \mathbb{R})$ que admitem ao menos uma holonomia uniforme (coleção de isomorfismos lineares entre fibras de um mesmo conjunto estável/instável local, que são conjugados à aplicação B nestes conjuntos e que variam continuamente com as fibras). Apesar de já termos alguns resultados parciais com relação a esta conjectura, ela ainda se encontra em aberto. Nosso objetivo no presente trabalho é dar algumas contribuições ao problema da continuidade, com vistas a uma solução definitiva.

Para os resultados aqui apresentados, nós precisaremos de dois conceitos importantes. O primeiro deles é a *condição de 2-invariância* de um cociclo linear munido de uma holonomia estável uniforme: de forma sucinta, esse cociclo admite duas secções $M \rightarrow \mathbb{P}\mathbb{R}^2$ contínuas, transversais e invariantes tanto pelo cociclo quanto pela holonomia (com suas versões projetivas). E o outro conceito é a *condição de equiproximidade* de uma sequência de holonomias instáveis: os isomorfismos lineares que compõem cada holonomia estão em uma vizinhança uniforme da identidade, desde que as fibras estejam suficientemente próximas. Essa condição é suficiente para concluirmos a existência de uma subsequência convergente dessas holonomias instáveis (uma versão do famoso Teorema de Arzelà–Ascoli para holonomias), o que nos permite abrir mão de que as holonomias precisam

¹Versão estendida de acordo com o Art. 4 da Resolução 02/2019 do Programa de Pós-Graduação em Matemática da UFMG.

convergir para que o cociclo linear em questão seja um ponto de continuidade para os expoentes de Lyapunov. Dito isso, mostra-se neste trabalho que, dada uma sequência de cociclos lineares admitindo uma holonomia estável uniforme e uma holonomia instável (tanto uniforme quanto não-uniforme) de modo que os cociclos lineares convirjam (com holonomia estável) para um cociclo linear que satisfaz a condição de 2-invariância, e a sequência formada pelas holonomias instáveis satisfazem a condição de equiproximidade, então há a convergência dos expoentes de Lyapunov. Prova-se ainda que, substituindo a hipótese da sequência de holonomias instáveis satisfazerem a condição de equiproximidade por uma convergência uniforme em um conjunto de medida positiva, podemos afirmar que, se os expoentes de Lyapunov do cociclo linear limite forem não-nulos, então todo expoente de Lyapunov da sequência passa a ser também não-nulo a partir de algum elemento. Esse é o passo inicial para se mostrar a continuidade.

Palavras-chave: expoentes de Lyapunov; problema da continuidade; conjectura de Viana; cociclos lineares; homeomorfismo hiperbólico.

Abstract

Marcelo Viana have conjectured that Lyapunov exponents are continuous in the space of linear cocycles admitting either uniform s -holonomies or u -holonomies, provided that the base map is a hyperbolic homeomorphism over a compact metric space endowed with an ergodic probability measure with local product structure and full support. Despite it has already been some partial results, such conjecture remains unproven. Our goal in the present work is to contribute to the continuity problem of Lyapunov exponents. For so, we take a linear cocycle admitting a uniform s -holonomy and also satisfying the *2-invariance condition* (that means to admit two continuous sections to the projective space such that they are transversal and invariant by both the linear cocycle and the holonomy in their projective forms). In Theorem A it is proven that, given a sequence of linear cocycles admitting uniform holonomies (both of them) such that the linear cocycles converge (with s -holonomies) to the one fixed before, and that the sequence of u -holonomies satisfy the *equiproximity condition* (that is, the linear isomorphisms that compose the holonomies are in a uniform neighbourhood of the identity, provided that the fibres are close enough), then the Lyapunov exponents converge. Theorem B is a stronger result since we change the uniform u -holonomy for a non-uniform one. And finally, it is shown in Theorem C that we can assure non-zero Lyapunov exponents eventually in the sequence of linear cocycles converging to one with non-zero Lyapunov exponents simply by replacing the equiproximity condition on the u -holonomies by uniform convergence in a positive-measure set.

Keywords: Lyapunov exponents; continuity problem; Viana's conjecture; linear cocycles; hyperbolic homeomorphisms.

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Introduction

The study of Lyapunov exponents originated from the doctoral thesis of A.M. Lyapunov [Lya92] about the stability theory of differential equations in the late 19th century. It was only in the 60's that the works of Furstenberg and Kesten [FK60], and Oseledets [Ose68] brought the notion of Lyapunov exponents to the field of ergodic theory. Since then, it grew into a very broad and active area. Some of these results are reproduced in Theorem 1.2.1 and Theorem 3.1.1. We focus only on the discrete-time case in this study, although there is an entirely parallel theory for the continuous one.

Roughly speaking, Lyapunov exponents describe the behaviour of a given *cocycle over a measure-preserving dynamical system*, and they are related to the study of dependence on initial conditions. One may view the Lyapunov exponents as real functions that take values in the space of $\mathrm{SL}(2, \mathbb{R})$ -cocycles. Thus it is important to understand how do these functions depend on their underlying data. For instance, if this dependence occurs in a continuous way, it follows that small perturbations on the initial condition does not alter its dynamics.

We focus in this work on linear cocycles from a compact metric space to $\mathrm{SL}(2, \mathbb{R})$. In addition, we consider in the base a hyperbolic homeomorphism preserving an equilibrium state for a Hölder-continuous potential. In such setting, a result of Bochi [Boc02] establishes that the Lyapunov exponents are C^0 -continuous on a given cocycle only when one of the following is true: (1) all Lyapunov exponents are equal; or (2) the cocycle is *hyperbolic*. In other words, discontinuity of Lyapunov exponents is quite common among linear cocycles over invertible base maps.

However, if we restrict the space of linear cocycle to those that admit *invariant holonomies*, then there are some results proving continuity of Lyapunov exponents. In fact, Backes, Brown and Butler [BBB18] have shown that Lyapunov exponents vary continuously in the space of linear cocycles with uniform invariant holonomies. On the other hand, Freijo and Marin [FM21] have moved further proving that one does not need both holonomies to be uniform. Instead it is sufficient to consider linear cocycles admitting one uniform and another non-uniform invariant holonomy. The latter is a weaker class of invariant holonomies.

Symbolically, we may summarise both results with the following diagram:

$$(A_k, H^{s,k}, H^{u,k}) \xrightarrow{C^0} (A, H^s, H^u) \implies \lambda_+(A_k) \rightarrow \lambda_+(A),$$

where we are assuming that all s -holonomies are uniform and the u -holonomies could be taken either uniform (as in [BBB18]) or non-uniform (as in [FM21]).

A few years ago Viana and Yang [VY19] proved another case of continuity without asking for the existence of invariant holonomies directly. Instead they assumed that the linear cocycle A met some regularity conditions and is constant along local stable sets. Their result states that, if A does not admit *invariant sections*, then it is a C^0 -continuity point of the Lyapunov exponents. In this case, it is possible to construct uniform s -holonomies. Considering this, Viana has conjectured that it is enough to ask for a single uniform holonomy in order to get continuity.

Conjecture (Viana, [Via20, Conjecture 6.3]). *Consider a hyperbolic homeomorphism over a compact metric space endowed with an ergodic probability measure with local product structure and full support. The Lyapunov exponents vary continuously in the space of linear cocycles admitting either uniform s -holonomies or uniform u -holonomies.*

$$\begin{array}{ccc} (A_k, H^{s,k}) \xrightarrow{C^0} (A, H^s) & & \\ \text{or} & \implies & \lambda_+(A_k) \rightarrow \lambda_+(A) \\ (A_k, H^{u,k}) \xrightarrow{C^0} (A, H^u) & & \end{array}$$

In the present work we make some contributions to the continuity problem described above. Our start point is in cocycles that admits two invariant sections, in contrast with the Viana–Yang result. Besides, in order to reduce the regularity conditions, we still ask for the existence of u -holonomies, but only for the converging sequence of linear cocycles. In fact such u -holonomies do not need to be convergent, neither the limit cocycle need to admit a u -holonomy.

In Theorem A we state that, if the converging sequence of cocycles admits u -holonomies that are uniformly close to the identity map when the base points are near enough (this condition will be called *equiproximity*), then the Lyapunov exponents also converge.

$$\begin{array}{ccc} A \text{ satisfy 2-invariance} & & \\ (A_k, H^{s,k}) \xrightarrow{C^0} (A, H^s) & \implies & \lambda_+(A_k) \rightarrow \lambda_+(A). \\ \text{and} & & \\ \{H^{u,k}\}_{k \in \mathbb{N}^*} \text{ equiproximate} & & \end{array}$$

However, we prove in fact a stronger version of this result in Theorem B. Indeed, we only need to ask for the u -holonomies to be non-uniform instead of uniform. Our main technical contribution in such proof is Theorem 4.1.2 (Arzelà–Ascoli for holonomies), stating that every equiproximate sequence of u -holonomies admits a convergent subsequence.

In the other hand, instead of asking for the sequence of u -holonomies to be equiproximate in Theorem C, we ask that such sequence converges uniformly in a positive-measure set. Our conclusion is that, if the upper Lyapunov exponent of A is positive, then it is also true for every linear cocycle after some point in the sequence.

Chapter 1

Preliminaries and Statements

1.1 Linear and projective cocycles

Linear cocycles are the basic object in the study of Lyapunov exponents. We define them briefly in this section and introduce some examples. Our emphasis is on the two-dimensional case.

For our setting, let (M, dist) be a compact metric space, and also take $f: M \rightarrow M$ to be a continuous map. Let us consider a continuous function $B: M \rightarrow \text{GL}(2, \mathbb{R})$ with value in the linear group $\text{GL}(2, \mathbb{R})$ of invertible 2×2 matrices with real coefficients.

The *linear cocycle* defined by B over f is the skew-product

$$\begin{aligned} F_B: M \times \mathbb{R}^2 &\longrightarrow M \times \mathbb{R}^2 \\ (x, v) &\longmapsto (f(x), B(x) \cdot v). \end{aligned} \quad (1.1)$$

On the occasion where the base map f is fixed, we usually denote the cocycle by B itself. Notice that $F_B^n(x, v) = (f^n(x), B^n(x) \cdot v)$ for every $n \in \mathbb{N}$, where

$$B^n(x) = B(f^{n-1}(x)) \cdots B(f(x))B(x).$$

It is worth to mention that, in the case where the base map f is invertible, the linear cocycle F_B as above is also invertible. Indeed, for every $n \in \mathbb{N}^*$ and $(x, v) \in M \times \mathbb{R}^2$, we have

$$F_B^{-n}(x, v) = (f^{-n}(x), B^{-n}(x) \cdot v),$$

where

$$B^{-n}(x) = B(f^{-n}(x))^{-1} \cdots B(f^{-1}(x))^{-1} = B^n(f^{-n}(x))^{-1}.$$

Example 1.1.1 (*Random matrices*). Let $X = \text{GL}(2, \mathbb{R})$ and $M = X^{\mathbb{Z}}$. Take $f: M \rightarrow M$ to be the shift map on X and then consider the mapping $B: M \rightarrow X$ as the projection on the zeroth coordinate:

$$B: (g_j)_{j \in \mathbb{Z}} \longmapsto g_0.$$

Therefore, the linear cocycle defined by B over f is given by

$$F_B((g_j)_{j \in \mathbb{Z}}, v) = ((g_{j+1})_{j \in \mathbb{Z}}, g_0 \cdot v).$$

This type of cocycle is called a *locally constant linear cocycle*. Observe that the n th iterate of B is $B^n((g_j)_{j \in \mathbb{Z}}) = g_{n-1} \cdots g_1 g_0$.

Example 1.1.2 (*Derivative cocycle*). Let $f: M \rightarrow M$ be a C^1 endomorphism on the d -dimensional torus $\mathbb{T}^d$. Choosing some bases in the tangent plane, we may view the derivative $Df(x) = B(x)$ at each point $x \in \mathbb{T}^d$ as a matrix. Hence, the tangent map

$$\begin{aligned} Df: \quad \mathbb{T}\mathbb{T}^d &\longrightarrow \mathbb{T}\mathbb{T}^d \\ (x, v) &\longmapsto (f(x), Df(x) \cdot v) \end{aligned}$$

is a linear cocycle.

Eventually we are going to need some properties derived by compactness. On this occasion we can define a new type of cocycles considering the projection $\mathbb{P}\mathbb{R}^2$ of the Euclidean plane.

The *projective cocycle* defined by B over f is the skew-product

$$\begin{aligned} \mathbb{P}F_B: \quad M \times \mathbb{P}\mathbb{R}^2 &\longrightarrow M \times \mathbb{P}\mathbb{R}^2 \\ (x, [v]) &\longmapsto (f(x), [B(x) \cdot v]), \end{aligned} \tag{1.2}$$

where $[v]$ is the point in $\mathbb{P}\mathbb{R}^2$ that represents the class of a non-zero vector $v \in \mathbb{R}^2$. Finally, we denote the action of the projective cocycle in the fibres by

$$\mathbb{P}B(x) \cdot [v] = [B(x) \cdot v].$$

1.2 Lyapunov exponents

Given a linear cocycle defined by $B: M \rightarrow \text{GL}(2, \mathbb{R})$ over f , we can measure its asymptotic behaviour through orbits computing the following limits:

$$\lambda_+(x) = \lim_{n \in \mathbb{N}^*} \frac{1}{n} \log \|B^n(x)\| \quad \text{and} \quad \lambda_-(x) = \lim_{n \in \mathbb{N}^*} \frac{1}{n} \log \|B^n(x)^{-1}\|^{-1}, \tag{1.3}$$

where we take the operator norm in $\text{GL}(2, \mathbb{R})$. Whenever these limits exist, we call them (extremal) *Lyapunov exponents* of the cocycle B at $x \in M$.

As we have mentioned, the ergodic theory of the Lyapunov exponents was initiated by the following theorem of Furstenberg and Kesten [FK60].

Theorem 1.2.1 (Furstenberg–Kesten). *Take a continuous map $f: M \rightarrow M$ over a compact metric space endowed with an f -invariant probability measure μ . Given a continuous linear cocycle $B: M \rightarrow \text{GL}(2, \mathbb{R})$ over f , the numbers $\lambda_+(x)$ and $\lambda_-(x)$ defined in Equation (1.3) exist for μ -almost every $x \in M$. Moreover, the functions $\lambda_{\pm}$ are f -invariant and μ -integrable with*

$$\int \lambda_+ d\mu = \lim_{n \in \mathbb{N}^*} \frac{1}{n} \int \log \|B^n(x)\| d\mu(x)$$

and

$$\int \lambda_- d\mu = \lim_{n \in \mathbb{N}^*} \frac{1}{n} \int \log \|B^n(x)^{-1}\|^{-1} d\mu(x).$$

This theorem assures the existence of the Lyapunov exponents $\lambda_{\pm}$, as functions of the point x , in a full-measure subset of the space M . Observe that such result is a direct consequence of the Kingman Subadditive Ergodic Theorem [Kin68].

If the dynamical system (f, μ) in the base is ergodic, then the Lyapunov exponents $\lambda_{\pm}$ are constant in a full-measure subset. As a consequence, the underlying data to these functions is the linear cocycle B , and we can denote them as $\lambda_{\pm}(B)$.

Notice that the conclusion of Theorem 1.2.1 for the ergodic case implies that the function $B \mapsto \lambda_+(B)$ is always upper semicontinuous in the C^0 topology. Similarly, the function $B \mapsto \lambda_-(B)$ is lower semicontinuous. Therefore, if $\lambda_+(B) = \lambda_-(B)$, then B is a C^0 -continuity point for the Lyapunov exponents.

We show now that, for the continuity problem of Lyapunov exponents, it is sufficient to focus on $\mathrm{SL}(2, \mathbb{R})$ -valued cocycles instead of $\mathrm{GL}(2, \mathbb{R})$ -valued ones. Observe here that the special linear group $\mathrm{SL}(2, \mathbb{R})$ of 2×2 matrices with real coefficients and determinant ± 1 is a subgroup of $\mathrm{GL}(2, \mathbb{R})$.

The main reason of this reduction is that every $\mathrm{SL}(2, \mathbb{R})$ -valued linear cocycle B satisfies $\lambda_+(B) + \lambda_-(B) = 0$. Hence, we only need to check the Lyapunov exponent λ_+ in order to prove that the given cocycle is a continuity point. The proof of the reduction goes as follow.

Given a linear cocycle $B: M \rightarrow \mathrm{GL}(2, \mathbb{R})$, consider the real-valued functions $s_B, g_B: M \rightarrow \mathbb{R}$ defined by

$$s_B(x) = \mathrm{sgn}(\det B(x)) \quad \text{and} \quad g_B(x) = s_B(x)|\det B(x)|^{\frac{1}{2}}.$$

Consider also the linear cocycle $A: M \rightarrow \mathrm{SL}(2, \mathbb{R})$ that satisfies

$$B(x) = g_B(x)A(x) \quad \text{for every } x \in M.$$

Notice that, by continuity of B and compactness of M , functions s_B and g_B are continuous, and they also depend continuously on the cocycle. Thus, since

$$\lambda_+(B) = \lambda_+(A) + \int \log |g_B| d\mu,$$

we conclude that B is a continuity point if and only if A is a continuity point.

Moreover, $\lambda_+(B) = \lambda_-(B)$ if and only if $\lambda_+(A) = 0 = \lambda_-(A)$. From now on we will assume that the cocycles are always $\mathrm{SL}(2, \mathbb{R})$ -valued.

1.3 Hyperbolic homeomorphisms

In this section we fix the type of base map that we are going to consider in the entire work. We still ask for (M, dist) to be a compact metric space.

Following [Via08], we say that $f: M \rightarrow M$ is a *hyperbolic homeomorphism* if there are constants $\varepsilon > 0$, $\delta > 0$, $\lambda_s < 1$ and $\lambda_u < 1$ such that the *local stable* and *local unstable sets*

$$\begin{aligned} W_{\mathrm{loc}}^s(x) &= \{y \in M : \mathrm{dist}(f^n(x), f^n(y)) \leq \varepsilon \text{ for every } n \in \mathbb{N}\} \quad \text{and} \\ W_{\mathrm{loc}}^u(x) &= \{y \in M : \mathrm{dist}(f^{-n}(x), f^{-n}(y)) \leq \varepsilon \text{ for every } n \in \mathbb{N}\} \end{aligned}$$

satisfy:

- (a) $\text{dist}(f(x), f(y)) \leq \lambda_s \text{dist}(x, y)$ if $y \in W_{\text{loc}}^s(x)$;
- (b) $\text{dist}(f^{-1}(x), f^{-1}(y)) \leq \lambda_u \text{dist}(x, y)$ if $y \in W_{\text{loc}}^u(x)$; and
- (c) if $\text{dist}(x, y) < 2\delta$, then $W_{\text{loc}}^s(x) \cap W_{\text{loc}}^u(y)$ consists in a single point $[x, y]$, and this point varies continuously with (x, y) .

This last condition (c) implies that, for any point $x \in M$, the map

$$\begin{aligned} \Phi_x: \quad W_{\text{loc}}^s(x) \times W_{\text{loc}}^u(x) &\longrightarrow M \\ (y, z) &\longmapsto [y, z] \end{aligned}$$

is a homeomorphism onto a neighbourhood V_x of x .

In particular, the class of hyperbolic homeomorphisms contains some classic examples in Dynamical Systems, like Anosov diffeomorphisms, the natural extension of expanding maps, and the two-sided subshift of finite type. This last item is explained in details in Section 2.1.

In the remaining of this section we describe the type of measures that we consider in our statements. First, it is required a geometric property that allows us to split the measure into local stable and unstable sets. It is mostly important, for instance, in the Hopf argument.

We say that an f -invariant probability measure μ has *local product structure* if, for every $x \in M$, there are a continuous function $\rho_x: V_x \rightarrow (0, \infty)$, and probability measures μ_x^s on $W_{\text{loc}}^s(x)$ and μ_x^u on $W_{\text{loc}}^u(x)$ such that

$$\mu(E) = \int (\chi_E \circ \Phi_x) \rho_x d(\mu_x^s \times \mu_x^u) \text{ for any measurable set } E \subset V_x.$$

That is,

$$\mu|_{V_x} = \rho_x d(\mu_x^s \times \mu_x^u)$$

if one identifies V_x with $W_{\text{loc}}^s(x) \times W_{\text{loc}}^u(x)$ through the map Φ_x .

It is also required to the measure μ be supported in the whole space M . In this case we say that μ has *full support*. And finally we ask that μ is f -ergodic.

We would like to remark that it is sufficient that a hyperbolic homeomorphism f be *topologically mixing* in order to admit a probability measure as we described. In fact, if f is topologically mixing, then every equilibrium state for some Hölder-continuous potential satisfies these properties (*cf.* Ruelle Theorem, for instance, in [VO16, Theorem 12.1]). Therefore we are going to suppose from now on that our dynamical system (M, f) is topologically mixing.

1.4 Theorem A: a case of convergence

Let $f: M \rightarrow M$ be a hyperbolic homeomorphism over a compact metric space.

An important tool used to prove continuity of Lyapunov exponents was introduced by Bonatti, Gómez-Mont and Viana [BGV03]. We are going to define it in the next paragraph.

Definition 1 (Uniform holonomy). A (uniform) *u-holonomy* for a linear cocycle B over a hyperbolic homeomorphism f is a collection of linear isomorphisms $H_{y,z}^u: \mathbb{R}^2 \rightarrow \mathbb{R}^2$, defined for every $y, z \in W_{\text{loc}}^u(x)$ and $x \in M$, which has the following properties:

$$(U.1) \quad H_{z,w}^u \circ H_{y,z}^u = H_{y,w}^u \text{ and } H_{y,y}^u = \text{id};$$

$$(U.2) \quad H_{f^{-1}(y), f^{-1}(z)}^u = B^{-1}(z) \circ H_{y,z}^u \circ B(f^{-1}(y));$$

$$(U.3) \quad (y, z) \mapsto H_{y,z}^u \text{ is continuous.}$$

We define a (uniform) *s-holonomy* for B over f similarly for points in the same local stable set.

It is important to inform that uniform holonomies exist for every α -Hölder continuous linear cocycle B over a hyperbolic homeomorphism satisfying the *fibre-bunching* condition; that is, exists an $N \in \mathbb{N}^*$ such that:

$$\sup_{x \in M} \|B^N(x)\| \|B^N(x)^{-1}\| \min\{\lambda_s, \lambda_u\}^{\alpha N} < 1. \quad (1.4)$$

This is a well-known result based in [BGV03, Lemme 1.12].

We would like to remark that (uniform) *s-holonomies* for a cocycle *do not need* to be unique (see, for instance, [KS16]). For this reason we denote the space of continuous linear cocycles $M \rightarrow \text{SL}(2, \mathbb{R})$ with *s-holonomies* by $\mathcal{H}_s^0(2, M)$. Thus, each element of the space $\mathcal{H}_s^0(2, M)$ is a pair (B, H^s) where H^s denotes an *s-holonomy* for B over f .

Considering Ω^s as the set of pairs of points in the same local stable set, we endow the space $\mathcal{H}_s^0(2, M)$ with the topology induced from the inclusion:

$$\mathcal{H}_s^0(2, M) \hookrightarrow C^0(M; \text{SL}(2, \mathbb{R})) \times C^0(\Omega^s; \text{SL}(2, \mathbb{R}))$$

where the spaces on the right are endowed with the C^0 and the uniform topologies, respectively. In terms of convergence, this means that a sequence $\{(B_k, H^{s,k})\}_{k \in \mathbb{N}^*}$ converges to some (B, H^s) in $\mathcal{H}_s^0(2, M)$ if

$$(U.1') \quad B_k \rightarrow B \text{ in the } C^0 \text{ topology; and}$$

$$(U.2') \quad H^{s,k} \rightarrow H^s \text{ uniformly on each local stable set.}$$

Now we define a condition in a given element of the space $\mathcal{H}_s^0(2, M)$ that is a hypothesis of each theorem we prove in this text.

Definition 2 (2-invariance). A linear cocycle $(B, H^s) \in \mathcal{H}_s^0(2, M)$ is said to satisfy the *2-invariance condition* whenever it admits two continuous sections $\xi_1, \xi_2: M \rightarrow \mathbb{P}\mathbb{R}^2$ which have the following properties, for each $x \in M$:

$$\mathbb{P}B\text{-invariance: } \mathbb{P}B(x) \cdot \xi_i(x) = \xi_i(f(x)) \text{ for } i = 1, 2;$$

Transversality: Each pair of non-zero vectors v_1, v_2 so that $\mathbb{P}v_i = \xi_i(x)$ ($i = 1, 2$) forms a basis for $\mathbb{R}^2$; and

$\mathbb{P}H^s$ -invariance: $\mathbb{P}H_{y,z}^s \cdot \xi_i(y) = \xi_i(z)$ for $i = 1, 2$ and every pair $y, z \in W_{\text{loc}}^s(x)$.

Before we get to the result of this subsection, we need to present a condition in a given sequence of (uniform) u -holonomies that is required in the statement.

Definition 3 (Equiproximity). A sequence $\{H^{u,k}\}_{k \in \mathbb{N}^*}$ of (uniform) u -holonomies is said to satisfy the *equiproximity condition* if, for every $\epsilon > 0$, one can choose a $\delta > 0$ with the property:

$$y \in W_{\text{loc}}^u(x) \text{ with } \text{dist}(x, y) < \delta \Rightarrow \|H_{x,y}^{u,k} - \text{id}\| < \epsilon \text{ for all } k \in \mathbb{N}^*.$$

We can finally state the following.

Theorem A. Consider a hyperbolic homeomorphism $f: M \rightarrow M$ over a compact metric space endowed with an equilibrium state μ for some Hölder-continuous potential. Let $(A, H^s) \in \mathcal{H}_s^0(2, M)$ satisfying the 2-invariance condition. If one takes a sequence $\{(A_k, H^{s,k}, H^{u,k})\}_{k \in \mathbb{N}^*}$ of continuous linear cocycles with uniform holonomies such that:

- (1) $\{(A_k, H^{s,k})\}_{k \in \mathbb{N}^*}$ converges to (A, H^s) in $\mathcal{H}_s^0(2, M)$; and
- (2) $\{H^{u,k}\}_{k \in \mathbb{N}^*}$ satisfies the equiproximity condition,

then $\{\lambda_+(A_k)\}_{k \in \mathbb{N}^*}$ converges to $\lambda_+(A)$.

We would like to remark that the above theorem *is not* contemplated in [BBB18]. Indeed, we are not asking the convergence of u -holonomies and neither the existence of such holonomy for A , as in the cited paper.

In the sequence we present an example of how to construct a sequence of u -holonomies satisfying the equiproximity condition but not converging.

Example 1.4.1. Let $f: M \rightarrow M$ be a hyperbolic homeomorphism over a compact metric space. Consider a sequence $\{A_k\}_{k \in \mathbb{N}^*}$ of linear cocycles $M \rightarrow \text{SL}(2, \mathbb{R})$ over f converging in the C^0 -topology to a linear cocycle $A: M \rightarrow \text{SL}(2, \mathbb{R})$. Recall that invariant holonomies do not need to be unique. Let us suppose that each cocycle A_k admits two u -holonomies $H^{1,k}$ and $H^{2,k}$ such that those sequences are convergent but their limits are distinct. Then we can construct, for each $y \in W_{\text{loc}}^u(x)$, another sequence $\{H_{x,y}^k\}_{k \in \mathbb{N}^*}$ such that:

$$H_{x,y}^k = \begin{cases} H_{x,y}^{1,k}, & \text{if } k \text{ is even;} \\ H_{x,y}^{2,k}, & \text{if } k \text{ is odd.} \end{cases}$$

It is clear that the sequence $\{H^k\}_{k \in \mathbb{N}^*}$ does not converge. However, it satisfies the equiproximity condition due to the fact that all convergent sequence does.

Remark 1.4.2. In [KN11, Theorem 5.5.3], the authors built an example of a constant β -fibre-bunched cocycle B on the torus and an open C^∞ -neighbourhood $\mathcal{U}$ such that every other cocycle in it is β -fibre-bunched and is cohomologous to B via a transfer map of class (strictly) less than C^β . Kalinin and Sadovskaya used such result in [KS16] to show that every cocycle in $\mathcal{U}$ admits two distinct invariant holonomies (either stable or unstable). We are working in a modification of [KN11, Theorem 5.5.3] in order to provide a proper example for the result of the present section, using the ideas of [KS16] and Example 1.4.1 above.

In view of Viana's conjecture, the previous example highlights that the necessary hypotheses on u -holonomies for continuity of Lyapunov exponents are weaker than what we ask for s -holonomies. This also prevails for Theorem B and Theorem C in the following sections.

In next section we present another result that extends Theorem A. This follows since every uniform invariant holonomy is also non-uniform trivially, as we are going to see.

1.5 Theorem B: a case of convergence for non-uniform holonomies

In order to improve the result in [BBB18], Freijo and Marin studied in [FM21] a weaker type of invariant holonomies introduced by Viana [Via08], which is described in the following.

Definition 4 (Non-uniform holonomy). A *non-uniform u -holonomy* for a cocycle $B: M \rightarrow \mathrm{SL}(2, \mathbb{R})$ over a hyperbolic homeomorphism f is given by a full-measure set M^u and a collection of linear isomorphisms $H_{y,z}^u: \mathbb{R}^2 \rightarrow \mathbb{R}^2$, defined for every $y, z \in W_{\mathrm{loc}}^u(x)$ if $x \in M^u$, which has the following properties:

$$(NU.1) \quad H_{z,w}^u \circ H_{y,z}^u = H_{y,w}^u \text{ and } H_{y,y}^u = \mathrm{id};$$

$$(NU.2) \quad H_{f^{-1}(y), f^{-1}(z)}^u = B^{-1}(z) \circ H_{y,z}^u \circ B(f^{-1}(y))$$

and there is an increasing sequence $\{\mathcal{D}_m^u\}_{m \in \mathbb{N}^*}$ of compact subsets (called u -holonomy blocks) such that $\cup \mathcal{D}_m^u = M^u$, in addition that each element satisfies $\mu(M \setminus \mathcal{D}_m^u) < \frac{1}{m}$,

$$(NU.3) \quad (y, z) \mapsto H_{y,z}^u \text{ is continuous when } x \in \mathcal{D}_m^u; \text{ and}$$

$$(NU.4) \quad \text{if } \lambda_+(B) = 0, \text{ then } \|H_{f^j(y), f^j(z)}^u\| \text{ is uniformly bounded for } x \in \mathcal{D}_m^u, \\ \text{and this holds for every } y, z \in W_{\mathrm{loc}}^u(x) \text{ and any } j \in \mathbb{N}.$$

For clarifying purposes, the continuity condition in item (NU.3) above states that, for every $m \in \mathbb{N}^*$ and every $\epsilon > 0$, there is a $\delta_m > 0$ such that

$$\|H_{y_1, z_1}^u - H_{y_2, z_2}^u\| < \epsilon$$

if there are $x_1, x_2 \in \mathcal{D}_m^u$ with $y_1, z_1 \in W_{\mathrm{loc}}^u(x_1)$, $y_2, z_2 \in W_{\mathrm{loc}}^u(x_2)$, and

$$\mathrm{dist}(y_1, y_2) < \delta_m, \quad \text{and} \quad \mathrm{dist}(z_1, z_2) < \delta_m.$$

Although the definition given above of non-uniform u -holonomy requires three elements (to say, M^u , H^u and $\{\mathcal{D}_m^u\}_{m \in \mathbb{N}^*}$), we omit the other two and denote such holonomy by H^u only. As proved in [FM21, Proposition 2], the existence of non-uniform invariant holonomies are provided for α -Hölder continuous linear cocycle B over a hyperbolic homeomorphism satisfying the *non-uniform fibre-bunching* condition. It is a weaker version of Equation (1.4):

$$\limsup_{n \in \mathbb{N}^*} \|B^n(x)\| \|B^n(x)^{-1}\| \min\{\lambda_s, \lambda_u\}^{\alpha n} < 1 \quad \text{for } \mu\text{-almost every } x,$$

or, equivalently,

$$\lambda_+(B) - \lambda_-(B) = 2\lambda_+(B) < \alpha \log \lambda_u.$$

We are going to present now a non-uniform version of the definition of equiproximity. The main difference relies on u -holonomy blocks.

Definition 5 (Equiproximity [revisited]). A sequence $\{H^{u,k}\}_{k \in \mathbb{N}^*}$ of non-uniform u -holonomies is said to satisfy the *equiproximity condition* if:

- (1) it is possible to consider the same u -holonomy blocks $\{\mathcal{D}_m^u\}_{m \in \mathbb{N}^*}$ for each element of $\{H^{u,k}\}_{k \in \mathbb{N}^*}$; and,
- (2) for every $\epsilon > 0$ and $m \in \mathbb{N}^*$, one can choose a $\delta_m > 0$ with the property:

$$\left. \begin{array}{l} z \in \mathcal{D}_m^u \quad \text{and} \\ x, y \in W_{\text{loc}}^u(z) \quad \text{with } \text{dist}(x, y) < \delta_m \end{array} \right\} \Rightarrow \|H_{x,y}^{u,k} - \text{id}\| < \epsilon \text{ for all } k \in \mathbb{N}^*.$$

We can now state the result of this section.

Theorem B. *Consider a hyperbolic homeomorphism $f: M \rightarrow M$ over a compact metric space endowed with an equilibrium state μ for some Hölder-continuous potential. Let $(A, H^s) \in \mathcal{H}_s^0(2, M)$ satisfying the 2-invariance condition. If one takes a sequence $\{(A_k, H^{s,k}, H^{u,k})\}_{k \in \mathbb{N}^*}$ of continuous linear cocycles with uniform s -holonomies and non-uniform u -holonomies such that:*

- (1) $\{(A_k, H^{s,k})\}_{k \in \mathbb{N}^*}$ converges to (A, H^s) in $\mathcal{H}_s^0(2, M)$; and
- (2) $\{H^{u,k}\}_{k \in \mathbb{N}^*}$ satisfies the equiproximity condition,

then $\{\lambda_+(A_k)\}_{k \in \mathbb{N}^}$ converges to $\lambda_+(A)$.*

We would like to remark (similar to the uniform case) that the result above is *not* fully contemplated in [FM21]. Indeed, we are not asking for the linear cocycle to be non-uniformly fibre-bunched, and neither for convergence of non-uniform u -holonomies.

Observe that, since every uniform holonomy is trivially non-uniform also, Theorem B above implies Theorem A stated in the previous section.

The proof of Theorem B is given by the following steps. In Subsection 2.2 we define a semiconjugation of hyperbolic homeomorphisms and subshifts, showing that Theorem B is a consequence of Theorem 2.1.1. Finally, this last result is proved in Section 5.1.

1.6 Theorem C: non-zero Lyapunov exponents

Recall we are considering a hyperbolic homeomorphism $f: M \rightarrow M$ defined over a compact metric space (M, dist) , and preserving a probability measure μ .

So far we have worked with u -holonomies defined everywhere, or almost everywhere. By contrast, for the results in this section we deal with a *restricted u -holonomy* in a subset of M . By this we mean a collection of linear isomorphisms, defined for every pair $y, z \in W_{\text{loc}}^u(x)$ where the point x is in the given subset, satisfying item (U.1) and item (U.3) of the uniform holonomy definition.

Theorem C. *Consider a hyperbolic homeomorphism $f: M \rightarrow M$ over a compact metric space endowed with an equilibrium state μ for a Hölder-continuous potential. Let $(A, H^s) \in \mathcal{H}_s^0(2, M)$ with non-zero Lyapunov exponents and satisfying the 2-invariance condition. If one takes a sequence $\{(A_k, H^{s,k}, H^{u,k})\}_{k \in \mathbb{N}^*}$ of continuous linear cocycles with uniform s -holonomies and non-uniform u -holonomies, and a set $\mathcal{D} \subset M$ with positive μ -measure such that:*

- (1) $\{(A_k, H^{s,k})\}_{k \in \mathbb{N}^*}$ converges to (A, H^s) in $\mathcal{H}_s^0(2, M)$; and
- (2) for each $q \in \mathcal{D}$, there exists a u -holonomy H^u for the cocycle A restricted to $W_{\text{loc}}^u(q)$ so that the sequence $\{H^{u,k}\}_{k \in \mathbb{N}^*}$ converges uniformly to H^u in such set,

then $\lambda_+(A_k) > 0$ for every sufficiently large $k \in \mathbb{N}^*$.

As one can notice, this result *is not* contemplated in [BBB18] nor in [FM21]. Indeed, we are not asking for existence of u -holonomies as in Definition 4, but just its restricted version. We also do not ask for convergence of such holonomies in a full-measure set.

The proof of Theorem C is given in the following steps. First we restate it for subshifts of finite type (see Section 2.1 to the definition) in Theorem 2.1.2. In Subsection 2.2 we show that every hyperbolic homeomorphism is semiconjugate to a subshift of finite type. Finally we prove Theorem 2.1.2 in Section 5.2.

We would like to make an observation concerning Theorem C and Bocker–Viana’s counterexample [BV17]. In such paper, Bocker and Viana provide a class of locally constant cocycles with non-zero Lyapunov exponents that are approximated in the Hölder topology by cocycles with null Lyapunov exponents (we discuss this more in Appendix A). Although the authors do not mention holonomies in the example, each linear cocycle in this sequence is non-uniformly fibre-bunched and hence admits non-uniform holonomies. Besides, we show in Section A.2 that these holonomies converge in a full-measure set.

Thus one could ask why Bocker–Viana’s counterexample does not contradict Theorem C above. In reality, this example does not befit the hypotheses of the theorem, since we need the s -holonomies to be uniform, not just non-uniform as we can construct in the example.

Chapter 2

Transferring the problem to subshifts of finite type

The purpose of this chapter is to enunciate a version of Theorems B and C in a suitable setting. In fact, the calculations in order to prove the results is simpler to manipulate in such setting.

2.1 Subshifts of finite type

In this section we introduce a particular class of hyperbolic homeomorphisms. It is a rich source of examples in Dynamical Systems and Ergodic Theory, and it came from the so called symbolic dynamics. In fact, there are two groups, namely the one-sided and the two-sided subshift of finite type. The latter is an invertible system whereas the former is not. We focus here on the two-sided case, but the other one is similar: one just needs to consider $\mathbb{N}$ instead of $\mathbb{Z}$ in definitions below.

Let $T = (t_{ij})_{i,j=0}^{\kappa-1}$ be a $\kappa \times \kappa$ matrix, for some fixed $\kappa \in \mathbb{N}^*$, such that each entry t_{ij} is either 0 or 1. We ask that each row and column of T has at least one non-zero entry. Such matrix is called the *transition matrix*. We define the (two-sided) *sequence space associated with T* as

$$\hat{\Sigma} = \{\hat{\mathbf{x}} = (\dots, x_{-1}; x_0, x_1, \dots) \in \{0, \dots, \kappa - 1\}^{\mathbb{Z}} : t_{x_j x_{j+1}} = 1 \text{ for } j \in \mathbb{Z}\};$$

that is, the set of bi-infinite admissible sequences with alphabet $\{0, \dots, \kappa - 1\}$.

We endow the space $\hat{\Sigma}$ with the ρ -metric $\text{dist}_\rho(\hat{\mathbf{x}}, \hat{\mathbf{y}}) = \rho^{N(\hat{\mathbf{x}}, \hat{\mathbf{y}})}$, for some $\rho \in (0, 1)$, where

$$N(\hat{\mathbf{x}}, \hat{\mathbf{y}}) = \sup\{N \in \mathbb{N} : x_n = y_n \text{ for every } |n| \leq N\}. \quad (2.1)$$

We can construct a topology on the space $\hat{\Sigma}$ using the following concept. Given numbers $j_1, \dots, j_k \in \mathbb{Z}$ and symbols $\alpha_1, \dots, \alpha_k \in \{0, \dots, \kappa - 1\}$, we define a *cylinder of rank k* as the subset

$$C_{j_1, \dots, j_k}^{\alpha_1, \dots, \alpha_k} = \{\hat{\mathbf{x}} \in \hat{\Sigma} : x_{j_i} = \alpha_i \text{ for } i = 1, \dots, k\}.$$

We set then that every cylinder is open and they form a basis for the topology. Such topology is equivalent to the one induced by the ρ -metric. We would like to remark that $\hat{\Sigma}$ is compact in such topology.

The dynamics in the sequence space will be given by the *left shift map* that is defined by

$$\begin{aligned} \hat{f}: \hat{\Sigma} &\longrightarrow \hat{\Sigma} \\ (\dots, x_{-1}; x_0, x_1, \dots) &\longmapsto (\dots, x_0; x_1, x_2, \dots). \end{aligned}$$

It is easy to see that $\hat{f}$ takes cylinders into cylinders. Additionally, it can be proven that $\hat{f}$ is a hyperbolic homeomorphism over $\hat{\Sigma}$ with contraction ρ and expansion $\sigma = \rho^{-1}$. In this case the local stable and unstable sets of a sequence $\hat{\mathbf{x}} \in \hat{\Sigma}$ are

$$W_{\text{loc}}^s(\hat{\mathbf{x}}) = \{\hat{\mathbf{y}} \in \hat{\Sigma} : x_j = y_j \text{ for every } j \in \mathbb{N}\}$$

and

$$W_{\text{loc}}^u(\hat{\mathbf{x}}) = \{\hat{\mathbf{y}} \in \hat{\Sigma} : x_{-j} = y_{-j} \text{ for every } j \in \mathbb{N}\}.$$

The last property of hyperbolic homeomorphisms follows directly. Indeed, consider sequences $\hat{\mathbf{x}}, \hat{\mathbf{y}} \in \hat{\Sigma}$ that are ρ -close to each other. This implies $x_0 = y_0$. Now, observe that the intersection of $W_{\text{loc}}^s(\hat{\mathbf{x}})$ with $W_{\text{loc}}^u(\hat{\mathbf{y}})$ consists in exactly one point: the sequence $\hat{\mathbf{z}} = [\hat{\mathbf{x}}, \hat{\mathbf{y}}]$ where

$$z_j = \begin{cases} y_j, & \text{if } j < 0, \\ x_j, & \text{if } j \geq 0, \end{cases}$$

that varies continuously with $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$.

The dynamical system $(\hat{\Sigma}, \hat{f})$ is called a *subshift of finite type*. When the transition matrix T has all entries equal 1 (that is, when all sequences are admissible), we usually say that $(\hat{\Sigma}, \hat{f})$ is a *full shift of κ symbols*.

There is a sufficient condition to determine whether a subshift of finite type is topologically mixing. Indeed, this is the case whenever the transition matrix T is *aperiodic*, meaning that one can find an $n \in \mathbb{N}^*$ such that every element of the matrix T^n is positive. For instance, every full shift is topologically mixing.

We end this section with the respectively versions of Theorems B and C in the subshift setting. We begin by the result equivalent to Theorem B.

Theorem 2.1.1. *Let $(\hat{\Sigma}, \hat{f})$ be a subshift of finite type and $\hat{\mu}$ be an $\hat{f}$ -ergodic probability measure on $\hat{\Sigma}$ that has local product structure and full support. Consider $(\hat{A}, H^s) \in \mathcal{H}_s^0(2, \hat{\Sigma})$ satisfying the 2-invariance condition. If one takes any sequence $\{(\hat{A}_k, H^{s,k}, H^{u,k})\}_{k \in \mathbb{N}^*}$ of linear cocycles with uniform s -holonomies and non-uniform u -holonomies such that*

- (1) $(\hat{A}_k, H^{s,k}) \rightarrow (\hat{A}, H^s)$ in $\mathcal{H}_s^0(2, \hat{\Sigma})$; and
- (2) $\{H^{u,k}\}_{k \in \mathbb{N}^*}$ satisfies the equiproximity condition,

then $\{\lambda_+(\hat{A}_k)\}_{k \in \mathbb{N}^}$ converges to $\lambda_+(\hat{A})$.*

The proof of such result is presented in Section 5.1. Next we present the result corresponding to Theorem C.

Theorem 2.1.2. *Let $(\hat{\Sigma}, \hat{f})$ be a subshift of finite type and $\hat{\mu}$ be an $\hat{f}$ -ergodic probability measure on $\hat{\Sigma}$ that has local product structure and full support. Consider $(\hat{A}, H^s) \in \mathcal{H}_s^0(2, \hat{\Sigma})$ with non-zero Lyapunov exponents and satisfying the 2-invariance condition. If one takes a sequence $\{(\hat{A}_k, H^{s,k}, H^{u,k})\}_{k \in \mathbb{N}^*}$ of linear cocycles with uniform s -holonomies and non-uniform u -holonomies, and a set $\hat{\mathcal{D}} \subset \hat{\Sigma}$ with positive $\hat{\mu}$ -measure such that:*

- (1) $(\hat{A}_k, H^{s,k}) \rightarrow (\hat{A}, H^s)$ in $\mathcal{H}_s^0(2, \hat{\Sigma})$; and
- (2) *for each $\hat{\mathbf{q}} \in \hat{\mathcal{D}}$, there exists a u -holonomy H^u for the cocycle $\hat{A}$ restricted to $W_{\text{loc}}^u(\hat{\mathbf{q}})$ so that the sequence $\{H^{u,k}\}_{k \in \mathbb{N}^*}$ converges uniformly to H^u in such set,*

then $\lambda_+(\hat{A}_k) > 0$ for every sufficiently large $k \in \mathbb{N}^$.*

The proof of the result above is given in Section 5.2.

2.2 Translation: from hyperbolic homeomorphisms into subshifts of finite type

In this section we deduce Theorem B and Theorem C from Theorems 2.1.1 and 2.1.2, respectively. With this purpose, we need to carry our problem over from hyperbolic homeomorphisms into subshifts of finite type. This can be done by Markov partitions. Bowen [Bow70] has already proven the existence of Markov partitions in our setting.

Theorem 2.2.1 (Theorem 12 of [Bow70]). *Every hyperbolic homeomorphism has a Markov partition with diameter arbitrarily small.*

From now on, fix a hyperbolic homeomorphism $f: M \rightarrow M$ over a compact metric space (M, dist) endowed with an equilibrium state μ for a Hölder-continuous potential. Let $(A, H^s) \in \mathcal{H}_s^0(2, M)$ be a linear cocycle satisfying the 2-invariance condition. For now we do not distinct between Theorem B and Theorem C; this is done later. Finally, take a sequence $(A_k, H^{s,k}) \rightarrow (A, H^s)$ in $\mathcal{H}_s^0(2, M)$.

Now, since f has a Markov partition by Theorem 2.2.1, there are a subshift of finite type $(\hat{f}, \hat{\Sigma})$ and a topological semiconjugation $h: \hat{\Sigma} \rightarrow M$ such that the following diagram commutes:

$$\begin{array}{ccc} \hat{\Sigma} & \xrightarrow{\hat{f}} & \hat{\Sigma} \\ \downarrow h & \circlearrowleft & \downarrow h \\ M & \xrightarrow{f} & M \end{array}$$

that is, $h \circ \hat{f} = f \circ h$.

Using this semiconjugation, we can define new maps $\hat{A}: \hat{\Sigma} \rightarrow \text{SL}(2, \mathbb{R})$ and $\hat{A}_k: \hat{\Sigma} \rightarrow \text{SL}(2, \mathbb{R})$ (for each $k \in \mathbb{N}^*$) by

$$\hat{A}(\hat{\mathbf{x}}) = A(h(\hat{\mathbf{x}})) \quad \text{and} \quad \hat{A}_k(\hat{\mathbf{x}}) = A_k(h(\hat{\mathbf{x}})). \quad (2.2)$$

It is easy to see that such maps $\hat{A}$ and $\hat{A}_k$ are linear cocycles over $\hat{f}$. Besides, as $A_k \rightarrow A$, we also have that $\hat{A}_k \rightarrow \hat{A}$ in the C^0 topology.

Again, by the use of h , we can define new collections $\hat{H}^s$ and $\hat{H}^{s,k}$ (for each $k \in \mathbb{N}^*$) of isomorphisms by

$$\begin{aligned} \hat{H}_{\hat{\mathbf{y}}, \hat{\mathbf{z}}}^s &= H_{h(\hat{\mathbf{y}}), h(\hat{\mathbf{z}})}^s \\ \text{and} \quad &\text{for every } \hat{\mathbf{y}}, \hat{\mathbf{z}} \in W_{\text{loc}}^s(\hat{\mathbf{x}}) \text{ with } \hat{\mathbf{x}} \in \hat{\Sigma}. \\ \hat{H}_{\hat{\mathbf{y}}, \hat{\mathbf{z}}}^{s,k} &= H_{h(\hat{\mathbf{y}}), h(\hat{\mathbf{z}})}^{s,k} \end{aligned} \quad (2.3)$$

Lemma 2.2.2. *The collection $\hat{H}^s$ (and also $\hat{H}^{s,k}$ for each $k \in \mathbb{N}^*$) is well defined and a uniform s -holonomy for $\hat{A}$ (respectively, $\hat{A}_k$) over $\hat{f}$.*

Proof. In order to prove that $\hat{H}^s$ is well defined, we only need to verify that h preserves local stable sets. Indeed, given $\hat{\mathbf{x}} \in \hat{\Sigma}$ and $\hat{\mathbf{y}}, \hat{\mathbf{z}} \in W_{\text{loc}}^s(\hat{\mathbf{x}})$, we know that

$$\text{dist}_\rho(\hat{f}^n(\hat{\mathbf{y}}), \hat{f}^n(\hat{\mathbf{z}})) \leq \rho \text{ for all } n \in \mathbb{N}.$$

Since h is continuous, we can decrease ρ if necessary to ensure that

$$\text{dist}(f^n \circ h(\hat{\mathbf{y}}), f^n \circ h(\hat{\mathbf{z}})) = \text{dist}(h \circ \hat{f}^n(\hat{\mathbf{y}}), h \circ \hat{f}^n(\hat{\mathbf{z}})) < \varepsilon \text{ for all } n \in \mathbb{N},$$

concluding that $h(\hat{\mathbf{z}}) \in W_{\text{loc}}^s(h(\hat{\mathbf{y}}))$, as we wanted.

Now, we need to verify all the three conditions in the definition of uniform s -holonomies.

(i) Using the fact that H^s is an s -holonomy, we get:

$$\hat{H}_{\hat{\mathbf{z}}, \hat{\mathbf{w}}}^s \circ \hat{H}_{\hat{\mathbf{y}}, \hat{\mathbf{z}}}^s = H_{h(\hat{\mathbf{z}}), h(\hat{\mathbf{w}})}^s \circ H_{h(\hat{\mathbf{y}}), h(\hat{\mathbf{z}})}^s = H_{h(\hat{\mathbf{y}}), h(\hat{\mathbf{w}})}^s = \hat{H}_{\hat{\mathbf{y}}, \hat{\mathbf{w}}}^s$$

$$\text{and } \hat{H}_{\hat{\mathbf{y}}, \hat{\mathbf{y}}}^s = H_{h(\hat{\mathbf{y}}), h(\hat{\mathbf{y}})}^s = \text{id}.$$

(ii) Using also the fact that h is a semiconjugation, we get:

$$\begin{aligned} \hat{H}_{\hat{f}(\hat{\mathbf{y}}), \hat{f}(\hat{\mathbf{z}})}^s &= H_{h \circ \hat{f}(\hat{\mathbf{y}}), h \circ \hat{f}(\hat{\mathbf{z}})}^s = H_{f \circ h(\hat{\mathbf{y}}), f \circ h(\hat{\mathbf{z}})}^s \\ &= A(h(\hat{\mathbf{y}})) \circ H_{h(\hat{\mathbf{y}}), h(\hat{\mathbf{z}})}^s \circ A(h(\hat{\mathbf{z}}))^{-1} \\ &= \hat{A}(\hat{\mathbf{y}}) \circ \hat{H}_{\hat{\mathbf{y}}, \hat{\mathbf{z}}}^s \circ \hat{A}(\hat{\mathbf{z}})^{-1}. \end{aligned}$$

(iii) Given an $\epsilon > 0$, choose a $\delta_1 > 0$ satisfying

$$\|H_{y_1, z_1}^s - H_{y_2, z_2}^s\| < \epsilon$$

if $y_1, z_1 \in W_{\text{loc}}^s(x_1)$, $y_2, z_2 \in W_{\text{loc}}^s(x_2)$ for some $x_1, x_2 \in M$ and such that $\text{dist}(y_1, y_2) < \delta_1$, $\text{dist}(z_1, z_2) < \delta_1$. Now choose a $\delta_2 > 0$ in a manner

that $\text{dist}_\rho(\hat{\mathbf{y}}, \hat{\mathbf{z}}) < \delta_2$ implies $\text{dist}(h(\hat{\mathbf{y}}), h(\hat{\mathbf{z}})) < \delta_1$. It is sufficient to take $\delta = \min\{\delta_1, \delta_2\}$ in order to conclude:

$$\begin{aligned} & \text{if } \text{dist}_\rho(\hat{\mathbf{y}}_1, \hat{\mathbf{y}}_2) < \delta \text{ and } \text{dist}_\rho(\hat{\mathbf{z}}_1, \hat{\mathbf{z}}_2) < \delta, \\ & \text{then } \|\hat{H}_{\hat{\mathbf{y}}_1, \hat{\mathbf{z}}_1}^s - \hat{H}_{\hat{\mathbf{y}}_2, \hat{\mathbf{z}}_2}^s\| = \|H_{h(\hat{\mathbf{y}}_1), h(\hat{\mathbf{z}}_1)}^s - H_{h(\hat{\mathbf{y}}_2), h(\hat{\mathbf{z}}_2)}^s\| < \epsilon. \end{aligned}$$

Therefore, the map $(\hat{\mathbf{y}}, \hat{\mathbf{z}}) \mapsto \hat{H}_{\hat{\mathbf{y}}, \hat{\mathbf{z}}}^s$ is continuous.

The proof for $\hat{H}^{s,k}$ ($k \in \mathbb{N}^*$) is strictly the same. $\square$

An important fact of the new s -holonomies $\{\hat{H}^{s,k}\}_{k \in \mathbb{N}}$ is that, since by hypothesis $H^{s,k} \rightarrow H^s$ uniformly in each local stable set, we also have that $\hat{H}^{s,k} \rightarrow \hat{H}^s$ uniformly in each local stable set. Thus we conclude that $(\hat{A}_k, \hat{H}^{s,k}) \rightarrow (\hat{A}, \hat{H}^s)$ in $\mathcal{H}_s^0(2, \hat{\Sigma})$.

A significant question that may arise is if the cocycle $(\hat{A}, \hat{H}^s) \in \mathcal{H}_s^0(2, \hat{\Sigma})$ constructed above preserves the 2-invariance condition. We answer this question in the next lemma.

Lemma 2.2.3. *Let us consider two linear cocycles $(A, H^s) \in \mathcal{H}_s^0(2, M)$ and $(\hat{A}, \hat{H}^s) \in \mathcal{H}_s^0(2, \hat{\Sigma})$ related by Equations (2.2) and (2.3). If (A, H^s) satisfies the 2-invariance condition, then $(\hat{A}, \hat{H}^s)$ also satisfies.*

Proof. Let us consider the two invariant sections $\xi_1, \xi_2: M \rightarrow \mathbb{P}\mathbb{R}^2$ as in the definition of 2-invariance condition for (A, H^s) . We can define another sections $\hat{\xi}_1, \hat{\xi}_2: \hat{\Sigma} \rightarrow \mathbb{P}\mathbb{R}^2$ by $\hat{\xi}_i(\hat{\mathbf{x}}) = \xi_i(h(\hat{\mathbf{x}}))$ for $i = 1, 2$, where $h: \hat{\Sigma} \rightarrow M$ is the semiconjugation.

We need to check the three properties on the definition of 2-invariance condition. Then, given any $\hat{\mathbf{x}} \in \hat{\Sigma}$,

$\mathbb{P}\hat{A}$ -invariance: for $i = 1, 2$, we have

$$\mathbb{P}\hat{A}(\hat{\mathbf{x}}) \cdot \hat{\xi}_i(\hat{\mathbf{x}}) = \mathbb{P}A(h(\hat{\mathbf{x}})) \cdot \xi_i(h(\hat{\mathbf{x}})) = \xi_i(f(h(\hat{\mathbf{x}}))) = \xi_i(h(\hat{f}(\hat{\mathbf{x}}))) = \hat{\xi}_i(\hat{f}(\hat{\mathbf{x}}));$$

Transversality: by the 2-invariance condition of (A, H^s) , any pair of vectors $v_1, v_2 \in \mathbb{R}^2$ such that $\mathbb{P}v_i = \xi_i(h(\hat{\mathbf{x}})) = \hat{\xi}_i(\hat{\mathbf{x}})$, for $i = 1, 2$, is a basis of $\mathbb{R}^2$;

$\mathbb{P}\hat{H}^s$ -invariance: for $i = 1, 2$ and every $\hat{\mathbf{y}}, \hat{\mathbf{z}} \in W_{\text{loc}}^s(\hat{\mathbf{x}})$, we have

$$\mathbb{P}\hat{H}_{\hat{\mathbf{y}}, \hat{\mathbf{z}}}^s \cdot \hat{\xi}_i(\hat{\mathbf{y}}) = \mathbb{P}H_{h(\hat{\mathbf{y}}), h(\hat{\mathbf{z}})}^s \cdot \xi_i(h(\hat{\mathbf{y}})) = \xi_i(h(\hat{\mathbf{z}})) = \hat{\xi}_i(\hat{\mathbf{z}}).$$

This proves the 2-invariance condition for $(\hat{A}, \hat{H}^s)$, as we wanted. $\square$

We argue that we can define a measure $\hat{\mu}$ on $\hat{\Sigma}$ as the pullback of μ by the semiconjugation h . Indeed, since μ is an equilibrium state for a Hölder-continuous potential, we know that the set of points where h is not injective, that is, the set $\{x \in M: \#h^{-1}(x) > 1\}$ has zero μ -measure. As h is surjective, it follows that the measure $\hat{\mu}$ is well defined. It can be proven that $\hat{\mu}$ is $\hat{f}$ -ergodic, has local product structure and has full support.

Consider two linear cocycles $B: M \rightarrow \text{SL}(2, \mathbb{R})$ and $\hat{B}: \hat{\Sigma} \rightarrow \text{SL}(2, \mathbb{R})$, which are related by the formula $\hat{B} = B \circ h$. A pertinent question is whether or not the Lyapunov exponents of B differ from the ones of $\hat{B}$, but the answer is *no*. Indeed, for $\hat{\mu}$ -almost every $\hat{\mathbf{x}}$,

$$\begin{aligned} \lambda_+(\hat{B}) &= \lim_{n \in \mathbb{N}^*} \frac{1}{n} \log \|\hat{B}^n(\hat{\mathbf{x}})\| \\ &= \lim_{n \in \mathbb{N}^*} \frac{1}{n} \log \|\hat{B}(\hat{f}^{n-1}(\hat{\mathbf{x}})) \circ \dots \circ \hat{B}(\hat{f}(\hat{\mathbf{x}})) \circ \hat{B}(\hat{\mathbf{x}})\| \\ &= \lim_{n \in \mathbb{N}^*} \frac{1}{n} \log \|B(h \circ \hat{f}^{n-1}(\hat{\mathbf{x}})) \circ \dots \circ B(h \circ \hat{f}(\hat{\mathbf{x}})) \circ B(h(\hat{\mathbf{x}}))\| \\ &= \lim_{n \in \mathbb{N}^*} \frac{1}{n} \log \|B(f^{n-1} \circ h(\hat{\mathbf{x}})) \circ \dots \circ B(f \circ h(\hat{\mathbf{x}})) \circ B(h(\hat{\mathbf{x}}))\| \\ &= \lim_{n \in \mathbb{N}^*} \frac{1}{n} \log \|B^n(h(\hat{\mathbf{x}}))\| = \lambda_+(B). \end{aligned} \quad (2.4)$$

We made the calculations only for the upper Lyapunov exponent since one is the opposite of the other.

From now on we need to split into two cases, since Theorem B and Theorem C have different hypotheses.

Case 1) A convergence result

Our goal now is to show how to conclude Theorem B from Theorem 2.1.1. There is another hypothesis left to analyse.

In this direction, let $H^{u,k}$ be a non-uniform u -holonomy for A_k over f (for each $k \in \mathbb{N}^*$). Suppose that their sequence $\{H^{u,k}\}_{k \in \mathbb{N}^*}$ satisfies the equiproximity condition. We need to show that each $\hat{A}_k$ admits a non-uniform u -holonomy $\hat{H}^{u,k}$ over $\hat{f}$ in a manner that $\{\hat{H}^{u,k}\}_{k \in \mathbb{N}^*}$ also satisfies the equiproximity condition.

However, in a similar way as we have done for s -holonomies, we can construct new collections $\hat{H}^u$ and $\hat{H}^{u,k}$ (for each $k \in \mathbb{N}^*$) of isomorphisms by the use of the semiconjugation h :

$$\begin{aligned} \hat{H}_{\hat{\mathbf{y}}, \hat{\mathbf{z}}}^u &= H_{h(\hat{\mathbf{y}}), h(\hat{\mathbf{z}})}^u \text{ for every } \hat{\mathbf{y}}, \hat{\mathbf{z}} \in W_{\text{loc}}^u(\hat{\mathbf{x}}) \text{ if } \hat{\mathbf{x}} \in h^{-1}(M^u) \\ &\text{and} \\ \hat{H}_{\hat{\mathbf{y}}, \hat{\mathbf{z}}}^{u,k} &= H_{h(\hat{\mathbf{y}}), h(\hat{\mathbf{z}})}^{u,k} \text{ for every } \hat{\mathbf{y}}, \hat{\mathbf{z}} \in W_{\text{loc}}^u(\hat{\mathbf{x}}) \text{ if } \hat{\mathbf{x}} \in h^{-1}(M^{u,k}). \end{aligned} \quad (2.5)$$

The same kind of computation did in the proof of Lemma 2.2.2 can also prove the following.

Lemma 2.2.4. *The collection $\hat{H}^u$ (and also $\hat{H}^{u,k}$ for each $k \in \mathbb{N}^*$) is well defined and it is a non-uniform u -holonomy for $\hat{A}$ (respectively, $\hat{A}_k$) over $\hat{f}$ with set $h^{-1}(M^u)$ (respectively, $h^{-1}(M^{u,k})$).*

We need to check if the new non-uniform u -holonomy sequence $\{\hat{H}^{u,k}\}_{k \in \mathbb{N}^*}$ preserves the equiproximity condition. This is achieved through continuity of the semiconjugation h .

Lemma 2.2.5. *Consider two sequences $\{H^{u,k}\}_{k \in \mathbb{N}^*}$ and $\{\hat{H}^{u,k}\}_{k \in \mathbb{N}^*}$ of non-uniform u -holonomies related by Equation (2.5). If $\{H^{u,k}\}_{k \in \mathbb{N}^*}$ satisfies the equiproximity condition, then $\{\hat{H}^{u,k}\}_{k \in \mathbb{N}^*}$ still satisfies.*

Proof. Fix some $\epsilon > 0$ and $m \in \mathbb{N}^*$, and let $\delta_m > 0$ be the one provided by the equiproximity condition of $\{H^{u,k}\}_{k \in \mathbb{N}^*}$. Since h is uniformly continuous, there is a $\delta'_m > 0$ such that $\text{dist}(h(\hat{\mathbf{x}}), h(\hat{\mathbf{y}})) < \delta_m$ whenever $\text{dist}_\rho(\hat{\mathbf{x}}, \hat{\mathbf{y}}) < \delta'_m$. Consider then a point $\hat{\mathbf{z}} \in h^{-1}(\mathcal{D}_m^u)$ and take any $\hat{\mathbf{x}}, \hat{\mathbf{y}} \in W_{\text{loc}}^u(\hat{\mathbf{z}})$ with $\text{dist}_\rho(\hat{\mathbf{x}}, \hat{\mathbf{y}}) < \delta'_m$. It is easy to see that $\|\hat{H}_{\hat{\mathbf{x}}, \hat{\mathbf{y}}}^{u,k} - \text{id}\| = \|H_{h(\hat{\mathbf{x}}), h(\hat{\mathbf{y}})}^{u,k} - \text{id}\| < \epsilon$ for every $k \in \mathbb{N}^*$. In conclusion, $\{\hat{H}^{u,k}\}_{k \in \mathbb{N}^*}$ also satisfies the equiproximity condition. $\square$

Observe then that we are in Theorem 2.1.1 hypotheses. Such result concludes that $\lambda_+(\hat{A}_k) \rightarrow \lambda_+(\hat{A})$. Therefore we can infer by Equation (2.4) that we also have $\lambda_+(A_k) \rightarrow \lambda_+(A)$. This completes the proof of Theorem B.

In the following we deal with the other case.

Case 2) Non-zero Lyapunov exponents

Our goal here is to show how to conclude Theorem C assuming Theorem 2.1.2. Recall we are assuming that A has non-zero Lyapunov exponents. Hence, by Equation (2.4), $\lambda_\pm(\hat{A})$ are as well away from zero.

There are a few more hypotheses of the statements that we have not considered yet. Indeed, we need to prove the existence of a set $\hat{\mathcal{D}}$ with positive $\hat{\mu}$ -measure, and such that every point $\hat{\mathbf{q}} \in \hat{\mathcal{D}}$ has the property: there is a u -holonomy $\hat{H}^u$ for $\hat{A}$ restricted to $W_{\text{loc}}^u(\hat{\mathbf{q}})$ such that $\{\hat{H}^{u,k}\}_{k \in \mathbb{N}^*}$ converges uniformly to $\hat{H}^u$ in such set.

Recall that the set $\mathcal{D}$ has positive μ -measure, and also that $\hat{\mu}$ is the pullback of μ by the semiconjugation h . In view of this, it is sufficient to show that the set $\hat{\mathcal{D}}$ of points $\hat{\mathbf{q}}$ with $q = h(\hat{\mathbf{q}}) \in \mathcal{D}$ satisfies our needs.

Let us fix some $\hat{\mathbf{q}} \in \hat{\mathcal{D}}$ for now. With arguments similar to the ones used to prove Lemma 2.2.2, we can show that the new collections $\hat{H}^u|_{W_{\text{loc}}^u(\hat{\mathbf{q}})}$ and $\hat{H}^{u,k}|_{W_{\text{loc}}^u(\hat{\mathbf{q}})}$ of isomorphisms

$$\begin{aligned} \hat{H}_{\hat{\mathbf{x}}, \hat{\mathbf{y}}}^u &= H_{h(\hat{\mathbf{x}}), h(\hat{\mathbf{y}})}^u \\ \text{and} \quad &\text{for every } (\hat{\mathbf{x}}, \hat{\mathbf{y}}) \in W_{\text{loc}}^u(\hat{\mathbf{q}}). \\ \hat{H}_{\hat{\mathbf{x}}, \hat{\mathbf{y}}}^{u,k} &= H_{h(\hat{\mathbf{x}}), h(\hat{\mathbf{y}})}^{u,k} \end{aligned}$$

are u -holonomies for $\hat{A}$ and $\hat{A}_k$, respectively, over $\hat{f}$. In addition, it is possible to prove that $\hat{H}^{u,k} \rightarrow \hat{H}^u$ uniformly on $W_{\text{loc}}^u(\hat{\mathbf{q}})$. Since we take an arbitrary point $\hat{\mathbf{q}} \in \hat{\mathcal{D}}$, we conclude that the set $\hat{\mathcal{D}} = h^{-1}(\mathcal{D})$ has the property we need.

Observe that we are in the hypotheses of Theorem 2.1.2. Such result states that every linear cocycle $\hat{A}_k$ has non-zero Lyapunov exponents for all sufficiently large $k \in \mathbb{N}^*$. By the use of Equation (2.4), we can conclude the same but for every A_k . Therefore the proof of Theorem C is completed.

Our objective from this point is to prove Theorems 2.1.1 and 2.1.2. In order to do this we need some other results that are used in the cited proofs. For

instance, in the next chapter we establish a sufficient condition to the continuity of Lyapunov exponents.

Chapter 3

A sufficient condition to continuity

The present chapter is dedicated to exhibit a sufficient condition to the continuity problem of Lyapunov exponents. Such condition is used to prove both Theorem 2.1.1 in Section 5.1 and Theorem 2.1.2 in Section 5.2.

Throughout this chapter we work with the subshift setting. Let $(\hat{\Sigma}, \hat{f})$ be a subshift of finite type, and let $\hat{\mu}$ be an $\hat{f}$ -invariant probability measure.

3.1 Oseledets Theorem

The next result is a refinement of the Furstenberg–Kesten Theorem (as in Theorem 1.2.1). Indeed, given a linear cocycle $\hat{B}: \hat{\Sigma} \rightarrow \text{SL}(2, \mathbb{R})$, its conclusion is formulated in terms of the norms of the images $\hat{B}^n(\hat{\mathbf{x}}) \cdot v$, for every non-zero vector $v \in \mathbb{R}^2$, rather than the norms of the matrices $\hat{B}^n(\hat{\mathbf{x}})$ themselves.

Theorem 3.1.1 (Oseledets). *Consider a subshift of finite type $(\hat{\Sigma}, \hat{f})$ and an $\hat{f}$ -invariant probability measure $\hat{\mu}$. Given a linear cocycle $\hat{B}: \hat{\Sigma} \rightarrow \text{SL}(2, \mathbb{R})$ over $\hat{f}$, its extremal Lyapunov exponents satisfy, for $\hat{\mu}$ -almost every $\hat{\mathbf{x}} \in \hat{\Sigma}$:*

1. *Either $\lambda_-(\hat{B}) = 0 = \lambda_+(\hat{B})$ and*

$$\lim_{n \in \mathbb{Z}} \frac{1}{n} \log \|\hat{B}^n(\hat{\mathbf{x}}) \cdot v\| = \lambda_{\pm}(\hat{B}) \text{ for all } v \in \mathbb{R}^2;$$

2. *Or $\lambda_-(\hat{B}) < 0 < \lambda_+(\hat{B})$ and there is a decomposition $\mathbb{R}^2 = E_{\hat{\mathbf{x}}}^u \oplus E_{\hat{\mathbf{x}}}^s$ such that*

$$\begin{aligned} \lim_{n \in \mathbb{N}^*} \frac{1}{n} \log \|\hat{B}^n(\hat{\mathbf{x}}) \cdot v\| &= \begin{cases} \lambda_-(\hat{B}) & \text{if } v \in E_{\hat{\mathbf{x}}}^s \setminus \{0\} \\ \lambda_+(\hat{B}) & \text{if } v \in \mathbb{R}^2 \setminus E_{\hat{\mathbf{x}}}^s \end{cases} \\ \lim_{n \in \mathbb{N}^*} \frac{1}{n} \log \|\hat{B}^{-n}(\hat{\mathbf{x}}) \cdot v\|^{-1} &= \begin{cases} \lambda_+(\hat{B}) & \text{if } v \in E_{\hat{\mathbf{x}}}^u \setminus \{0\} \\ \lambda_-(\hat{B}) & \text{if } v \in \mathbb{R}^2 \setminus E_{\hat{\mathbf{x}}}^u. \end{cases} \end{aligned}$$

The direct sum decomposition $\mathbb{R}^2 = E_{\hat{\mathbf{x}}}^u \oplus E_{\hat{\mathbf{x}}}^s$ is $F_{\hat{B}}$ -invariant, meaning that $\hat{B}(\hat{\mathbf{x}}) \cdot E_{\hat{\mathbf{x}}}^u = E_{\hat{f}(\hat{\mathbf{x}})}^u$ and $\hat{B}(\hat{\mathbf{x}}) \cdot E_{\hat{\mathbf{x}}}^s = E_{\hat{f}(\hat{\mathbf{x}})}^s$. Moreover, the angle between these two lines decreases subexponentially along orbits:

$$\lim_{n \in \mathbb{Z}} \frac{1}{n} \log |\sin \angle(E_{\hat{f}^n(\hat{\mathbf{x}})}^u, E_{\hat{f}^n(\hat{\mathbf{x}})}^s)| = 0.$$

We would like to remark that the above theorem could be formulated in a more general setting, like in Theorem 1.2.1.

3.2 Invariant su -states

In the present section we discuss a type of measures in the product space $\hat{\Sigma} \times \mathbb{P}\mathbb{R}^2$ that are invariant by holonomies in the following sense. First, let us consider a linear cocycle $\hat{B}: \hat{\Sigma} \rightarrow \mathrm{SL}(2, \mathbb{R})$ admitting s - and u -holonomies $H^{s, \hat{B}}$ and $H^{u, \hat{B}}$, respectively (both holonomies can be either uniform or non-uniform). Recall that $F_{\hat{B}}$ yields a projective cocycle $\mathbb{P}F_{\hat{B}}$, as in Equation (1.2).

Take any probability measure $\hat{m}_{\hat{B}}$ on $\hat{\Sigma} \times \mathbb{P}\mathbb{R}^2$ projecting down to $\hat{\mu}$ on $\hat{\Sigma}$. We define a *disintegration* of $\hat{m}_{\hat{B}}$ along the fibres $\{\hat{\mathbf{x}}\} \times \mathbb{P}\mathbb{R}^2$ ($\hat{\mathbf{x}} \in \hat{\Sigma}$) as a measurable family $\{\hat{m}_{\hat{B}, \hat{\mathbf{x}}}\}_{\hat{\mathbf{x}} \in \hat{\Sigma}}$ composed by probability measures supported on the projective plane $\mathbb{P}\mathbb{R}^2$ and such that

$$\hat{m}_{\hat{B}}(E) = \int \hat{m}_{\hat{B}, \hat{\mathbf{x}}}(\{t \in \mathbb{P}\mathbb{R}^2: (\hat{\mathbf{x}}, t) \in E\}) d\hat{\mu}(\hat{\mathbf{x}}),$$

for any measurable set $E \subset \hat{\Sigma} \times \mathbb{P}\mathbb{R}^2$. The existence of such disintegration is due to the work of Rokhlin (we refer to [Rok49]).

It is important to point out that a disintegration of a given measure is unique up to a zero measure set. Observe also that the following both sentences are equivalent:

- (a) $\hat{m}_{\hat{B}}$ is $\mathbb{P}F_{\hat{B}}$ -invariant; and
- (b) $\hat{m}_{\hat{B}}$ admits a $\mathbb{P}\hat{B}$ -invariant disintegration $\{\hat{m}_{\hat{B}, \hat{\mathbf{x}}}\}_{\hat{\mathbf{x}} \in \hat{\Sigma}}$ along the fibres, meaning that $\mathbb{P}\hat{B}(\hat{\mathbf{x}})_* \hat{m}_{\hat{B}, \hat{\mathbf{x}}} = \hat{m}_{\hat{B}, \hat{f}(\hat{\mathbf{x}})}$ for $\hat{\mu}$ -almost every $\hat{\mathbf{x}} \in \hat{\Sigma}$.

We say that a disintegration of $\hat{m}_{\hat{B}}$ is u -invariant with respect to $H^{u, \hat{B}}$ if:

$$(\mathbb{P}H_{\hat{\mathbf{x}}, \hat{\mathbf{y}}}^{u, \hat{B}})_* \hat{m}_{\hat{B}, \hat{\mathbf{x}}} = \hat{m}_{\hat{B}, \hat{\mathbf{y}}} \quad \text{for } \hat{\mu}\text{-almost every } \hat{\mathbf{x}} \in \hat{\Sigma} \text{ and any } \hat{\mathbf{y}} \in W_{\mathrm{loc}}^u(\hat{\mathbf{x}}).$$

We define an *s*-invariant disintegration in an analogous way.

Now, a probability measure $\hat{m}_{\hat{B}}$ on $\hat{\Sigma} \times \mathbb{P}\mathbb{R}^2$ projecting down to $\hat{\mu}$ is called a *u-state* for $(\hat{B}, H^{u, \hat{B}})$ if it is $\mathbb{P}F_{\hat{B}}$ -invariant and if it also admits a u -invariant disintegration with respect to $H^{u, \hat{B}}$. We would like to remark that such holonomy could be either uniform or non-uniform in this definition. The notion of *s-states* for $(\hat{B}, H^{s, \hat{B}})$ is similar.

Finally, we say that $\hat{m}_{\hat{B}}$ is an *su-state* for $(\hat{B}, H^{s, \hat{B}}, H^{u, \hat{B}})$ whenever it is both an *s*-state and a *u*-state. This kind of measures has a particular feature that is summarised in the following result extracted from [FM21] that was inspired by the ideas of [AV10, Proposition 4.8].

Proposition 3.2.1 (Proposition 5 of [FM21]). *Consider a subshift of finite type $(\hat{\Sigma}, \hat{f})$ and an $\hat{f}$ -ergodic probability measure $\hat{\mu}$ that has local product structure and full support. Let $\hat{B}: \hat{\Sigma} \rightarrow \mathrm{SL}(2, \mathbb{R})$ be a linear cocycle admitting a uniform*

s -holonomy and a u -holonomy (either uniform or non-uniform). If $\hat{m}_{\hat{B}}$ is an su -state for $\hat{B}$ on $\hat{\Sigma} \times \mathbb{P}\mathbb{R}^2$, then it admits a continuous disintegration which is invariant by the projective cocycle $\mathbb{P}\hat{B}$ and also by both holonomies. Remark that, if the u -holonomy is uniform, then such holonomy invariance is on every point; but if it is non-uniform, then the invariance is on $\hat{\mu}$ -almost every point.

Although the holonomies of the previous result do not come from the non-uniform fibre-bunching condition, as in [FM21], we do not need to give a different proof here since it follows strictly using the holonomies defined in the present work. The same is true for the next proposition.

We would like to remark that, given a linear cocycle admitting both holonomies, we cannot expect to find an su -state for it. Nevertheless, there is one sufficient condition to the existence of su -states. Such result is called the Invariance Principle, and it was first stated by Ledrappier [Led86] and has been extended for other settings since. One of those extensions is due to Freijo and Marin in [FM21], and it goes as follows.

Proposition 3.2.2 (Invariance Principle, Theorem 1 of [FM21]). *Consider the same hypotheses of Proposition 3.2.1 above. If the linear cocycle has null Lyapunov exponents, then every $\mathbb{P}\hat{B}$ -invariant probability measure projecting down to $\hat{\mu}$ is an su -state.*

Remark 3.2.3. In the proof of the Invariance Principle cited above it is used a property of the non-uniform holonomies H^u that came from the non-uniform fibre-bunching condition. Indeed, it is needed that the norms $\|H_{f^j(y), f^j(z)}^u\|$ are uniformly bounded for all iterates of points $y, z \in W_{\text{loc}}^u(x)$, where x is in a u -holonomy block. Thus we added such property in Item (NU.4) at Definition 4 of non-uniform holonomy.

In the following we present a straightforward result about linear cocycles in general (though we opt to state it in our setting) that will be very useful in the rest of the text. It was partially based on [Led84, Proposition 5.1]. Observe that its final conclusion on the existence of a u -state contrast with our discussion above about su -states.

Proposition 3.2.4. *Consider a subshift of finite type $(\hat{\Sigma}, \hat{f})$ and an $\hat{f}$ -ergodic probability measure $\hat{\mu}$. Every cocycle $\hat{B}: \hat{\Sigma} \rightarrow \text{SL}(2, \mathbb{R})$ admits a $\mathbb{P}F_{\hat{B}}$ -ergodic probability measure $\hat{m}_{\hat{B}}$ in $\hat{\Sigma} \times \mathbb{P}\mathbb{R}^2$ that projects down to $\hat{\mu}$, and satisfies the equation:*

$$\lambda_+(\hat{B}) = \int \Phi_{\hat{B}} d\hat{m}_{\hat{B}}, \quad \text{where} \quad \Phi_{\hat{B}}(\hat{\mathbf{x}}, [v]) = \log \frac{\|\hat{B}(\hat{\mathbf{x}}) \cdot v\|}{\|v\|}. \quad (3.1)$$

Furthermore, if $\hat{B}$ also admits a u -holonomy, then we can assume that the measure $\hat{m}_{\hat{B}}$ is a u -state.

Proof. Our first step is to prove the existence of the measure satisfying Equation (3.1). Let us deal with the case $\lambda_+(\hat{B}) = 0$ first. Notice that, for all $n \in \mathbb{N}^*$

and any $(\hat{\mathbf{x}}, [v]) \in \hat{\Sigma} \times \mathbb{P}\mathbb{R}^2$, we have

$$\sum_{j=0}^{n-1} \Phi_{\hat{B}} \circ \mathbb{P}F_{\hat{B}}^j(\hat{\mathbf{x}}, [v]) = \log \frac{\|\hat{B}^n(\hat{\mathbf{x}}) \cdot v\|}{\|v\|} \in [-\log \|\hat{B}^n(\hat{\mathbf{x}})^{-1}\|, \log \|\hat{B}^n(\hat{\mathbf{x}})\|].$$

Besides, by Theorem 1.2.1, for $\hat{\mu}$ -almost every $\hat{\mathbf{x}} \in \hat{\Sigma}$,

$$\lim_{n \in \mathbb{N}^*} \frac{1}{n} \log \|\hat{B}^n(\hat{\mathbf{x}})\| = \lim_{n \in \mathbb{N}^*} \frac{1}{n} \log \|\hat{B}^n(\hat{\mathbf{x}})^{-1}\| = \lambda_+(\hat{B}).$$

Thus, given any $\mathbb{P}F_{\hat{B}}$ -invariant (in particular, $\mathbb{P}F_{\hat{B}}$ -ergodic projecting down to $\hat{\mu}$) probability measure $\hat{m}_{\hat{B}}$, the two equations above give us that

$$\lim_{n \in \mathbb{N}} \frac{1}{n} \sum_{j=0}^{n-1} \Phi_{\hat{B}} \circ \mathbb{P}F_{\hat{B}}^j(\hat{\mathbf{x}}, [v]) \in [-\lambda_+(\hat{B}), \lambda_+(\hat{B})] = \{0\},$$

for $\hat{m}_{\hat{B}}$ -almost every $(\hat{\mathbf{x}}, [v]) \in \hat{\Sigma} \times \mathbb{P}\mathbb{R}^2$. Owing to Birkhoff Ergodic Theorem, it follows that $\int \Phi_{\hat{B}} d\hat{m}_{\hat{B}} = 0 = \lambda_+(\hat{B})$, as we wanted.

It remains to analyse when $\lambda_+(\hat{B}) > 0$. In this case, Theorem 3.1.1 asserts that exists a $\hat{B}$ -invariant splitting $\mathbb{R}^2 = E_{\hat{\mathbf{x}}}^u \oplus E_{\hat{\mathbf{x}}}^s$ defined at $\hat{\mu}$ -almost every $\hat{\mathbf{x}} \in \hat{\Sigma}$, and such that

$$\lim_{n \in \mathbb{Z}} \frac{1}{n} \log \|\hat{B}^n(\hat{\mathbf{x}}) \cdot v\| = \begin{cases} \lambda_+(\hat{B}) & \text{if } v \in E_{\hat{\mathbf{x}}}^u \setminus \{0\}, \\ \lambda_-(\hat{B}) & \text{if } v \in E_{\hat{\mathbf{x}}}^s \setminus \{0\}. \end{cases}$$

Let $\hat{m}_{\hat{B}}$ be a probability measure in $\hat{\Sigma} \times \mathbb{P}\mathbb{R}^2$ projecting down to $\hat{\mu}$ whose conditional probability along each fibre $\{\hat{\mathbf{x}}\} \times \mathbb{P}\mathbb{R}^2$ is supported on $E_{\hat{\mathbf{x}}}^u$. Explicitly, $\hat{m}_{\hat{B}} = \int \delta_{(\hat{\mathbf{x}}, E_{\hat{\mathbf{x}}}^u)} d\hat{\mu}(\hat{\mathbf{x}})$. Observe that $\hat{m}_{\hat{B}}$ is $\mathbb{P}F_{\hat{B}}$ -ergodic by construction. Finally, Birkhoff Ergodic Theorem implies that

$$\begin{aligned} \int \Phi_{\hat{B}} d\hat{m}_{\hat{B}} &= \int \lim_{n \in \mathbb{N}^*} \frac{1}{n} \sum_{j=0}^{n-1} \Phi_{\hat{B}} \circ \mathbb{P}F_{\hat{B}}^j d\hat{m}_{\hat{B}} \\ &= \iint \lim_{n \in \mathbb{N}^*} \frac{1}{n} \log \frac{\|\hat{B}^n(\hat{\mathbf{x}}) \cdot v\|}{\|v\|} d\delta_{E_{\hat{\mathbf{x}}}^u}(v) d\hat{\mu}(\hat{\mathbf{x}}) \\ &= \int \lambda_+(\hat{B}) d\hat{\mu} = \lambda_+(\hat{B}), \end{aligned} \tag{3.2}$$

concluding this part.

Our last step is to prove that, if $\hat{B}$ admits a u -holonomy (either uniform or non-uniform), then the measure $\hat{m}_{\hat{B}}$ constructed above is a u -state. In order to do that we make use of the u -Invariance Principle established in [VY19, Theorem 6.1]. Therefore we need to check if

$$\lim_{n \in \mathbb{N}^*} \frac{1}{n} \log \|D\hat{B}^n(\hat{\mathbf{x}}) \cdot v\| \leq 0 \quad \text{for } \hat{m}_{\hat{B}}\text{-almost every } (\hat{\mathbf{x}}, [v]) \in \hat{\Sigma} \times \mathbb{P}\mathbb{R}^2.$$

Initially, observe that, for each $\hat{\mathbf{x}} \in \hat{\Sigma}$, the operator $D\hat{B}(\hat{\mathbf{x}}): \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is a projection such that

$$\|D\hat{B}(\hat{\mathbf{x}}) \cdot v\| = |\det \hat{B}(\hat{\mathbf{x}})| \left(\frac{\|v\|}{\|\hat{B}(\hat{\mathbf{x}}) \cdot v\|} \right)^2 = \left(\frac{\|v\|}{\|\hat{B}(\hat{\mathbf{x}}) \cdot v\|} \right)^2 \quad (3.3)$$

for all $v \in \mathbb{R}^2$. Thus, defining the continuous function $\varphi_{\hat{B}}: \hat{\Sigma} \times \mathbb{P}\mathbb{R}^2 \rightarrow \mathbb{R}$ by

$$\varphi_{\hat{B}}(\hat{\mathbf{x}}, [v]) = \log \|D\hat{B}(\hat{\mathbf{x}}) \cdot v\|,$$

then Equations (3.2) and (3.3) allow us to obtain that

$$\int \varphi_{\hat{B}} d\hat{m}_{\hat{B}} = -2 \int \log \frac{\|\hat{B}(\hat{\mathbf{x}}) \cdot v\|}{\|v\|} d\hat{m}_{\hat{B}}(\hat{\mathbf{x}}, [v]) = -2 \int \Phi_{\hat{B}} d\hat{m}_{\hat{B}} = -2\lambda_+(\hat{B}).$$

Finally, using Birkhoff Ergodic Theorem on $\varphi_{\hat{B}}$ in addition to the ergodicity of $\hat{m}_{\hat{B}}$ we conclude, for $\hat{m}_{\hat{B}}$ -almost every $(\hat{\mathbf{x}}, [v]) \in \hat{\Sigma} \times \mathbb{P}\mathbb{R}^2$, that

$$\begin{aligned} \lim_{n \in \mathbb{N}} \frac{1}{n} \log \|D\hat{B}^n(\hat{\mathbf{x}}) \cdot v\| &= \lim_{n \in \mathbb{N}} \frac{1}{n} \sum_{j=0}^{n-1} \varphi_{\hat{B}} \circ \mathbb{P}F_{\hat{B}}^j(\hat{\mathbf{x}}, [v]) \\ &= \int \varphi_{\hat{B}} d\hat{m}_{\hat{B}} = -2\lambda_+(\hat{B}) \leq 0, \end{aligned}$$

as we wanted. This completes the proof. $\square$

3.3 Diagonal cocycles

Recall we are considering a subshift of finite type $(\hat{\Sigma}, \hat{f})$ and an $\hat{f}$ -invariant probability measure $\hat{\mu}$. Our goal in this section is to prove that every linear cocycle satisfying the 2-invariance condition can be diagonalised. We also discuss some properties that arises from this process. Once again, the results of this section could be stated in a more general setting, like hyperbolic homeomorphisms.

Proposition 3.3.1. *Consider a subshift of finite type $(\hat{\Sigma}, \hat{f})$, and a linear cocycle $(\hat{B}, H^s) \in \mathcal{H}_s^0(2, \hat{\Sigma})$ satisfying the 2-invariance condition. There is a continuous change of coordinates that diagonalise the matrix representation of $\hat{B}$.*

Proof. Fix for now a point $\hat{\mathbf{x}} \in \hat{\Sigma}$. We denote the matrix $\hat{B}(\hat{\mathbf{x}})$ in $\text{SL}(2, \mathbb{R})$ by $\begin{pmatrix} a(\hat{\mathbf{x}}) & b(\hat{\mathbf{x}}) \\ c(\hat{\mathbf{x}}) & d(\hat{\mathbf{x}}) \end{pmatrix}$. Also, consider two unitary vectors $E_1(\hat{\mathbf{x}}), E_2(\hat{\mathbf{x}})$ such that $\mathbb{P}E_1(\hat{\mathbf{x}}) = \xi_1(\hat{\mathbf{x}})$, $\mathbb{P}E_2(\hat{\mathbf{x}}) = \xi_2(\hat{\mathbf{x}})$ and $\langle E_1(\hat{\mathbf{x}}); E_2(\hat{\mathbf{x}}) \rangle > 0$. Recall that such maps $\xi_1, \xi_2: \hat{\Sigma} \rightarrow \mathbb{P}\mathbb{R}^2$ are the invariant sections given by the 2-invariance condition of the cocycle $\hat{B}$.

We would like to observe that, with this construction, the $\mathbb{P}\hat{B}$ -invariance of sections guarantees that

$$\frac{\hat{B}(\hat{\mathbf{x}}) \cdot E_i(\hat{\mathbf{x}})}{\|\hat{B}(\hat{\mathbf{x}}) \cdot E_i(\hat{\mathbf{x}})\|} = E_i(\hat{f}(\hat{\mathbf{x}})) \quad \text{for } i = 1, 2. \quad (3.4)$$

Besides, we can affirm that the set $\mathcal{E}(\hat{\mathbf{x}}) = \{E_1(\hat{\mathbf{x}}), E_2(\hat{\mathbf{x}})\}$ is a basis for $\mathbb{R}^2$ by the transversality of sections.

If we denote the coordinates of each vector $E_i(\hat{\mathbf{x}})$ by $(X_i(\hat{\mathbf{x}}), Y_i(\hat{\mathbf{x}}))$ for $i = 1, 2$, then the transition matrix from the canonical basis to $\mathcal{E}(\hat{\mathbf{x}})$ is

$$R(\hat{\mathbf{x}}) = \frac{1}{\varphi(\hat{\mathbf{x}})} \begin{pmatrix} Y_2(\hat{\mathbf{x}}) & -X_2(\hat{\mathbf{x}}) \\ -Y_1(\hat{\mathbf{x}}) & X_1(\hat{\mathbf{x}}) \end{pmatrix},$$

where $\varphi(\hat{\mathbf{x}}) = X_1(\hat{\mathbf{x}})Y_2(\hat{\mathbf{x}}) - X_2(\hat{\mathbf{x}})Y_1(\hat{\mathbf{x}})$, and its inverse is

$$R(\hat{\mathbf{x}})^{-1} = \begin{pmatrix} X_1(\hat{\mathbf{x}}) & X_2(\hat{\mathbf{x}}) \\ Y_1(\hat{\mathbf{x}}) & Y_2(\hat{\mathbf{x}}) \end{pmatrix}.$$

Notice that the continuity of the sections implies that the change of coordinates R is also continuous when we vary the point $\hat{\mathbf{x}}$ in the base.

We must show then that matrix $R(\hat{f}(\hat{\mathbf{x}})) \hat{B}(\hat{\mathbf{x}}) R(\hat{\mathbf{x}})^{-1}$ is diagonal. In this regard, let us begin by observing that Equation (3.4) written in coordinates gives us:

$$\begin{cases} a(\hat{\mathbf{x}})X_i(\hat{\mathbf{x}}) + b(\hat{\mathbf{x}})Y_i(\hat{\mathbf{x}}) = \|\hat{B}(\hat{\mathbf{x}}) \cdot E_i(\hat{\mathbf{x}})\| X_i(\hat{f}(\hat{\mathbf{x}})) \\ c(\hat{\mathbf{x}})X_i(\hat{\mathbf{x}}) + d(\hat{\mathbf{x}})Y_i(\hat{\mathbf{x}}) = \|\hat{B}(\hat{\mathbf{x}}) \cdot E_i(\hat{\mathbf{x}})\| Y_i(\hat{f}(\hat{\mathbf{x}})) \end{cases} \quad \text{for } i = 1, 2.$$

Thus,

$$\begin{aligned} R(\hat{f}(\hat{\mathbf{x}})) \hat{B}(\hat{\mathbf{x}}) R(\hat{\mathbf{x}})^{-1} &= \frac{1}{\varphi(\hat{f}(\hat{\mathbf{x}}))} \begin{pmatrix} Y_2(\hat{f}(\hat{\mathbf{x}})) & -X_2(\hat{f}(\hat{\mathbf{x}})) \\ -Y_1(\hat{f}(\hat{\mathbf{x}})) & X_1(\hat{f}(\hat{\mathbf{x}})) \end{pmatrix} \begin{pmatrix} a(\hat{\mathbf{x}}) & b(\hat{\mathbf{x}}) \\ c(\hat{\mathbf{x}}) & d(\hat{\mathbf{x}}) \end{pmatrix} \begin{pmatrix} X_1(\hat{\mathbf{x}}) & X_2(\hat{\mathbf{x}}) \\ Y_1(\hat{\mathbf{x}}) & Y_2(\hat{\mathbf{x}}) \end{pmatrix} \\ &= \frac{1}{\varphi(\hat{f}(\hat{\mathbf{x}}))} \begin{pmatrix} Y_2(\hat{f}(\hat{\mathbf{x}})) & -X_2(\hat{f}(\hat{\mathbf{x}})) \\ -Y_1(\hat{f}(\hat{\mathbf{x}})) & X_1(\hat{f}(\hat{\mathbf{x}})) \end{pmatrix} \begin{pmatrix} a(\hat{\mathbf{x}})X_1(\hat{\mathbf{x}}) + b(\hat{\mathbf{x}})Y_1(\hat{\mathbf{x}}) & a(\hat{\mathbf{x}})X_2(\hat{\mathbf{x}}) + b(\hat{\mathbf{x}})Y_2(\hat{\mathbf{x}}) \\ c(\hat{\mathbf{x}})X_1(\hat{\mathbf{x}}) + d(\hat{\mathbf{x}})Y_1(\hat{\mathbf{x}}) & c(\hat{\mathbf{x}})X_2(\hat{\mathbf{x}}) + d(\hat{\mathbf{x}})Y_2(\hat{\mathbf{x}}) \end{pmatrix} \\ &= \frac{1}{\varphi(\hat{f}(\hat{\mathbf{x}}))} \begin{pmatrix} Y_2(\hat{f}(\hat{\mathbf{x}})) & -X_2(\hat{f}(\hat{\mathbf{x}})) \\ -Y_1(\hat{f}(\hat{\mathbf{x}})) & X_1(\hat{f}(\hat{\mathbf{x}})) \end{pmatrix} \begin{pmatrix} \|\hat{B}(\hat{\mathbf{x}}) \cdot E_1(\hat{\mathbf{x}})\| X_1(\hat{f}(\hat{\mathbf{x}})) & \|\hat{B}(\hat{\mathbf{x}}) \cdot E_2(\hat{\mathbf{x}})\| X_2(\hat{f}(\hat{\mathbf{x}})) \\ \|\hat{B}(\hat{\mathbf{x}}) \cdot E_1(\hat{\mathbf{x}})\| Y_1(\hat{f}(\hat{\mathbf{x}})) & \|\hat{B}(\hat{\mathbf{x}}) \cdot E_2(\hat{\mathbf{x}})\| Y_2(\hat{f}(\hat{\mathbf{x}})) \end{pmatrix} \\ &= \frac{1}{\varphi(\hat{f}(\hat{\mathbf{x}}))} \begin{pmatrix} \|\hat{B}(\hat{\mathbf{x}}) \cdot E_1(\hat{\mathbf{x}})\| \varphi(\hat{f}(\hat{\mathbf{x}})) & 0 \\ 0 & \|\hat{B}(\hat{\mathbf{x}}) \cdot E_2(\hat{\mathbf{x}})\| \varphi(\hat{f}(\hat{\mathbf{x}})) \end{pmatrix} \\ &= \begin{pmatrix} \|\hat{B}(\hat{\mathbf{x}}) \cdot E_1(\hat{\mathbf{x}})\| & 0 \\ 0 & \|\hat{B}(\hat{\mathbf{x}}) \cdot E_2(\hat{\mathbf{x}})\| \end{pmatrix}, \end{aligned}$$

as we wanted. $\square$

Remark 3.3.2. In order to construct the diagonalised matrix as above we only need the first two items in Definition 2; namely, the $\mathbb{P}\hat{B}$ -invariance and transversality. It is worth to mention that the last item (that is, the $\mathbb{P}H^s$ -invariance) is crucial to diagonalise the s -holonomy. This is done in Lemma 3.3.4.

It is worth to ask whether or not the process done at last proposition interferes in the continuity problem of Lyapunov exponents. For this matter, we need to check first that such change of coordinates preserves continuity. In other words, given a sequence $\{\hat{B}_k\}_{k \in \mathbb{N}^*}$ of linear cocycles converging in the C^0 -topology to a cocycle $\hat{B}$ that satisfies the 2-invariance condition, it is true that the new sequence formed by $R(\hat{f}(\hat{\mathbf{x}})) \hat{B}_k(\hat{\mathbf{x}}) R(\hat{\mathbf{x}})^{-1}$ still converges to the diagonal cocycle $R(\hat{f}(\hat{\mathbf{x}})) \hat{B}(\hat{\mathbf{x}}) R(\hat{\mathbf{x}})^{-1}$ in the same topology as before? The answer is *yes* because the transition map $R: \hat{\Sigma} \rightarrow \text{SL}(2, \mathbb{R})$ is continuous.

Now we point out that the mentioned process done in Proposition 3.3.1 does not alter the value of the Lyapunov exponents. Indeed, given any $\hat{x} \in \hat{\Sigma}$, if we consider $\hat{C}(\hat{\mathbf{x}}) = R(\hat{f}(\hat{\mathbf{x}})) \hat{B}(\hat{\mathbf{x}}) R(\hat{\mathbf{x}})^{-1}$, then

$$\begin{aligned} \hat{C}^n(\hat{\mathbf{x}}) &= (R(\hat{f}(\hat{f}^{n-1}(\hat{\mathbf{x}}))) \hat{B}(\hat{f}^{n-1}(\hat{\mathbf{x}})) R(\hat{f}^{n-1}(\hat{\mathbf{x}}))^{-1}) \cdots (R(\hat{f}(\hat{\mathbf{x}})) \hat{B}(\hat{\mathbf{x}}) R(\hat{\mathbf{x}})^{-1}) \\ &= R(\hat{f}^n(\hat{\mathbf{x}})) \hat{B}(\hat{f}^{n-1}(\hat{\mathbf{x}})) \cdots \hat{B}(\hat{f}(\hat{\mathbf{x}})) \hat{B}(\hat{\mathbf{x}}) R(\hat{\mathbf{x}})^{-1} \\ &= R(\hat{f}^n(\hat{\mathbf{x}})) \hat{B}^n(\hat{\mathbf{x}}) R(\hat{\mathbf{x}})^{-1} \end{aligned}$$

for each $n \in \mathbb{N}^*$. Notice that this defines a linear cocycle $\hat{C}: \hat{\Sigma} \rightarrow \text{GL}(2, \mathbb{R})$. Hence, we can calculate its Lyapunov exponents.

It is straightforward that, for $\hat{\mu}$ -almost every $\hat{\mathbf{x}} \in \hat{\Sigma}$, we have

$$\begin{aligned} \lambda_+(\hat{C}) &= \lim_{n \in \mathbb{N}^*} \frac{1}{n} \log \|\hat{C}^n(\hat{\mathbf{x}})\| = \lim_{n \in \mathbb{N}^*} \frac{1}{n} \log \|R(\hat{f}^n(\hat{\mathbf{x}})) \hat{B}^n(\hat{\mathbf{x}}) R(\hat{\mathbf{x}})^{-1}\| \\ &\leq \lim_{n \in \mathbb{N}^*} \frac{1}{n} \log (\|R(\hat{f}^n(\hat{\mathbf{x}}))\| \|\hat{B}^n(\hat{\mathbf{x}})\| \|R(\hat{\mathbf{x}})^{-1}\|) \\ &= \lim_{n \in \mathbb{N}^*} \frac{1}{n} \log \|R(\hat{f}^n(\hat{\mathbf{x}}))\| + \lim_{n \in \mathbb{N}^*} \frac{1}{n} \log \|\hat{B}^n(\hat{\mathbf{x}})\| + \lim_{n \in \mathbb{N}^*} \frac{1}{n} \log \|R(\hat{\mathbf{x}})^{-1}\| \\ &= 0 + \lambda_+(\hat{B}) + 0 = \lambda_+(\hat{B}). \end{aligned}$$

On the other hand, using the analogous expression

$$\hat{B}^n(\hat{\mathbf{x}}) = R(\hat{f}^n(\hat{\mathbf{x}}))^{-1} \hat{C}^n(\hat{\mathbf{x}}) R(\hat{\mathbf{x}})$$

for all $n \in \mathbb{N}^*$, the same argument concludes the other inequality. Therefore, $\lambda_+(\hat{C}) = \lambda_+(\hat{B})$. A similar calculation proves that this is also true for the lower Lyapunov exponent λ_- .

As one could observe, the above-defined cocycle $\hat{C}$ takes value in $\text{GL}(2, \mathbb{R})$. However, as we have already done in Section 1.2, we can reduce to the $\text{SL}(2, \mathbb{R})$ case: we can consider instead the linear cocycle $\hat{D}: \hat{\Sigma} \rightarrow \text{SL}(2, \mathbb{R})$ defined by

$$\hat{D}(\hat{\mathbf{x}}) = (\det(\hat{C}(\hat{\mathbf{x}}))^{-\frac{1}{2}} \hat{C}(\hat{\mathbf{x}})) = \begin{pmatrix} \sqrt{\frac{\|\hat{B}(\hat{\mathbf{x}}) \cdot E_1(\hat{\mathbf{x}})\|}{\|\hat{B}(\hat{\mathbf{x}}) \cdot E_2(\hat{\mathbf{x}})\|}} & 0 \\ 0 & \sqrt{\frac{\|\hat{B}(\hat{\mathbf{x}}) \cdot E_2(\hat{\mathbf{x}})\|}{\|\hat{B}(\hat{\mathbf{x}}) \cdot E_1(\hat{\mathbf{x}})\|}} \end{pmatrix}.$$

Up to a change in the indexes, we can assume $\|\hat{B}(\hat{\mathbf{x}}) \cdot E_1(\hat{\mathbf{x}})\| \geq \|\hat{B}(\hat{\mathbf{x}}) \cdot E_2(\hat{\mathbf{x}})\|$ for all $\hat{\mathbf{x}} \in \hat{\Sigma}$. Finally, like in the above-mentioned section, we conclude that continuity of $\lambda_{\pm}(\hat{D})$ implies continuity of $\lambda_{\pm}(\hat{B})$.

Therefore, every linear cocycle $\hat{B}$ satisfying the 2-invariance condition can be considered in the diagonal form:

$$\hat{B}(\hat{\mathbf{x}}) = \begin{pmatrix} a(\hat{\mathbf{x}}) & 0 \\ 0 & a(\hat{\mathbf{x}})^{-1} \end{pmatrix} \quad \text{where } a(\hat{\mathbf{x}}) \geq 1 \text{ for every } \hat{\mathbf{x}} \in \hat{\Sigma}. \quad (3.5)$$

The following result concerns the preservation of holonomies in the diagonalisation process.

Lemma 3.3.3. *Consider a linear cocycle $\hat{B}: \hat{\Sigma} \rightarrow \text{SL}(2, \mathbb{R})$ admitting an invariant s -holonomy H^s (it could be either uniform or non-uniform). The linear cocycle defined by $\hat{C}(\hat{\mathbf{x}}) = R(\hat{f}(\hat{\mathbf{x}}))\hat{B}(\hat{\mathbf{x}})R(\hat{\mathbf{x}})^{-1}$ also admits an invariant s -holonomy. This is also true if we consider u -holonomies instead.*

Proof. We are going to focus on the stable case, as the other is similar. Let us define, for each pair of points $\hat{\mathbf{x}}, \hat{\mathbf{y}} \in \hat{\Sigma}$ in the same local stable set, the linear isomorphism:

$$\hat{H}_{\hat{\mathbf{x}}, \hat{\mathbf{y}}}^s = R(\hat{\mathbf{y}})H_{\hat{\mathbf{x}}, \hat{\mathbf{y}}}^s R(\hat{\mathbf{x}})^{-1}. \quad (3.6)$$

Since H^s is an s -holonomy, we have:

(i) if $\hat{\mathbf{x}}, \hat{\mathbf{y}}, \hat{\mathbf{z}} \in \hat{\Sigma}$ are in the same local stable set, then

$$\begin{aligned} \hat{H}_{\hat{\mathbf{x}}, \hat{\mathbf{z}}}^s &= R(\hat{\mathbf{z}})H_{\hat{\mathbf{x}}, \hat{\mathbf{z}}}^s R(\hat{\mathbf{x}})^{-1} = R(\hat{\mathbf{z}})H_{\hat{\mathbf{y}}, \hat{\mathbf{z}}}^s H_{\hat{\mathbf{x}}, \hat{\mathbf{y}}}^s R(\hat{\mathbf{x}})^{-1} \\ &= R(\hat{\mathbf{z}})H_{\hat{\mathbf{y}}, \hat{\mathbf{z}}}^s R(\hat{\mathbf{y}})^{-1} R(\hat{\mathbf{y}})H_{\hat{\mathbf{x}}, \hat{\mathbf{y}}}^s R(\hat{\mathbf{x}})^{-1} = \hat{H}_{\hat{\mathbf{y}}, \hat{\mathbf{z}}}^s \hat{H}_{\hat{\mathbf{x}}, \hat{\mathbf{y}}}^s \end{aligned}$$

and

$$\hat{H}_{\hat{\mathbf{x}}, \hat{\mathbf{x}}}^s = R(\hat{\mathbf{x}})H_{\hat{\mathbf{x}}, \hat{\mathbf{x}}}^s R(\hat{\mathbf{x}})^{-1} = R(\hat{\mathbf{x}})R(\hat{\mathbf{x}})^{-1} = \text{id};$$

(ii) if $\hat{\mathbf{x}}, \hat{\mathbf{y}} \in \hat{\Sigma}$ are in the same local stable set, then

$$\begin{aligned} \hat{H}_{\hat{f}(\hat{\mathbf{x}}), \hat{f}(\hat{\mathbf{y}})}^s &= R(\hat{f}(\hat{\mathbf{y}}))H_{\hat{f}(\hat{\mathbf{x}}), \hat{f}(\hat{\mathbf{y}})}^s R(\hat{f}(\hat{\mathbf{x}}))^{-1} \\ &= R(\hat{f}(\hat{\mathbf{y}}))\hat{B}(\hat{\mathbf{y}})H_{\hat{\mathbf{x}}, \hat{\mathbf{y}}}^s \hat{B}(\hat{\mathbf{x}})^{-1} R(\hat{f}(\hat{\mathbf{x}}))^{-1} \\ &= R(\hat{f}(\hat{\mathbf{y}}))\hat{B}(\hat{\mathbf{y}})R(\hat{\mathbf{y}})^{-1} R(\hat{\mathbf{y}})H_{\hat{\mathbf{x}}, \hat{\mathbf{y}}}^s R(\hat{\mathbf{x}})^{-1} R(\hat{\mathbf{x}})\hat{B}(\hat{\mathbf{x}})^{-1} R(\hat{f}(\hat{\mathbf{x}}))^{-1} \\ &= \hat{C}(\hat{\mathbf{y}})\hat{H}_{\hat{\mathbf{x}}, \hat{\mathbf{y}}}^s \hat{C}(\hat{\mathbf{x}})^{-1}; \end{aligned}$$

(iii) since the transition map $R: \hat{\Sigma} \rightarrow \text{SL}(2, \mathbb{R})$ is continuous, we can conclude that $(\hat{\mathbf{x}}, \hat{\mathbf{y}}) \mapsto \hat{H}_{\hat{\mathbf{x}}, \hat{\mathbf{y}}}^s$ vary continuously in the same manner as H^s . That is, on each local stable set if H^s is uniform, or on each s -holonomy block if H^s is non-uniform.

This completes the proof. $\square$

We would like to prove a property of the s -holonomies constructed above that is important in the rest of the text.

Lemma 3.3.4. *Let $(\hat{B}, H^s)$ be a linear cocycle satisfying the 2-invariance condition. Every linear isomorphism $\hat{H}_{\hat{\mathbf{x}}, \hat{\mathbf{y}}}^s$ defined at Equation (3.6) preserves the coordinate axes. In other words, it admits a diagonal matrix representation. This property holds for both uniform and non-uniform holonomies.*

Proof. The method is the same as done in Proposition 3.3.1.

Fix for now a pair $\hat{\mathbf{x}}, \hat{\mathbf{y}} \in \hat{\Sigma}$ such that the linear isomorphism $H_{\hat{\mathbf{x}}, \hat{\mathbf{y}}}^s$ is defined. We denote its matrix in $\text{SL}(2, \mathbb{R})$ by

$$\begin{pmatrix} a(\hat{\mathbf{x}}, \hat{\mathbf{y}}) & b(\hat{\mathbf{x}}, \hat{\mathbf{y}}) \\ c(\hat{\mathbf{x}}, \hat{\mathbf{y}}) & d(\hat{\mathbf{x}}, \hat{\mathbf{y}}) \end{pmatrix}.$$

Finally, let $\mathcal{E}(\hat{\mathbf{x}}) = \{E_1(\hat{\mathbf{x}}), E_2(\hat{\mathbf{x}})\}$ and $R(\hat{\mathbf{x}})$ be as in the proof of Proposition 3.3.1.

We would like to observe that the holonomy-invariance of sections guarantees:

$$\frac{H_{\hat{\mathbf{x}}, \hat{\mathbf{y}}}^s \cdot E_i(\hat{\mathbf{x}})}{\|H_{\hat{\mathbf{x}}, \hat{\mathbf{y}}}^s \cdot E_i(\hat{\mathbf{x}})\|} = E_i(\hat{\mathbf{y}}) \quad \text{for } i = 1, 2.$$

Once again, this can be written in coordinates as:

$$\begin{cases} a(\hat{\mathbf{x}}, \hat{\mathbf{y}})X_i(\hat{\mathbf{x}}) + b(\hat{\mathbf{x}}, \hat{\mathbf{y}})Y_i(\hat{\mathbf{x}}) = \|H_{\hat{\mathbf{x}}, \hat{\mathbf{y}}}^s \cdot E_i(\hat{\mathbf{x}})\|X_i(\hat{\mathbf{y}}) \\ c(\hat{\mathbf{x}}, \hat{\mathbf{y}})X_i(\hat{\mathbf{x}}) + d(\hat{\mathbf{x}}, \hat{\mathbf{y}})Y_i(\hat{\mathbf{x}}) = \|H_{\hat{\mathbf{x}}, \hat{\mathbf{y}}}^s \cdot E_i(\hat{\mathbf{x}})\|Y_i(\hat{\mathbf{y}}) \end{cases} \quad \text{for } i = 1, 2.$$

Finally, the same computation did in the proof of Proposition 3.3.1 shows that matrix $R(\hat{\mathbf{y}}) H_{\hat{\mathbf{x}}, \hat{\mathbf{y}}}^s R(\hat{\mathbf{x}})^{-1}$ is diagonal. Indeed, it assumes the form:

$$\hat{H}_{\hat{\mathbf{x}}, \hat{\mathbf{y}}}^s = R(\hat{\mathbf{y}}) H_{\hat{\mathbf{x}}, \hat{\mathbf{y}}}^s R(\hat{\mathbf{x}})^{-1} = \begin{pmatrix} \|H_{\hat{\mathbf{x}}, \hat{\mathbf{y}}}^s \cdot E_1(\hat{\mathbf{x}})\| & 0 \\ 0 & \|H_{\hat{\mathbf{x}}, \hat{\mathbf{y}}}^s \cdot E_2(\hat{\mathbf{x}})\| \end{pmatrix}.$$

This proves the preservation of coordinate axes. $\square$

We say that $(\hat{B}, H^s)$ is a *diagonal cocycle* when the matrix representation of both $\hat{B}$ and every $H_{\hat{\mathbf{x}}, \hat{\mathbf{y}}}^s$ is diagonal. Notice that Proposition 3.3.1 together with Lemma 3.3.4 above show that every linear cocycle satisfying the 2-invariance condition can be considered a diagonal cocycle. And this is what we are going to do from now on.

We end off this section with another direct but interesting result concerning the disintegration of measures invariant by projective cocycles in its diagonal form.

Lemma 3.3.5. *Consider a subshift of finite type $(\hat{\Sigma}, \hat{f})$ and an $\hat{f}$ -ergodic probability measure $\hat{\mu}$. Let $\hat{B}: \hat{\Sigma} \rightarrow \text{SL}(2, \mathbb{R})$ be a diagonal cocycle with non-zero Lyapunov exponents. Every $\mathbb{P}F_{\hat{B}}$ -invariant probability measure $\hat{m}_{\hat{B}}$ projecting down to $\hat{\mu}$ has constant disintegration.*

Proof. As $\lambda_+(\hat{B}) > 0$ we can define probability measures $\hat{m}^u$ and $\hat{m}^s$ supported on the Oseledets splitting (see Theorem 3.1.1). However, since we are considering $\hat{B}$ in its diagonal form, that decomposition is the trivial one. Thus the measures $\hat{m}^u$ and $\hat{m}^s$ are of the form:

$$\hat{m}^u = \int \delta_{(\hat{\mathbf{x}}, a)} d\hat{\mu}(\hat{\mathbf{x}}) \quad \text{and} \quad \hat{m}^s = \int \delta_{(\hat{\mathbf{x}}, r)} d\hat{\mu}(\hat{\mathbf{x}}), \quad (3.7)$$

where the projective points $a, r \in \mathbb{P}\mathbb{R}^2$ represent the coordinate axes. Observe that neither a nor r depends on the point $\hat{\mathbf{x}} \in \hat{\Sigma}$. Indeed, the diagonal form of $\hat{B}$ implies the preservation of those axes.

On the other hand, since the measure $\hat{m}_{\hat{B}}$ is $\mathbb{P}F_{\hat{B}}$ -invariant, and as it projects down to $\hat{\mu}$, we know that $\hat{m}_{\hat{B}}$ is in fact a convex combination of $\hat{m}^u$ and $\hat{m}^s$. What we have discussed so far means that exists a constant $0 \leq \beta \leq 1$ such that the collection $\{\hat{m}_{\hat{B}, \hat{\mathbf{x}}}\}_{\hat{\mathbf{x}} \in \hat{\Sigma}}$ with $\hat{m}_{\hat{B}, \hat{\mathbf{x}}} = \beta \delta_a + (1 - \beta) \delta_r$ is a disintegration of $\hat{m}_{\hat{B}}$. $\square$

3.4 Main proposition

In this section we prove a sufficient condition to convergence of Lyapunov exponents. It will be used to show both Theorems 2.1.1 and 2.1.2 in Sections 5.1 and 5.2, respectively.

Proposition 3.4.1. *Let $(\hat{\Sigma}, \hat{f})$ be a subshift of finite type and $\hat{\mu}$ be an $\hat{f}$ -ergodic probability measure on $\hat{\Sigma}$ that has local product structure and full support. Consider a diagonal cocycle $(\hat{A}, H^s) \in \mathcal{H}_s^0(2, \hat{\Sigma})$, and a sequence $\{(\hat{A}_k, H^{s,k})\}_{k \in \mathbb{N}^*}$ converging to $(\hat{A}, H^s)$ in $\mathcal{H}_s^0(2, \hat{\Sigma})$. Finally, choose a point $\hat{\mathbf{q}} \in \hat{\Sigma}$ containing all symbols in its non-negative entries. If, for each $k \in \mathbb{N}^*$,*

- (i) *there are projective u -holonomies h^u and $h^{u,k}$ restricted to $W_{\text{loc}}^u(\hat{\mathbf{q}})$ such that $\{h^{u,k}\}_{k \in \mathbb{N}^*}$ converges uniformly to h^u in $W_{\text{loc}}^u(\hat{\mathbf{q}})$; and*
- (ii) *the measure $\hat{m}_k$ (given by Proposition 3.2.4 for $\hat{A}_k$) admits a continuous disintegration that is invariant by the projective s -holonomies and also the u -holonomies of the previous item,*

then $\{\lambda_+(\hat{A}_k)\}_{k \in \mathbb{N}^}$ converges to $\lambda_+(\hat{A})$.*

Proof. Since null Lyapunov exponents imply continuity, we can assume from now on that $\lambda_+(\hat{A}) > 0$. Remember that we are considering the cocycle $\hat{A}$ in the diagonal form, like in Equation (3.5). We claim that one can take a periodic point $\hat{\mathbf{p}} \in \hat{\Sigma}$ such that the matrix $\hat{A}^\ell(\hat{\mathbf{p}})$ is hyperbolic, where ℓ is the period of $\hat{\mathbf{p}}$.

Indeed, for every ℓ -periodic point $\hat{\mathbf{x}} \in \hat{\Sigma}$, the logarithm of the eigenvalues of the matrix $\hat{A}^\ell(\hat{\mathbf{x}})$ are:

$$\frac{1}{\ell} \sum_{k=0}^{\ell-1} \log a(\hat{f}^k(\hat{\mathbf{x}})) \quad \text{and} \quad -\frac{1}{\ell} \sum_{k=0}^{\ell-1} \log a(\hat{f}^k(\hat{\mathbf{x}})). \quad (3.8)$$

Since the map $\hat{f}$ is transitive and ergodic with respect to $\hat{\mu}$, we conclude that it has dense periodic points. It follows that measures concentrated on periodic orbits are dense in the set of all $\hat{f}$ -invariant measures. Suppose, for contradiction, that $\hat{A}^\ell(\hat{\mathbf{x}})$ is not hyperbolic for every periodic point $\hat{\mathbf{x}}$ with period ℓ . Thus, each map of Equation (3.8) vanishes in the set of periodic points, implying that $\int \log a(\hat{\mathbf{x}}) d\mu'(\hat{\mathbf{x}}) = 0$ for every $\hat{f}$ -invariant measure μ' . Observe that this means null Lyapunov exponents of $\hat{A}$ for every $\hat{f}$ -invariant measure, contradicting our assumption of $\lambda_+(\hat{A}) > 0$. We conclude the existence of an ℓ -periodic point $\hat{\mathbf{p}} \in \hat{\Sigma}$ such that $\hat{A}^\ell(\hat{\mathbf{p}})$ is hyperbolic, as we have claimed.

Since the cocycle $\hat{A}$ is in its diagonal form, the matrix $\hat{A}^\ell(\hat{\mathbf{p}})$ is also diagonal. This means that the eigenspaces E_1 and E_2 of this matrix coincide with the coordinate axes. Consider then the projective points $a = \mathbb{P}E_1$ and $r = \mathbb{P}E_2$; that is, in terms of the dynamics yielded by the cocycle in the projective plane, a is an attractor and r is a repeller. It is easy to see that a is associated with the upper Lyapunov exponent ($E_1 = E^u$ of the Oseledets splitting), whereas r is associated with the lower Lyapunov exponent ($E_2 = E^s$). With that being said, notice that the sets

$$\{(\hat{\mathbf{x}}, a) : \hat{\mathbf{x}} \in \hat{\Sigma}\} \quad \text{and} \quad \{(\hat{\mathbf{x}}, r) : \hat{\mathbf{x}} \in \hat{\Sigma}\} \quad (3.9)$$

are ε -separated for some $\varepsilon > 0$.

We would like to remark that since hyperbolicity is an open condition, and as the sequence $\{\hat{A}_k\}_{k \in \mathbb{N}^*}$ converges to the mapping $\hat{A}$, it is true that exists a $k_0 \in \mathbb{N}^*$ such that the matrix $\hat{A}_k^\ell(\hat{\mathbf{p}})$ is also hyperbolic for every $k \geq k_0$. Fix for now any of these values k . We can consider projective points $a_k, r_k \in \mathbb{P}\mathbb{R}^2$ associated with the eigenspaces of matrix $\hat{A}_k^\ell(\hat{\mathbf{p}})$ in a manner that $a_k \rightarrow a$ and $r_k \rightarrow r$.

We want to construct continuous maps $\hat{\mathbf{x}} \mapsto a_{\hat{\mathbf{x}}}^k$ and $\hat{\mathbf{x}} \mapsto r_{\hat{\mathbf{x}}}^k$ so that $a_{\hat{\mathbf{p}}}^k = a_k$ and $r_{\hat{\mathbf{p}}}^k = r_k$. In order to this we are going to use the projective s - and u -holonomies $h^{s,k} = \mathbb{P}H^{s,k}$ and $h^{u,k}|_{W_{\text{loc}}^u(\hat{\mathbf{q}})}$, where the last one is defined for points in the local unstable set of $\hat{\mathbf{q}}$.

The other property of $\hat{\mathbf{q}}$ we have assumed is that it contains all symbols in its non-negative entries. This implies that one can take a $j \in \mathbb{N}$ in order for $\hat{\mathbf{p}}$ and $\hat{f}^j(\hat{\mathbf{q}})$ to be in the same cylinder, which will be denoted by $\mathcal{C}_{\hat{\mathbf{p}}}$. For the sake of simplicity we are going to consider $j = 0$. Therefore, given any point $\hat{\mathbf{x}} \in \mathcal{C}_{\hat{\mathbf{p}}}$, we will define projective points $a_{\hat{\mathbf{x}}}^k$ and $r_{\hat{\mathbf{x}}}^k$ by:

$$a_{\hat{\mathbf{x}}}^k = h_{\hat{\mathbf{y}}, \hat{\mathbf{x}}}^{s,k} \circ h_{\hat{\mathbf{z}}, \hat{\mathbf{y}}}^{u,k} \circ h_{\hat{\mathbf{p}}, \hat{\mathbf{z}}}^{s,k}(a_k) \quad \text{and} \quad r_{\hat{\mathbf{x}}}^k = h_{\hat{\mathbf{y}}, \hat{\mathbf{x}}}^{s,k} \circ h_{\hat{\mathbf{z}}, \hat{\mathbf{y}}}^{u,k} \circ h_{\hat{\mathbf{p}}, \hat{\mathbf{z}}}^{s,k}(r_k),$$

where $\hat{\mathbf{y}} = [\hat{\mathbf{x}}, \hat{\mathbf{q}}]$ and $\hat{\mathbf{z}} = [\hat{\mathbf{p}}, \hat{\mathbf{q}}]$ as in Figure 3.1(a).

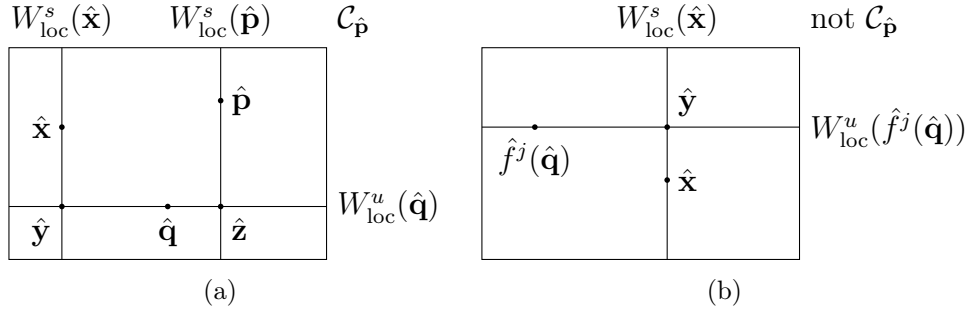


Figure 3.1: Points of intersection in the definition of maps $\hat{\mathbf{x}} \mapsto a_{\hat{\mathbf{x}}}^k$ and $\hat{\mathbf{x}} \mapsto r_{\hat{\mathbf{x}}}^k$ in: (a) the cylinder of $\hat{\mathbf{p}}$; or (b) any other cylinder.

Now, let $\hat{\mathbf{x}} \in \hat{\Sigma} \setminus \mathcal{C}_{\hat{\mathbf{p}}}$. Since the point $\hat{\mathbf{q}}$ contains all symbols in its non-negative entries, there is a $j \in \mathbb{N}^*$ such that $\hat{f}^j(\hat{\mathbf{q}})$ lies in the same cylinder as $\hat{\mathbf{x}}$. We can then define the projective points $a_{\hat{\mathbf{x}}}^k$ and $r_{\hat{\mathbf{x}}}^k$ by:

$$a_{\hat{\mathbf{x}}}^k = h_{\hat{\mathbf{y}}, \hat{\mathbf{x}}}^{s,k} \circ h_{\hat{f}^j(\hat{\mathbf{q}}), \hat{\mathbf{y}}}^{u,k} (\mathbb{P}\hat{A}_k^j(\hat{\mathbf{q}}) \cdot a_{\hat{\mathbf{q}}}^k) \quad \text{and} \quad r_{\hat{\mathbf{x}}}^k = h_{\hat{\mathbf{y}}, \hat{\mathbf{x}}}^{s,k} \circ h_{\hat{f}^j(\hat{\mathbf{q}}), \hat{\mathbf{y}}}^{u,k} (\mathbb{P}\hat{A}_k^j(\hat{\mathbf{q}}) \cdot r_{\hat{\mathbf{q}}}^k),$$

where $\hat{\mathbf{y}} = [\hat{\mathbf{x}}, \hat{f}^j(\hat{\mathbf{q}})]$ as in Figure 3.1(b).

However, one may observe that the u -holonomies above are taken at points outside $W_{\text{loc}}^u(\hat{\mathbf{q}})$, but in the local unstable set of some iterate of $\hat{\mathbf{q}}$. This is not a problem at all, because we can define those u -holonomies from what we already know. Indeed, since $\hat{\mathbf{q}}$ has all symbols in its non-negative entries, for each symbol $s \in \{0, \dots, \kappa-1\}$, one can take a $j(s) \in \mathbb{N}^*$ such that $\hat{f}^{j(s)}(\hat{\mathbf{q}})$ relies in the cylinder $[0; s]$. Thus, for every symbol s and points $\hat{\mathbf{x}}, \hat{\mathbf{y}} \in W_{\text{loc}}^u(\hat{f}^{j(s)}(\hat{\mathbf{q}}))$, we can define:

$$h_{\hat{\mathbf{x}}, \hat{\mathbf{y}}}^u = \mathbb{P}\hat{A}^{j(s)}(\hat{f}^{-j(s)}(\hat{\mathbf{y}})) \circ h_{\hat{f}^{-j(s)}(\hat{\mathbf{x}}), \hat{f}^{-j(s)}(\hat{\mathbf{y}})}^u \circ \mathbb{P}\hat{A}^{-j(s)}(\hat{\mathbf{x}}); \quad \text{and}$$

$$h_{\hat{\mathbf{x}}, \hat{\mathbf{y}}}^{u,k} = \mathbb{P}\hat{A}_k^{j(s)}(\hat{f}^{-j(s)}(\hat{\mathbf{y}})) \circ h_{\hat{f}^{-j(s)}(\hat{\mathbf{x}}), \hat{f}^{-j(s)}(\hat{\mathbf{y}})}^{u,k} \circ \mathbb{P}\hat{A}_k^{-j(s)}(\hat{\mathbf{x}}) \quad \text{for all } k \in \mathbb{N}^*,$$

since $\hat{f}^{-j(s)}(\hat{\mathbf{x}}), \hat{f}^{-j(s)}(\hat{\mathbf{y}}) \in W_{\text{loc}}^u(\hat{\mathbf{q}})$.

Notice that $a_{\hat{\mathbf{p}}}^k = a_k$ and $r_{\hat{\mathbf{p}}}^k = r_k$ as we wanted. In addition, the maps $\hat{\mathbf{x}} \mapsto a_{\hat{\mathbf{x}}}^k$ and $\hat{\mathbf{x}} \mapsto r_{\hat{\mathbf{x}}}^k$ are continuous due to continuity of the holonomies involved in their definitions. We have also that $a_{\hat{\mathbf{x}}}^k \rightarrow a$ uniformly in $\hat{\Sigma}$. Indeed, one can verify this using the facts:

- (1) $a_k \rightarrow a$;
- (2) a is invariant by u -holonomies (due to Proposition 3.2.4) and by s -holonomies (due to Lemma 3.3.4);
- (3) $H^{s,k} \rightarrow H^s$ uniformly in $\hat{\Sigma}$; and
- (4) $h^{u,k} \rightarrow h^u$ uniformly in $W_{\text{loc}}^u(\hat{\mathbf{q}})$.

On the other hand, the r case does not follow directly as above. This happens since we cannot assume that r is invariant by u -holonomies. However, since the holonomies converge uniformly, it is possible to verify that $\{r_{\hat{\mathbf{x}}}^k\}_{k \in \mathbb{N}^*}$ converges to a point $r(\hat{\mathbf{x}}) \in \mathbb{P}\mathbb{R}^2$ varying continuously with $\hat{\mathbf{x}} \in \hat{\Sigma}$.

Claim. The limit map $r(\cdot)$ is $\mathbb{P}\hat{A}$ -invariant.

Fix for now a point $\hat{\mathbf{x}} \in \mathcal{C}_{\hat{\mathbf{p}}}$. Observe that, by definition of $\{r_{\hat{\mathbf{x}}}^k\}_{k \in \mathbb{N}^*}$, the convergences that we have, along with the fact that r is invariant by s -holonomies (due to Lemma 3.3.4), imply:

$$r(\hat{\mathbf{x}}) = h_{\hat{\mathbf{y}}, \hat{\mathbf{x}}}^s \circ h_{\hat{\mathbf{z}}, \hat{\mathbf{y}}}^u \circ h_{\hat{\mathbf{p}}, \hat{\mathbf{z}}}^s(r) = h_{\hat{\mathbf{y}}, \hat{\mathbf{x}}}^s \circ h_{\hat{\mathbf{z}}, \hat{\mathbf{y}}}^u(r),$$

where $\hat{\mathbf{y}} = [\hat{\mathbf{x}}, \hat{\mathbf{q}}]$ and $\hat{\mathbf{z}} = [\hat{\mathbf{p}}, \hat{\mathbf{q}}]$ as in Figure 3.1(a). In particular, we have $r(\hat{\mathbf{q}}) = h_{\hat{\mathbf{z}}, \hat{\mathbf{q}}}^u(r)$.

Now, let us compute $r(\hat{f}(\hat{\mathbf{x}}))$. For this we assume that $\hat{f}(\hat{\mathbf{x}})$ does not lie in $\mathcal{C}_{\hat{\mathbf{p}}}$, since the other case is simpler. Similarly as before, we have:

$$r(\hat{f}(\hat{\mathbf{x}})) = h_{\hat{\mathbf{w}}, \hat{f}(\hat{\mathbf{x}})}^s \circ h_{\hat{f}(\hat{\mathbf{q}}), \hat{\mathbf{w}}}^u(\mathbb{P}\hat{A}(\hat{\mathbf{q}}) \cdot r(\hat{\mathbf{q}})),$$

where $\hat{\mathbf{w}} = [\hat{f}(\hat{\mathbf{x}}), \hat{f}(\hat{\mathbf{q}})] = \hat{f}(\hat{\mathbf{y}})$ and thus

$$\begin{aligned} r(\hat{f}(\hat{\mathbf{x}})) &= (\mathbb{P}\hat{A}(\hat{\mathbf{x}}) \circ h_{\hat{\mathbf{y}}, \hat{\mathbf{x}}}^s \circ \mathbb{P}\hat{A}^{-1}(\hat{\mathbf{y}}))(\mathbb{P}\hat{A}(\hat{\mathbf{y}}) \circ h_{\hat{\mathbf{q}}, \hat{\mathbf{y}}}^u \circ \mathbb{P}\hat{A}(\hat{\mathbf{q}})^{-1})(\mathbb{P}\hat{A}(\hat{\mathbf{q}}) \cdot r(\hat{\mathbf{q}})) \\ &= \mathbb{P}\hat{A}(\hat{\mathbf{x}}) \circ h_{\hat{\mathbf{y}}, \hat{\mathbf{x}}}^s \circ h_{\hat{\mathbf{q}}, \hat{\mathbf{y}}}^u(r(\hat{\mathbf{q}})) \\ &= \mathbb{P}\hat{A}(\hat{\mathbf{x}}) \circ h_{\hat{\mathbf{y}}, \hat{\mathbf{x}}}^s \circ h_{\hat{\mathbf{q}}, \hat{\mathbf{y}}}^u \circ h_{\hat{\mathbf{z}}, \hat{\mathbf{q}}}^u(r) \\ &= \mathbb{P}\hat{A}(\hat{\mathbf{x}}) \circ h_{\hat{\mathbf{y}}, \hat{\mathbf{x}}}^s \circ h_{\hat{\mathbf{z}}, \hat{\mathbf{y}}}^u(r) \\ &= \mathbb{P}\hat{A}(\hat{\mathbf{x}})(r(\hat{\mathbf{x}})). \end{aligned}$$

The case when $\hat{\mathbf{x}} \notin \mathcal{C}_{\hat{\mathbf{p}}}$ is similar. This proves our claim.

The cocycle invariance of our previous claim allows us to conclude the next one. Also, recall that $(\hat{f}, \hat{\mu})$ is an ergodic system.

Claim. The limit map $r(\cdot)$ is constant and equals r .

Let us define the following probability measure on $\hat{\Sigma} \times \mathbb{P}\mathbb{R}^2$:

$$\hat{m}_r(E) = \int \delta_{(\hat{\mathbf{x}}, r(\hat{\mathbf{x}}))}(E) d\hat{\mu}(\hat{\mathbf{x}}),$$

for every measurable set $E \subset \hat{\Sigma} \times \mathbb{P}\mathbb{R}^2$. Notice that since the limit map $r(\cdot)$ is $\mathbb{P}\hat{A}$ -invariant, such probability measure $\hat{m}_r$ is $\mathbb{P}\hat{A}$ -invariant also.

Now, Lemma 3.3.5 states that every $\mathbb{P}\hat{A}$ -invariant probability measure projecting down to $\hat{\mu}$ has constant disintegration. Then it applies to $\hat{m}_r$ and hence, since disintegrations are essentially unique, we conclude that the limit map $r(\cdot)$ is constant (everywhere since it is continuous).

There are only two possibilities: $r(\cdot)$ equals either r or a . This is because every $\mathbb{P}\hat{A}$ -invariant probability measure projecting down to $\hat{\mu}$ is a convex combination of $\hat{m}^s$ and $\hat{m}^u$. However, we already know that $r(\hat{\mathbf{p}}) = r$, so it follows that $r(\hat{\mathbf{x}}) \equiv r$, as we have claimed.

Before we continue let us summarise what we have so far. For each $k \in \mathbb{N}^*$ there are continuous maps $\hat{\mathbf{x}} \mapsto a_{\hat{\mathbf{x}}}^k$ and $\hat{\mathbf{x}} \mapsto r_{\hat{\mathbf{x}}}^k$. Besides, such maps converge uniformly: $a_{\hat{\mathbf{x}}}^k \rightarrow a$ and $r_{\hat{\mathbf{x}}}^k \rightarrow r$ in $\hat{\Sigma}$.

Let us take the $\mathbb{P}F_{\hat{A}_k}$ -ergodic probability measure $\hat{m}_k$ in $\hat{\Sigma} \times \mathbb{P}\mathbb{R}^2$ as in the hypotheses. By assumption, the measure $\hat{m}_k$ admits a continuous disintegration $\{\hat{m}_{\hat{\mathbf{x}}}^k\}_{\hat{\mathbf{x}} \in \hat{\Sigma}}$ that is invariant by $\mathbb{P}\hat{A}_k$ and both holonomies $h^{s,k}$ and $h^{u,k}|_{W_{\text{loc}}^u(\hat{\mathbf{q}})}$. This implies that $(\mathbb{P}\hat{A}_k)_* \hat{m}_{\hat{\mathbf{x}}}^k = \hat{m}_{f(\hat{\mathbf{x}})}^k$ for all $\hat{\mathbf{x}} \in \hat{\Sigma}$. It is easy to check from it that $\text{supp } \hat{m}_{\hat{\mathbf{p}}}^k \subset \{r_k, a_k\}$. In turn, with the holonomy-invariance we can conclude that $\text{supp } \hat{m}_{\hat{\mathbf{x}}}^k \subset \{r_{\hat{\mathbf{x}}}^k, a_{\hat{\mathbf{x}}}^k\}$ for each $\hat{\mathbf{x}} \in \hat{\Sigma}$, and also that the cardinality of these sets is constant. Notice that this holds for every $k \geq k_0$.

We will prove now that $\#\text{supp } \hat{m}_{\hat{\mathbf{p}}}^k = 1$ for every $k \in \mathbb{N}^*$ sufficiently large. In fact, let us suppose by contradiction that $\#\text{supp } \hat{m}_{\hat{\mathbf{p}}}^k = 2$ for infinitely many values of $k \geq k_0$. The set of such values k will be denoted by $\mathbb{N}'$.

Let us consider at first that *there is* a $k \in \mathbb{N}'$ such that maps $\hat{\mathbf{x}} \mapsto r_{\hat{\mathbf{x}}}^k$ and $\hat{\mathbf{x}} \mapsto a_{\hat{\mathbf{x}}}^k$ are $\mathbb{P}\hat{A}_k$ -invariant. It means that we can construct two $F_{\hat{A}_k}$ -invariant probability measures:

$$\hat{m}^{u,k} = \int \delta_{(\hat{\mathbf{x}}, a_{\hat{\mathbf{x}}}^k)} d\hat{\mu}(\hat{\mathbf{x}}) \quad \text{and} \quad \hat{m}^{s,k} = \int \delta_{(\hat{\mathbf{x}}, r_{\hat{\mathbf{x}}}^k)} d\hat{\mu}(\hat{\mathbf{x}}).$$

Now, since $\text{supp } \hat{m}_{\hat{\mathbf{x}}}^k = \{a_{\hat{\mathbf{x}}}^k, r_{\hat{\mathbf{x}}}^k\}$ for every $\hat{\mathbf{x}} \in \hat{\Sigma}$, it follows that the probability measure $\hat{m}_k$ is a convex combination of $\hat{m}^{u,k}$ and $\hat{m}^{s,k}$, contradicting the ergodicity of that measure.

That being so, *there is no* $k \in \mathbb{N}'$ such that the maps $\hat{\mathbf{x}} \mapsto a_{\hat{\mathbf{x}}}^k$ and $\hat{\mathbf{x}} \mapsto r_{\hat{\mathbf{x}}}^k$ are invariant. It means that for every $k \in \mathbb{N}'$ we can find an $\hat{\mathbf{x}}^k \in \hat{\Sigma}$ such that $\mathbb{P}\hat{A}_k(\hat{\mathbf{x}}^k) \cdot a_{\hat{\mathbf{x}}^k}^k = r_{f(\hat{\mathbf{x}}^k)}^k$. Since the space $\hat{\Sigma}$ is compact there is an accumulation point $\hat{\mathbf{x}}'$ of the sequence $\{\hat{\mathbf{x}}^k\}_{k \in \mathbb{N}'}$. Due to the convergences we already have one can choose a sufficiently large $k \in \mathbb{N}'$ in order that

$$\text{dist}(r, r_{f(\hat{\mathbf{x}}^k)}^k) < \frac{\varepsilon}{2} \quad \text{and} \quad \text{dist}(\mathbb{P}\hat{A}_k(\hat{\mathbf{x}}^k) \cdot a_{\hat{\mathbf{x}}^k}^k, \mathbb{P}\hat{A}(\hat{\mathbf{x}}') \cdot a) < \frac{\varepsilon}{2},$$

where $\varepsilon > 0$ is the constant that separates the sets given by Equation (3.9). However, it follows that

$$\text{dist}(r, a) \leq \text{dist}(r, r_{\hat{f}(\hat{\mathbf{x}}^k)}^k) + \text{dist}(\mathbb{P}\hat{A}_k(\hat{\mathbf{x}}^k) \cdot a_{\hat{\mathbf{x}}^k}^k, \mathbb{P}\hat{A}(\hat{\mathbf{x}}') \cdot a) < \varepsilon,$$

contradicting the ε -separability of the referred sets.

Therefore, we conclude that $\# \text{supp } \hat{m}_{\mathbf{p}}^k = 1$ for every k sufficiently large, as we wanted. Thus we end up with two possibilities: either $\hat{m}_k = \hat{m}^{u,k}$ or $\hat{m}_k = \hat{m}^{s,k}$ for every $k \in \mathbb{N}^*$ large enough. Let us consider the following measures:

$$\hat{m}^u = \int \delta_{(\hat{\mathbf{x}}, a)} d\hat{\mu}(\hat{\mathbf{x}}) \quad \text{and} \quad \hat{m}^s = \int \delta_{(\hat{\mathbf{x}}, r)} d\hat{\mu}(\hat{\mathbf{x}}).$$

These measures are such that $\lambda_+(\hat{A}) = \int \Phi d\hat{m}^u$ and $\lambda_-(\hat{A}) = \int \Phi d\hat{m}^s$ by Proposition 3.2.4.

Claim. $\hat{m}^{u,k} \rightarrow \hat{m}^u$ or $\hat{m}^{s,k} \rightarrow \hat{m}^s$ in the weak-* topology.

Before we prove it, note that it implies the convergence of the sequence $\{\int \Phi_k d\hat{m}_k\}_{k \in \mathbb{N}^*}$ to either $\int \Phi d\hat{m}^s$ or $\int \Phi d\hat{m}^u$. In the first case we conclude that $\lambda_+(\hat{A}_k) \rightarrow \lambda_-(\hat{A})$, which only makes sense if the Lyapunov exponents of $\hat{A}$ are null. This clearly contradicts our assumptions. Therefore, we are in the second case, where we conclude that $\lambda_+(\hat{A}_k) \rightarrow \lambda_+(\hat{A})$, as we wanted.

Now we are going to prove our previous claim. In fact, we show the convergence $\hat{m}^{u,k} \rightarrow \hat{m}^u$ in the weak-* topology. The remaining one follows with a similar argument.

Let $\varphi: \hat{\Sigma} \times \mathbb{P}\mathbb{R}^2 \rightarrow \mathbb{R}$ be a continuous function. Given any $k \in \mathbb{N}'$, define the function $\psi_k: \hat{\Sigma} \rightarrow \mathbb{R}$ by $\psi_k(\hat{\mathbf{x}}) = \varphi(\hat{\mathbf{x}}, a_{\hat{\mathbf{x}}}^k)$, and note that $|\psi_k| \leq |\varphi|$. In a similar way we define the function $\psi: \hat{\Sigma} \rightarrow \mathbb{R}$ by $\psi(\hat{\mathbf{x}}) = \varphi(\hat{\mathbf{x}}, a)$. As $a_{\hat{\mathbf{x}}}^k \rightarrow a$ uniformly in $\hat{\Sigma}$, we conclude that $\psi_k \rightarrow \psi$ uniformly as well. Thus the Dominated Convergence Theorem gives us that

$$\int \varphi d\hat{m}^{u,k} = \int \psi_k d\hat{\mu} \rightarrow \int \psi d\hat{\mu} = \int \varphi d\hat{m}^u.$$

It follows that $\hat{m}^{u,k} \rightarrow \hat{m}^u$ in the weak-* topology as we have claimed. $\square$

Chapter 4

Auxiliary results

In this chapter we present a few more results used to show both Theorem 2.1.1 and Theorem 2.1.2. Once again, we are considering a subshift of finite type $(\hat{\Sigma}, \hat{f})$ and an $\hat{f}$ -ergodic probability measure $\hat{\mu}$ that has local product structure and full support.

4.1 Arzelà–Ascoli Theorem for holonomies

In this section we are going to establish a sufficient condition to the existence of a uniformly convergent subsequence of holonomies. It is based on the well known result from Real Analysis called Arzelà–Ascoli Theorem. Such condition holds in a more general context, but we choose to state it in our setting for simplicity.

Let us fix for now a point $\hat{\mathbf{x}} \in \hat{\Sigma}$. We denote by $\mathcal{H}_{\hat{\mathbf{x}}}^u$ the space of all maps $W_{\text{loc}}^u(\hat{\mathbf{x}}) \times \mathbb{P}\mathbb{R}^2 \rightarrow \mathbb{P}\mathbb{R}^2$ that are continuous and are the projection of a linear transformation in the second coordinate. We would like to remark that every map $\varphi(\hat{\mathbf{y}}, t) = h_{\hat{\mathbf{x}}, \hat{\mathbf{y}}}^u(t) = \mathbb{P}H_{\hat{\mathbf{x}}, \hat{\mathbf{y}}}^u(t)$ that comes from a projective u -holonomy is in $\mathcal{H}_{\hat{\mathbf{x}}}^u$.

Observe that $\mathcal{H}_{\hat{\mathbf{x}}}^u$ is a subspace of the complete space of continuous maps, $C^0(W_{\text{loc}}^u(\hat{\mathbf{x}}) \times \mathbb{P}\mathbb{R}^2, \mathbb{P}\mathbb{R}^2)$. We can endow this space with the uniform topology and so $\mathcal{H}_{\hat{\mathbf{x}}}^u$ inherits it by inclusion. Additionally, it can be proven that $\mathcal{H}_{\hat{\mathbf{x}}}^u$ is in fact a closed subspace and hence it is also complete. In sum, a sequence of maps $\{\mathfrak{h}_k\}_{k \in \mathbb{N}} \subset \mathcal{H}_{\hat{\mathbf{x}}}^u$ converges in this topology to some $\mathfrak{h} \in \mathcal{H}_{\hat{\mathbf{x}}}^u$ if and only if, for every $\varepsilon > 0$, one can take a $k_0 \in \mathbb{N}$ such that

$$\text{dist}(\mathfrak{h}_k(\hat{\mathbf{y}}, t), \mathfrak{h}(\hat{\mathbf{y}}, t)) < \varepsilon \quad \text{for all } \hat{\mathbf{y}} \in W_{\text{loc}}^u(\hat{\mathbf{x}}), t \in \mathbb{P}\mathbb{R}^2, \text{ and } k \geq k_0.$$

Let us define the concept of *equicontinuity* in the space $\mathcal{H}_{\hat{\mathbf{x}}}^u$. A set $\mathcal{H} \subset \mathcal{H}_{\hat{\mathbf{x}}}^u$ is said to be *equicontinuous* at a point $(\hat{\mathbf{y}}_1, t_1) \in W_{\text{loc}}^u(\hat{\mathbf{x}}) \times \mathbb{P}\mathbb{R}^2$ if, for every $\varepsilon > 0$, there is $\delta > 0$ such that

$$\text{dist}(\mathfrak{h}(\hat{\mathbf{y}}_1, t_1), \mathfrak{h}(\hat{\mathbf{y}}_2, t_2)) < \varepsilon$$

for all $\mathfrak{h} \in \mathcal{H}$, all $\hat{\mathbf{y}}_2 \in W_{\text{loc}}^u(\hat{\mathbf{x}})$ with $\text{dist}_\rho(\hat{\mathbf{y}}_1, \hat{\mathbf{y}}_2) < \delta$, and all $t_2 \in \mathbb{P}\mathbb{R}^2$ with $\text{dist}(t_1, t_2) < \delta$.

The main tool that will be used in the proof of the holonomy version of Arzelà–Ascoli Theorem is [Dix84, Theorem 6.3.1], that we transcribe here. Such

result gives us a sufficient condition to the existence of a uniformly convergent subsequence.

Proposition 4.1.1 (Ascoli Theorem, [Dix84]). *Let $\mathcal{X}$ be a compact metric space, $\mathcal{Y}$ a complete metric space, and $\mathcal{F}$ an equicontinuous subset of the space $C^0(\mathcal{X}, \mathcal{Y})$. Assume that, for each $x \in \mathcal{X}$, the set $\mathcal{F}_x = \{f(x) \in \mathcal{Y} : f \in \mathcal{F}\}$ has compact closure in $\mathcal{Y}$. Thus $\mathcal{F}$ has compact closure in the metric space $C^0(\mathcal{X}, \mathcal{Y})$.*

Since non-uniform holonomies for a cocycle *may not* be unique we denote the space of continuous linear cocycles $M \rightarrow \mathrm{SL}(2, \mathbb{R})$ with non-uniform u -holonomies by $\mathcal{H}_u^0(2, M)$. Hence, each element of the space $\mathcal{H}_u^0(2, M)$ is a pair (B, H^u) where H^u denotes a non-uniform u -holonomy for B over f . Recall that the definition of such holonomy comprises an increasing sequence $\{\mathcal{D}_m^u\}_{m \in \mathbb{N}^*}$ of u -holonomy blocks (check Definition 4 for further information).

Finally we can state the following.

Theorem 4.1.2 (Arzelà–Ascoli for holonomies). *Consider a sequence of linear cocycles $\{(\hat{A}_k, H^{u,k})\}_{k \in \mathbb{N}^*}$ in $\mathcal{H}_u^0(2, \hat{\Sigma})$ such that $\{\hat{A}_k\}_{k \in \mathbb{N}^*}$ converges in the C^0 -topology to a map $\hat{A} : \hat{\Sigma} \rightarrow \mathrm{SL}(2, \mathbb{R})$, and the sequence $\{H^{u,k}\}_{k \in \mathbb{N}^*}$ satisfies the equiproximity condition (Definition 5).*

For $\hat{\mu}$ -almost every point $\hat{\mathbf{x}} \in \hat{\Sigma}$ there is an infinite subset $\mathbb{N}' \subset \mathbb{N}^$ such that the sequence $\{\mathfrak{h}_k\}_{k \in \mathbb{N}'}$ formed by maps $\mathfrak{h}_k(\hat{\mathbf{y}}, t) = \mathbb{P}H_{\hat{\mathbf{x}}, \hat{\mathbf{y}}}^{u,k}(t)$ converges uniformly to some $\mathfrak{h} : W_{\mathrm{loc}}^u(\hat{\mathbf{x}}) \times \mathbb{P}\mathbb{R}^2 \rightarrow \mathbb{P}\mathbb{R}^2$. Furthermore, this limit map generates a projective u -holonomy for $\hat{A}$ restricted to the set $W_{\mathrm{loc}}^u(\hat{\mathbf{x}})$.*

Proof. In order to make use of Proposition 4.1.1, it is necessary to check if our hypotheses befit those in that result. This is what we are going to do now.

Fix any $m \in \mathbb{N}^*$ and a point $\hat{\mathbf{x}}$ in the u -holonomy block $\mathcal{D}_m^u$. According to our discussion earlier in this section, the space $\mathcal{H}_{\hat{\mathbf{x}}}^u$ endowed with the uniform topology is complete. It is easy to check that $\mathcal{X} = W_{\mathrm{loc}}^u(\hat{\mathbf{x}}) \times \mathbb{P}\mathbb{R}^2$ and $\mathcal{Y} = \mathbb{P}\mathbb{R}^2$ are compact (in particular, complete) spaces. Besides, we have also mentioned that $\mathcal{H}_{\hat{\mathbf{x}}}^u$ is a closed subspace of $C^0(\mathcal{X}, \mathcal{Y})$.

For each $k \in \mathbb{N}^*$, define the map $\mathfrak{h}_k : \mathcal{X} \rightarrow \mathcal{Y}$ by $\mathfrak{h}_k(\hat{\mathbf{y}}, t) = \mathbb{P}H_{\hat{\mathbf{x}}, \hat{\mathbf{y}}}^{u,k}(t)$. Now consider the set $\mathcal{H} = \{\mathfrak{h}_k : k \in \mathbb{N}^*\} \subset \mathcal{H}_{\hat{\mathbf{x}}}^u$. We are going to show at first that $\mathcal{H}$ is equicontinuous. To this purpose let $\varepsilon_0 = 1$ and take $\delta_{m,0} > 0$ from the equiproximity condition. Observe that definition and compactness of $W_{\mathrm{loc}}^u(\hat{\mathbf{x}})$ implies the existence of a natural number $N = N(\hat{\mathbf{x}}, \delta_{m,0})$ such that $\mathrm{dist}_\rho(\hat{f}^{-N}(\hat{\mathbf{x}}), \hat{f}^{-N}(\hat{\mathbf{y}})) < \delta_{m,0}$ for every $\hat{\mathbf{y}} \in W_{\mathrm{loc}}^u(\hat{\mathbf{x}})$.

Fix for now $(\hat{\mathbf{y}}_1, t_1) \in \mathcal{X}$. By the properties of u -holonomies we have:

$$H_{\hat{f}^{-N}(\hat{\mathbf{x}}), \hat{f}^{-N}(\hat{\mathbf{y}}_1)}^{u,k} = \hat{A}_k^{-N}(\hat{\mathbf{y}}_1) H_{\hat{\mathbf{x}}, \hat{\mathbf{y}}_1}^{u,k} \hat{A}_k^N(\hat{f}^{-N}(\hat{\mathbf{x}}))$$

hence

$$\begin{aligned} H_{\hat{\mathbf{x}}, \hat{\mathbf{y}}_1}^{u,k} &= \hat{A}_k^N(\hat{f}^{-N}(\hat{\mathbf{y}}_1)) H_{\hat{f}^{-N}(\hat{\mathbf{x}}), \hat{f}^{-N}(\hat{\mathbf{y}}_1)}^{u,k} \hat{A}_k^{-N}(\hat{\mathbf{x}}) \\ \|H_{\hat{\mathbf{x}}, \hat{\mathbf{y}}_1}^{u,k}\| &\leq \underbrace{\|\hat{A}_k^N(\hat{f}^{-N}(\hat{\mathbf{y}}_1))\|}_{\text{bounded}} \underbrace{\|H_{\hat{f}^{-N}(\hat{\mathbf{x}}), \hat{f}^{-N}(\hat{\mathbf{y}}_1)}^{u,k}\|}_{<2} \underbrace{\|\hat{A}_k^{-N}(\hat{\mathbf{x}})\|}_{\text{bounded}} \end{aligned}$$

for each $k \in \mathbb{N}^*$, by the equiproximity condition. Since this upper bound depends neither on k nor on $\hat{\mathbf{y}}_1$, we can take a constant $L > 1$ such that

$$\sup\{\|H_{\hat{\mathbf{x}},\hat{\mathbf{y}}}^{u,k}\| : k \in \mathbb{N}^* \text{ and } \hat{\mathbf{y}} \in W_{\text{loc}}^u(\hat{\mathbf{x}})\} \leq L.$$

We can finally prove equicontinuity on $\mathcal{H}$. For so, given any $\varepsilon > 0$ take $\delta_m > 0$ again from the equiproximity condition, but for the constant $\frac{\varepsilon}{2L}$. Now, for every $\hat{\mathbf{y}}_2 \in W_{\text{loc}}^u(\hat{\mathbf{x}})$ with $\text{dist}(\hat{\mathbf{y}}_1, \hat{\mathbf{y}}_2) < \delta_m$, we have

$$H_{\hat{\mathbf{x}},\hat{\mathbf{y}}_2}^{u,k} = H_{\hat{\mathbf{y}}_1,\hat{\mathbf{y}}_2}^{u,k} H_{\hat{\mathbf{x}},\hat{\mathbf{y}}_1}^{u,k}$$

hence

$$\|H_{\hat{\mathbf{x}},\hat{\mathbf{y}}_1}^{u,k} - H_{\hat{\mathbf{x}},\hat{\mathbf{y}}_2}^{u,k}\| \leq \|H_{\hat{\mathbf{y}}_1,\hat{\mathbf{y}}_2}^{u,k} - \text{id}\| \|H_{\hat{\mathbf{x}},\hat{\mathbf{y}}_1}^{u,k}\| < \frac{\varepsilon}{2L} L = \frac{\varepsilon}{2}$$

for each $k \in \mathbb{N}$, again by the equiproximity condition. Given also $t_2 \in \mathbb{P}\mathbb{R}^2$ with $\text{dist}(t_1, t_2) < \delta_m$, let us consider two unitary vectors $\vec{v}_1, \vec{v}_2 \in \mathbb{R}^2$ such that $\mathbb{P}\vec{v}_1 = t_1$, $\mathbb{P}\vec{v}_2 = t_2$, and $\langle \vec{v}_1, \vec{v}_2 \rangle > 0$. Decreasing δ_m if necessary, we can assume that $\|\vec{v}_1 - \vec{v}_2\| < \frac{\varepsilon}{2L}$. Thus,

$$\begin{aligned} \|H_{\hat{\mathbf{x}},\hat{\mathbf{y}}_1}^{u,k} \cdot \vec{v}_1 - H_{\hat{\mathbf{x}},\hat{\mathbf{y}}_2}^{u,k} \cdot \vec{v}_2\| &\leq \|H_{\hat{\mathbf{x}},\hat{\mathbf{y}}_1}^{u,k} \cdot \vec{v}_1 - H_{\hat{\mathbf{x}},\hat{\mathbf{y}}_1}^{u,k} \cdot \vec{v}_2\| + \|H_{\hat{\mathbf{x}},\hat{\mathbf{y}}_1}^{u,k} \cdot \vec{v}_2 - H_{\hat{\mathbf{x}},\hat{\mathbf{y}}_2}^{u,k} \cdot \vec{v}_2\| \\ &= \|H_{\hat{\mathbf{x}},\hat{\mathbf{y}}_1}^{u,k} \cdot (\vec{v}_1 - \vec{v}_2)\| + \|(H_{\hat{\mathbf{x}},\hat{\mathbf{y}}_1}^{u,k} - H_{\hat{\mathbf{x}},\hat{\mathbf{y}}_2}^{u,k}) \cdot \vec{v}_2\| \\ &\leq \underbrace{\|H_{\hat{\mathbf{x}},\hat{\mathbf{y}}_1}^{u,k}\|}_{\leq L} \underbrace{\|\vec{v}_1 - \vec{v}_2\|}_{< \frac{\varepsilon}{2L}} + \underbrace{\|H_{\hat{\mathbf{x}},\hat{\mathbf{y}}_1}^{u,k} - H_{\hat{\mathbf{x}},\hat{\mathbf{y}}_2}^{u,k}\|}_{< \frac{\varepsilon}{2}} \underbrace{\|\vec{v}_2\|}_{=1} < \varepsilon \end{aligned}$$

for all $k \in \mathbb{N}^*$. Observe that it implies the equicontinuity of the set $\mathcal{H}$ at the point $(\hat{\mathbf{y}}_1, t_1)$. However, as the choice of this point was arbitrary, we can conclude that $\mathcal{H}$ is indeed equicontinuous.

It only remains to check the last hypothesis of Proposition 4.1.1, that is, whether the set $\mathcal{H}_{\hat{\mathbf{y}},t} := \{\mathfrak{h}(\hat{\mathbf{y}}, t) \in \mathcal{Y} : \mathfrak{h} \in \mathcal{H}\}$ has compact closure in $\mathcal{Y}$ for each $(\hat{\mathbf{y}}, t) \in \mathcal{X}$. Nevertheless, since $\mathcal{Y}$ is compact, that condition is always satisfied, independently of the choice of $\mathcal{H}$. In consequence, the referred result states that the actual set $\mathcal{H}$ has compact closure in $C^0(\mathcal{X}, \mathcal{Y})$. For instance, there is a uniformly convergent subsequence $\{\mathfrak{h}_k\}_{k \in \mathbb{N}'} \subset \mathcal{H}$. Let us denote the limit map of such subsequence as $\mathfrak{h} \in C^0(\mathcal{X}, \mathcal{Y})$; symbolically,

$$\mathfrak{h}(\hat{\mathbf{y}}, r) = \lim_{k \in \mathbb{N}'} \mathfrak{h}_k(\hat{\mathbf{y}}, r) = \lim_{k \in \mathbb{N}'} \mathbb{P}H_{\hat{\mathbf{x}},\hat{\mathbf{y}}}^{u,k}(r) \quad \text{for every } (\hat{\mathbf{y}}, r) \in \mathcal{X}. \quad (4.1)$$

But, as we have mentioned that $\mathcal{H}_{\hat{\mathbf{x}}}^u$ is a closed subspace of $C^0(\mathcal{X}, \mathcal{Y})$, we can conclude that $\mathfrak{h}$ belongs in fact to $\mathcal{H}_{\hat{\mathbf{x}}}^u$, that is, it is continuous and it is a projection of a linear transformation on the second coordinate.

In the following, we are going to verify whether the limit map $\mathfrak{h}$, as in Equation (4.1), generates a projective u -holonomy for $\hat{A}: \hat{\Sigma} \rightarrow \text{SL}(2, \mathbb{R})$ restricted to the local unstable set of $\hat{\mathbf{x}} \in \hat{\Sigma}$. First we need to construct a collection of maps $h_{\hat{\mathbf{y}},\hat{\mathbf{z}}}^u: \mathbb{P}\mathbb{R}^2 \rightarrow \mathbb{P}\mathbb{R}^2$ depending on $\mathfrak{h}$. Indeed, for each $\hat{\mathbf{y}}, \hat{\mathbf{z}} \in W_{\text{loc}}^u(\hat{\mathbf{x}})$, let us define

$$h_{\hat{\mathbf{x}},\hat{\mathbf{y}}}^u = \mathfrak{h}(\hat{\mathbf{y}}, \cdot) = \lim_{k \in \mathbb{N}'} \mathbb{P}H_{\hat{\mathbf{x}},\hat{\mathbf{y}}}^{u,k} \quad \text{and} \quad h_{\hat{\mathbf{y}},\hat{\mathbf{z}}}^u = h_{\hat{\mathbf{x}},\hat{\mathbf{z}}}^u (h_{\hat{\mathbf{x}},\hat{\mathbf{y}}}^u)^{-1}.$$

We have to check the conditions involved in the definition of a restricted projective u -holonomy; that is, the projective version of items (U.1) and (U.3) of Definition 1. To this purpose, consider any $\hat{\mathbf{y}}, \hat{\mathbf{z}}, \hat{\mathbf{w}} \in W_{\text{loc}}^u(\hat{\mathbf{x}})$.

$$(U.1) \quad h_{\hat{\mathbf{x}}, \hat{\mathbf{x}}}^u = \mathfrak{h}(\hat{\mathbf{x}}, \cdot) = \lim_{k \in \mathbb{N}'} \mathbb{P} H_{\hat{\mathbf{x}}, \hat{\mathbf{x}}}^{u, k} = \lim_{k \in \mathbb{N}'} \text{id} = \text{id},$$

$$h_{\hat{\mathbf{y}}, \hat{\mathbf{y}}}^u = h_{\hat{\mathbf{x}}, \hat{\mathbf{y}}}^u (h_{\hat{\mathbf{x}}, \hat{\mathbf{y}}}^u)^{-1} = \text{id},$$

and

$$h_{\hat{\mathbf{y}}, \hat{\mathbf{z}}}^u = h_{\hat{\mathbf{x}}, \hat{\mathbf{z}}}^u (h_{\hat{\mathbf{x}}, \hat{\mathbf{y}}}^u)^{-1} = h_{\hat{\mathbf{x}}, \hat{\mathbf{z}}}^u (h_{\hat{\mathbf{x}}, \hat{\mathbf{w}}}^u)^{-1} h_{\hat{\mathbf{x}}, \hat{\mathbf{w}}}^u (h_{\hat{\mathbf{x}}, \hat{\mathbf{y}}}^u)^{-1} = h_{\hat{\mathbf{w}}, \hat{\mathbf{z}}}^u h_{\hat{\mathbf{y}}, \hat{\mathbf{w}}}^u;$$

and

$$(U.3) \quad (\hat{\mathbf{y}}, \hat{\mathbf{z}}) \mapsto h_{\hat{\mathbf{y}}, \hat{\mathbf{z}}}^u = h_{\hat{\mathbf{x}}, \hat{\mathbf{z}}}^u (h_{\hat{\mathbf{x}}, \hat{\mathbf{y}}}^u)^{-1} \text{ is continuous as } \hat{\mathbf{y}} \text{ and } \hat{\mathbf{z}} \text{ vary in } W_{\text{loc}}^u(\hat{\mathbf{x}}).$$

Therefore, we conclude that h^u is a projective u -holonomy for $\hat{A}$ restricted to $W_{\text{loc}}^u(\hat{\mathbf{x}})$, as we wanted. $\square$

4.2 Straightened-out cocycles

In this section we study a technique (introduced by Bonatti, Gómez-Mont and Viana in [BGV03, §1.4]) that makes a given cocycle constant along local stable sets. Recall we are considering a two-sided subshift of finite type $(\hat{\Sigma}, \hat{f})$. Since the sequence space $\hat{\Sigma}$ can be foliated through local sets, such technique goes along with the purpose of project our problem to the one-sided subshift of finite type. This is required in order to complete the proof of Theorem B, as done in Section 5.1.

Lemma 4.2.1. *Every linear cocycle $(\hat{B}, H^s) \in \mathcal{H}_s^0(2, \hat{\Sigma})$ can be straightened out to a linear cocycle $\hat{B}: \hat{\Sigma} \rightarrow \text{SL}(2, \mathbb{R})$ constant along local stable sets. Besides, if $\hat{B}$ admits a non-uniform u -holonomy, then the straightened-out cocycle $\tilde{B}$ also admits one.*

Proof. For each symbol $i = 0, \dots, \kappa - 1$ fix a point $\hat{\mathbf{p}}_i \in [0; i]$. Let us define then the following mapping:

$$\hat{g}: \begin{array}{ccc} \hat{\Sigma} & \longrightarrow & \hat{\Sigma} \\ \hat{\mathbf{x}} & \longmapsto & \hat{\mathbf{y}} \text{ that is the unique point in } W_{\text{loc}}^u(\hat{\mathbf{p}}_{x_0}) \cap W_{\text{loc}}^s(\hat{\mathbf{x}}). \end{array}$$

We would like to observe that $\hat{g}$ is constant along stable sets.

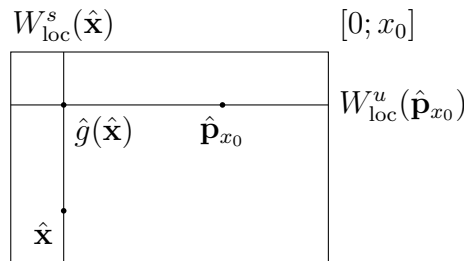


Figure 4.1: Local definition of the map $\hat{g}$ in a cylinder $[0; x_0]$.

Now we can construct the straightened-out cocycle $\tilde{B}: \hat{\Sigma} \rightarrow \text{SL}(2, \mathbb{R})$ by the formula

$$\tilde{B}(\hat{\mathbf{x}}) = H_{\hat{f}(\hat{\mathbf{x}}), \hat{g} \circ \hat{f}(\hat{\mathbf{x}})}^s \hat{B}(\hat{\mathbf{x}}) H_{\hat{g}(\hat{\mathbf{x}}), \hat{\mathbf{x}}}^s. \quad (4.2)$$

As expected, $\tilde{B}$ is a linear cocycle over $\hat{f}$. In order to show that it is constant along local stable sets let us take any pair $\hat{\mathbf{x}} \in \hat{\Sigma}$ and $\hat{\mathbf{y}} \in W_{\text{loc}}^s(\hat{\mathbf{x}})$. By the properties of s -holonomies and the mapping $\hat{g}$ defined above, we get:

$$\begin{aligned} \tilde{B}(\hat{\mathbf{y}}) &= H_{\hat{f}(\hat{\mathbf{y}}), \hat{g} \circ \hat{f}(\hat{\mathbf{y}})}^s \hat{B}(\hat{\mathbf{y}}) H_{\hat{g}(\hat{\mathbf{y}}), \hat{\mathbf{y}}}^s \\ &= H_{\hat{f}(\hat{\mathbf{y}}), \hat{g} \circ \hat{f}(\hat{\mathbf{y}})}^s \hat{B}(\hat{\mathbf{y}}) (H_{\hat{\mathbf{x}}, \hat{\mathbf{y}}}^s H_{\hat{g}(\hat{\mathbf{y}}), \hat{\mathbf{x}}}^s) \\ &= H_{\hat{f}(\hat{\mathbf{y}}), \hat{g} \circ \hat{f}(\hat{\mathbf{y}})}^s \left(H_{\hat{f}(\hat{\mathbf{x}}), \hat{f}(\hat{\mathbf{y}})}^s \hat{B}(\hat{\mathbf{x}}) \right) H_{\hat{g}(\hat{\mathbf{y}}), \hat{\mathbf{x}}}^s \\ &= H_{\hat{f}(\hat{\mathbf{x}}), \hat{g} \circ \hat{f}(\hat{\mathbf{y}})}^s \hat{B}(\hat{\mathbf{x}}) H_{\hat{g}(\hat{\mathbf{y}}), \hat{\mathbf{x}}}^s \\ &= H_{\hat{f}(\hat{\mathbf{x}}), \hat{g} \circ \hat{f}(\hat{\mathbf{x}})}^s \hat{B}(\hat{\mathbf{x}}) H_{\hat{g}(\hat{\mathbf{x}}), \hat{\mathbf{x}}}^s = \tilde{B}(\hat{\mathbf{x}}), \end{aligned}$$

as we wanted.

Finally, suppose that $\hat{B}$ admits a non-uniform u -holonomy H^u with blocks $\{\mathcal{D}_m^u\}_{m \in \mathbb{N}^*}$. Fix any $m \in \mathbb{N}^*$. For each $\hat{\mathbf{x}} \in \mathcal{D}_m^u$ and $\hat{\mathbf{y}} \in W_{\text{loc}}^u(\hat{\mathbf{y}})$, let us define the linear isomorphism $\tilde{H}_{\hat{\mathbf{x}}, \hat{\mathbf{y}}}^u: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ as

$$\tilde{H}_{\hat{\mathbf{x}}, \hat{\mathbf{y}}}^u = H_{\hat{\mathbf{y}}, \hat{g}(\hat{\mathbf{y}})}^s H_{\hat{\mathbf{x}}, \hat{\mathbf{y}}}^u H_{\hat{g}(\hat{\mathbf{x}}), \hat{\mathbf{x}}}^s.$$

We claim that the collection $\tilde{H}^u$ constructed this way is a non-uniform u -holonomy for the linear cocycle $\tilde{B}$ over $\hat{f}$ with the same u -holonomy blocks as H^u . Indeed, given any $\hat{\mathbf{x}} \in \mathcal{D}_m^u$ and $\hat{\mathbf{y}}, \hat{\mathbf{z}} \in W_{\text{loc}}^u(\hat{\mathbf{x}})$, we have

$$\begin{aligned} \text{(NU.1)} \quad \tilde{H}_{\hat{\mathbf{x}}, \hat{\mathbf{x}}}^u &= H_{\hat{\mathbf{x}}, \hat{g}(\hat{\mathbf{x}})}^s H_{\hat{\mathbf{x}}, \hat{\mathbf{x}}}^u H_{\hat{g}(\hat{\mathbf{x}}), \hat{\mathbf{x}}}^s = H_{\hat{\mathbf{x}}, \hat{g}(\hat{\mathbf{x}})}^s H_{\hat{g}(\hat{\mathbf{x}}), \hat{\mathbf{x}}}^s = \text{id}, \text{ and} \\ \tilde{H}_{\hat{\mathbf{x}}, \hat{\mathbf{y}}}^u &= H_{\hat{\mathbf{y}}, \hat{g}(\hat{\mathbf{y}})}^s H_{\hat{\mathbf{x}}, \hat{\mathbf{y}}}^u H_{\hat{g}(\hat{\mathbf{x}}), \hat{\mathbf{x}}}^s = H_{\hat{\mathbf{y}}, \hat{g}(\hat{\mathbf{y}})}^s (H_{\hat{\mathbf{z}}, \hat{\mathbf{y}}}^u H_{\hat{\mathbf{x}}, \hat{\mathbf{z}}}^u) H_{\hat{g}(\hat{\mathbf{x}}), \hat{\mathbf{x}}}^s \\ &= H_{\hat{\mathbf{y}}, \hat{g}(\hat{\mathbf{y}})}^s H_{\hat{\mathbf{z}}, \hat{\mathbf{y}}}^u (H_{\hat{g}(\hat{\mathbf{z}}), \hat{\mathbf{z}}}^s H_{\hat{\mathbf{z}}, \hat{g}(\hat{\mathbf{z}})}^s) H_{\hat{\mathbf{x}}, \hat{\mathbf{z}}}^u H_{\hat{g}(\hat{\mathbf{x}}), \hat{\mathbf{x}}}^s = \tilde{H}_{\hat{\mathbf{z}}, \hat{\mathbf{y}}}^u \tilde{H}_{\hat{\mathbf{x}}, \hat{\mathbf{z}}}^u; \\ \text{(NU.2)} \quad \tilde{H}_{\hat{f}^{-1}(\hat{\mathbf{x}}), \hat{f}^{-1}(\hat{\mathbf{y}})}^u &= H_{\hat{f}^{-1}(\hat{\mathbf{y}}), \hat{g} \circ \hat{f}^{-1}(\hat{\mathbf{y}})}^s H_{\hat{f}^{-1}(\hat{\mathbf{x}}), \hat{f}^{-1}(\hat{\mathbf{y}})}^u H_{\hat{g} \circ \hat{f}^{-1}(\hat{\mathbf{x}}), \hat{f}^{-1}(\hat{\mathbf{x}})}^s \\ &= H_{\hat{f}^{-1}(\hat{\mathbf{y}}), \hat{g} \circ \hat{f}^{-1}(\hat{\mathbf{y}})}^s (\hat{B}^{-1}(\hat{\mathbf{y}}) H_{\hat{\mathbf{x}}, \hat{\mathbf{y}}}^u \hat{B}^{-1}(\hat{\mathbf{x}})^{-1}) H_{\hat{g} \circ \hat{f}^{-1}(\hat{\mathbf{x}}), \hat{f}^{-1}(\hat{\mathbf{x}})}^s \\ &= H_{\hat{f}^{-1}(\hat{\mathbf{y}}), \hat{g} \circ \hat{f}^{-1}(\hat{\mathbf{y}})}^s \hat{B}^{-1}(\hat{\mathbf{y}}) (H_{\hat{g}(\hat{\mathbf{y}}), \hat{\mathbf{y}}}^s H_{\hat{\mathbf{y}}, \hat{g}(\hat{\mathbf{y}})}^s) H_{\hat{\mathbf{x}}, \hat{\mathbf{y}}}^u \hat{B}^{-1}(\hat{\mathbf{x}})^{-1} H_{\hat{g} \circ \hat{f}^{-1}(\hat{\mathbf{x}}), \hat{f}^{-1}(\hat{\mathbf{x}})}^s \\ &= \tilde{B}^{-1}(\hat{\mathbf{y}}) H_{\hat{\mathbf{y}}, \hat{g}(\hat{\mathbf{y}})}^s H_{\hat{\mathbf{x}}, \hat{\mathbf{y}}}^u (H_{\hat{g}(\hat{\mathbf{x}}), \hat{\mathbf{x}}}^s H_{\hat{\mathbf{x}}, \hat{g}(\hat{\mathbf{x}})}^s) \hat{B}^{-1}(\hat{\mathbf{x}})^{-1} H_{\hat{g} \circ \hat{f}^{-1}(\hat{\mathbf{x}}), \hat{f}^{-1}(\hat{\mathbf{x}})}^s \\ &= \tilde{B}^{-1}(\hat{\mathbf{y}}) \tilde{H}_{\hat{\mathbf{x}}, \hat{\mathbf{y}}}^u \tilde{B}^{-1}(\hat{\mathbf{x}})^{-1}; \text{ and} \end{aligned}$$

$$\text{(NU.3)} \quad (\hat{\mathbf{x}}, \hat{\mathbf{y}}) \mapsto \tilde{H}_{\hat{\mathbf{x}}, \hat{\mathbf{y}}}^u \text{ is continuous in } \mathcal{D}_m^u \text{ since it is a composition of continuous maps.}$$

Items (NU.1)–(NU.3) above characterise all properties of a non-uniform u -holonomy for $\tilde{B}$ over $\hat{f}$, as we claimed. $\square$

Remark 4.2.2. It is worth to mention that diagonal condition is preserved by the straightening-out process. Indeed, if we consider a diagonal cocycle $(\hat{B}, H^s)$, the linear isomorphism $\tilde{B}(\hat{\mathbf{x}})$ defined at Equation (4.2) is diagonal since it is the product of diagonal matrices.

The next result deals with invariant probability measures and u -states for the straightened-out cocycle.

Lemma 4.2.3. *In the setting of the previous lemma, every $\mathbb{P}F_{\tilde{B}}$ -invariant probability measure $\hat{m}$ in $\hat{\Sigma} \times \mathbb{P}\mathbb{R}^2$ that projects down to $\hat{\mu}$ can be transformed in a $\mathbb{P}F_{\tilde{B}}$ -invariant probability measure $\tilde{m}$ that also projects down to $\hat{\mu}$. Besides, if $\hat{m}$ is a u -state for $(\hat{B}, H^u)$, then $\tilde{m}$ is a u -state for $(\tilde{B}, \tilde{H}^u)$.*

Proof. Recall the mapping $\hat{g}$ constructed in the beginning of the last proof. We will use again the notation $h^s = \mathbb{P}H^s$ to the projective s -holonomy.

Now, given a disintegration $\{\hat{m}_{\hat{\mathbf{x}}}\}_{\hat{\mathbf{x}} \in \hat{\Sigma}}$ of $\hat{m}$, let us define the conditional measure

$$\tilde{m}_{\hat{\mathbf{x}}} = (h_{\hat{\mathbf{x}}, \hat{g}(\hat{\mathbf{x}})}^s)_* \hat{m}_{\hat{\mathbf{x}}}$$

for each $\hat{\mathbf{x}} \in \hat{\Sigma}$. Hence we can define the measure $\tilde{m} = \int \tilde{m}_{\hat{\mathbf{x}}} d\hat{\mu}(\hat{\mathbf{x}})$. Notice that it is direct from the construction that $\tilde{m}$ is a probability measure in $\hat{\Sigma} \times \mathbb{P}\mathbb{R}^2$ that projects down to $\hat{\mu}$.

Suppose at first that the disintegration $\{\hat{m}_{\hat{\mathbf{x}}}\}_{\hat{\mathbf{x}} \in \hat{\Sigma}}$ is $\mathbb{P}\hat{B}$ -invariant; that is,

$$\mathbb{P}\hat{B}(\hat{\mathbf{x}})_* \hat{m}_{\hat{\mathbf{x}}} = \hat{m}_{\hat{f}(\hat{\mathbf{x}})} \quad \text{for } \hat{\mu}\text{-almost every } \hat{\mathbf{x}} \in \hat{\Sigma}.$$

We need to show that the constructed disintegration $\{\tilde{m}_{\hat{\mathbf{x}}}\}_{\hat{\mathbf{x}} \in \hat{\Sigma}}$ is $\mathbb{P}\tilde{B}$ -invariant. In order to do it just notice that, for $\hat{\mu}$ -almost every $\hat{\mathbf{x}} \in \hat{\Sigma}$, we have:

$$\begin{aligned} \mathbb{P}\tilde{B}(\hat{\mathbf{x}})_* \tilde{m}_{\hat{\mathbf{x}}} &= \mathbb{P}(H_{\hat{f}(\hat{\mathbf{x}}), \hat{g} \circ \hat{f}(\hat{\mathbf{x}})}^s \hat{B}(\hat{\mathbf{x}}) H_{\hat{g}(\hat{\mathbf{x}}), \hat{\mathbf{x}}}^s)_* (h_{\hat{\mathbf{x}}, \hat{g}(\hat{\mathbf{x}})}^s)_* \hat{m}_{\hat{\mathbf{x}}} \\ &= \mathbb{P}(H_{\hat{f}(\hat{\mathbf{x}}), \hat{g} \circ \hat{f}(\hat{\mathbf{x}})}^s \hat{B}(\hat{\mathbf{x}}) H_{\hat{g}(\hat{\mathbf{x}}), \hat{\mathbf{x}}}^s H_{\hat{\mathbf{x}}, \hat{g}(\hat{\mathbf{x}})}^s)_* \hat{m}_{\hat{\mathbf{x}}} \\ &= (h_{\hat{f}(\hat{\mathbf{x}}), \hat{g} \circ \hat{f}(\hat{\mathbf{x}})}^s)_* \mathbb{P}\hat{B}(\hat{\mathbf{x}})_* \hat{m}_{\hat{\mathbf{x}}} \\ &= (h_{\hat{f}(\hat{\mathbf{x}}), \hat{g} \circ \hat{f}(\hat{\mathbf{x}})}^s)_* \hat{m}_{\hat{f}(\hat{\mathbf{x}})} = \tilde{m}_{\hat{f}(\hat{\mathbf{x}})}, \end{aligned}$$

as we needed. It follows that $\tilde{m}$ is $\mathbb{P}F_{\tilde{B}}$ -invariant.

Suppose at last that the disintegration $\{\hat{m}_{\hat{\mathbf{x}}}\}_{\hat{\mathbf{x}} \in \hat{\Sigma}}$ is u -invariant for the projective holonomy $h^u = \mathbb{P}H^u$; that is,

$$(h_{\hat{\mathbf{x}}, \hat{\mathbf{y}}}^u)_* \hat{m}_{\hat{\mathbf{x}}} = \hat{m}_{\hat{\mathbf{y}}} \quad \text{for } \hat{\mu}\text{-almost every } \hat{\mathbf{x}} \in \hat{\Sigma} \text{ and any } \hat{\mathbf{y}} \in W_{\text{loc}}^u(\hat{\mathbf{x}}).$$

We want to show that the constructed disintegration $\{\tilde{m}_{\hat{\mathbf{x}}}\}_{\hat{\mathbf{x}} \in \hat{\Sigma}}$ is u -invariant for $\tilde{h}^u = \mathbb{P}\tilde{H}^u$. Again, for $\hat{\mu}$ -almost every $\hat{\mathbf{x}} \in \hat{\Sigma}$ and any $\hat{\mathbf{y}} \in W_{\text{loc}}^u(\hat{\mathbf{x}})$, it is true that:

$$\begin{aligned} (\tilde{h}_{\hat{\mathbf{x}}, \hat{\mathbf{y}}}^u)_* \tilde{m}_{\hat{\mathbf{x}}} &= (h_{\hat{\mathbf{y}}, \hat{g}(\hat{\mathbf{y}})}^s h_{\hat{\mathbf{x}}, \hat{\mathbf{y}}}^u h_{\hat{g}(\hat{\mathbf{x}}), \hat{\mathbf{x}}}^s)_* (h_{\hat{\mathbf{x}}, \hat{g}(\hat{\mathbf{x}})}^s)_* \hat{m}_{\hat{\mathbf{x}}} \\ &= (h_{\hat{\mathbf{y}}, \hat{g}(\hat{\mathbf{y}})}^s h_{\hat{\mathbf{x}}, \hat{\mathbf{y}}}^u h_{\hat{g}(\hat{\mathbf{x}}), \hat{\mathbf{x}}}^s h_{\hat{\mathbf{x}}, \hat{g}(\hat{\mathbf{x}})}^s)_* \hat{m}_{\hat{\mathbf{x}}} \\ &= (h_{\hat{\mathbf{y}}, \hat{g}(\hat{\mathbf{y}})}^s)_* (h_{\hat{\mathbf{x}}, \hat{\mathbf{y}}}^u)_* \hat{m}_{\hat{\mathbf{x}}} \\ &= (h_{\hat{\mathbf{y}}, \hat{g}(\hat{\mathbf{y}})}^s)_* \hat{m}_{\hat{\mathbf{y}}} = \tilde{m}_{\hat{\mathbf{y}}}, \end{aligned}$$

as we wanted. It follows that $\tilde{m}$ is a u -state for $(\tilde{B}, \tilde{H}^u)$. $\square$

We end this section showing that the straightening-out process does not change the Lyapunov exponents. For this intend, let us consider a linear cocycle $(B, H^s) \in \mathcal{H}_s^0(2, \hat{\Sigma})$ and its straightened-out cocycle $\tilde{B}: \hat{\Sigma} \rightarrow \text{SL}(2, \mathbb{R})$, as in Equation (4.2). Then, for $\hat{\mu}$ -almost every $\hat{\mathbf{x}} \in \hat{\Sigma}$, we have:

$$\begin{aligned}
 \lambda_+(\tilde{B}) &= \lim_{n \in \mathbb{N}^*} \frac{1}{n} \log \|\tilde{B}^n(\hat{\mathbf{x}})\| \\
 &= \lim_{n \in \mathbb{N}^*} \frac{1}{n} \log \|H_{\hat{f}^n(\hat{\mathbf{x}}), \hat{g} \circ \hat{f}^n(\hat{\mathbf{x}})}^s \hat{B}^n(\hat{\mathbf{x}}) H_{\hat{g}(\hat{\mathbf{x}}), \hat{\mathbf{x}}}^s\| \\
 &= \lim_{n \in \mathbb{N}^*} \frac{1}{n} \left(\underbrace{\log \|H_{\hat{f}^n(\hat{\mathbf{x}}), \hat{g} \circ \hat{f}^n(\hat{\mathbf{x}})}^s\|}_{\text{bounded}} + \log \|\hat{B}^n(\hat{\mathbf{x}})\| + \underbrace{\log \|H_{\hat{g}(\hat{\mathbf{x}}), \hat{\mathbf{x}}}^s\|}_{\text{bounded}} \right) \\
 &= \lim_{n \in \mathbb{N}^*} \frac{1}{n} \log \|\hat{B}^n(\hat{\mathbf{x}})\| = \lambda_+(\hat{B})
 \end{aligned} \tag{4.3}$$

Therefore, it is not an issue for the continuity problem to make the straightening-out process on a linear cocycle.

4.3 Measures induced by u -states

This section is inspired by [BBB18, §4.3] and follows the same notations and definitions. Our goal is to understand somewhat more the dynamics yielded by the straightening-out process discussed on last section.

Let us pay some attention to the non-invertible case of the subshift setting. In other words, we are going to consider a one-sided subshift of finite type (Σ, f) as seen on Section 2.1.

Such dynamical system is somewhat related to the invertible case, that is, the two-sided subshift of finite type $(\hat{\Sigma}, \hat{f})$. Indeed, we can define the following projection:

$$\begin{aligned}
 P: \quad \hat{\Sigma} &\longrightarrow \Sigma \\
 (x_j)_{j \in \mathbb{Z}} &\longmapsto (x_j)_{j \in \mathbb{N}}.
 \end{aligned}$$

That is, the action of P is to drop out all negative entries of a sequence. Observe that, given any $\hat{\mathbf{x}} \in \hat{\Sigma}$, we have $W_{\text{loc}}^s(\hat{\mathbf{x}}) = P^{-1}(\mathbf{x})$, where $\mathbf{x} = P(\hat{\mathbf{x}})$.

As one can see in the next diagram, the projection P induces, from each cocycle $\hat{B}: \hat{\Sigma} \rightarrow \text{SL}(2, \mathbb{R})$ constant along local stable sets, a linear isomorphism $B: \Sigma \rightarrow \text{SL}(2, \mathbb{R})$ satisfying $\hat{B} = B \circ P$. We say that B is the *linear cocycle induced* from $\hat{B}$.

$$\begin{array}{ccc}
 \hat{\Sigma} & \xrightarrow{\hat{B}} & \text{SL}(2, \mathbb{R}) \\
 \downarrow P & \nearrow B & \\
 \Sigma & &
 \end{array}$$

From the last two paragraphs, it is direct to check that the diagrams below commute.

$$\begin{array}{ccc}
 \hat{\Sigma} & \xrightarrow{\hat{f}} & \hat{\Sigma} \\
 P \downarrow & & \downarrow P \\
 \Sigma & \xrightarrow{f} & \Sigma
 \end{array}
 \qquad
 \begin{array}{ccc}
 \hat{\Sigma} \times \mathbb{P}\mathbb{R}^2 & \xrightarrow{\mathbb{P}F_{\hat{B}}} & \hat{\Sigma} \times \mathbb{P}\mathbb{R}^2 \\
 P \times \text{id} \downarrow & & \downarrow P \times \text{id} \\
 \Sigma \times \mathbb{P}\mathbb{R}^2 & \xrightarrow{\mathbb{P}F_B} & \Sigma \times \mathbb{P}\mathbb{R}^2
 \end{array}$$

In addition, if we consider an $\hat{f}$ -ergodic probability measure $\hat{\mu}$ in $\hat{\Sigma}$, then we can define an f -ergodic probability measure μ in Σ as follows:

$$\mu(E) = \hat{\mu}(P^{-1}(E)) \quad \text{for every measurable set } E \subset \Sigma. \quad (4.4)$$

Observe that we can use the projection P to foliate the space $\hat{\Sigma}$ where the leaves are of the form $P^{-1}(\mathbf{x})$, for every $\mathbf{x} \in \Sigma$. The local product structure property of the measure $\hat{\mu}$ permits us to disintegrate it into conditional measures defined on each such leaf, as we can see in the next result.

For each $\mathbf{x}, \mathbf{y} \in \Sigma$ in the same cylinder $[0; i]$ we define the *unstable holonomy map* $h_{\mathbf{x}, \mathbf{y}}: P^{-1}(\mathbf{x}) \rightarrow P^{-1}(\mathbf{y})$ by the condition that $h_{\mathbf{x}, \mathbf{y}}(\hat{\mathbf{x}})$ is the unique point in the intersection of $W_{\text{loc}}^u(\hat{\mathbf{x}})$ with $P^{-1}(\mathbf{y})$.

Lemma 4.3.1 (Lemma 2.1 of [BV04]). *The measure $\hat{\mu}$ has a disintegration $\{\hat{\mu}_{\mathbf{x}}\}_{\mathbf{x} \in \Sigma}$ varying continuously with $\mathbf{x}$ in the weak-* topology. Besides, each unstable holonomy map*

$$h_{\mathbf{x}, \mathbf{y}}: (P^{-1}(\mathbf{x}), \hat{\mu}_{\mathbf{x}}) \longrightarrow (P^{-1}(\mathbf{y}), \hat{\mu}_{\mathbf{y}})$$

is absolutely continuous, with Jacobian $J_{\mathbf{x}, \mathbf{y}}$ depending continuously on $(\mathbf{x}, \mathbf{y})$.

That being so, we say that a probability measure m in $\Sigma \times \mathbb{P}\mathbb{R}^2$ is *induced by a u -state* if one can take:

- (a) $(\hat{B}, H^u) \in \mathcal{H}_u^0(2, \hat{\Sigma})$ constant along local stable sets; and
- (b) a u -state $\hat{m}$ for $(\hat{B}, H^u)$ such that $m = (P \times \text{id})_* \hat{m}$.

Such measure m is $\mathbb{P}F_B$ -invariant, where $B: \Sigma \rightarrow \text{SL}(2, \mathbb{R})$ is the linear cocycle induced from $\hat{B}$. Indeed, since $\mathbb{P}F_B \circ (P \times \text{id}) = (P \times \text{id}) \circ \mathbb{P}F_{\hat{B}}$ and by the $\mathbb{P}F_{\hat{B}}$ -invariance of $\hat{m}$, we have:

$$(\mathbb{P}F_B)_* m = (\mathbb{P}F_B)_* (P \times \text{id})_* \hat{m} = (P \times \text{id})_* (\mathbb{P}F_{\hat{B}})_* \hat{m} = (P \times \text{id})_* \hat{m} = m.$$

Let us explain briefly how to construct a measure m in Σ induced by a measure $\hat{m}$ in $\hat{\Sigma}$. Given a disintegration $\{\hat{m}_{\hat{\mathbf{x}}}\}_{\hat{\mathbf{x}} \in \hat{\Sigma}}$ of $\hat{m}$, we define for each $\mathbf{x} \in \Sigma$ the conditional measure

$$m_{\mathbf{x}} = \int_{P^{-1}(\mathbf{x})} \hat{m}_{\hat{\mathbf{x}}} d\hat{\mu}_{\mathbf{x}}(\hat{\mathbf{x}}) \quad (4.5)$$

and hence

$$m = \int_{\Sigma} m_{\mathbf{x}} d\mu(\mathbf{x}), \quad (4.6)$$

where $\mu = P_* \hat{\mu}$ as in Equation (4.4). It follows that, if $\hat{m}$ is a u -state for some linear cocycle constant along local stable sets, then m satisfies items (a)–(b) above.

In the next result we show explicitly where we use the equiproximity condition, that is crucial here. It is worth to notice that we are not asking for the limit measure $\hat{m}$ to be a u -state in such result.

Proposition 4.3.2. *Let $\hat{A}: \hat{\Sigma} \rightarrow \mathrm{SL}(2, \mathbb{R})$ be a diagonal cocycle constant along local stable sets and with non-zero Lyapunov exponents, and also let $\hat{m}$ be a $\mathbb{P}F_{\hat{A}}$ -invariant probability measure that projects down to $\hat{\mu}$. If one takes, for each $k \in \mathbb{N}^*$, a cocycle $(\hat{A}_k, H^{u,k}) \in \mathcal{H}_u^0(2, \hat{\Sigma})$ constant along local stable sets and admitting a u -state $\hat{m}_k$ such that:*

- $\hat{A}_k \rightarrow \hat{A}$ in the C^0 -topology;
- $\{H^{u,k}\}_{k \in \mathbb{N}^*}$ satisfies the equiproximity condition; and
- $\hat{m}_k \rightarrow \hat{m}$ in the weak-* topology,

then the measures m and $m_1, m_2, \dots$ induced by $\hat{m}$ and $\hat{m}_1, \hat{m}_2, \dots$, respectively, admit disintegrations that form an equicontinuous family.

Proof. We begin by showing that the measure m induced by $\hat{m}$ in the way of Equation (4.6) admits a continuous (in fact, constant) disintegration. In fact, this is due to Equation (4.5), since $\hat{m}$ admits a constant disintegration by Lemma 3.3.5. Thus, it only remains to prove the result for measures $\{m_k\}_{k \in \mathbb{N}^*}$.

Fix for now any $k \in \mathbb{N}^*$. Since $\hat{m}_k$ is a u -state for $(\hat{A}_k, H^{u,k})$, it admits a u -invariant disintegration $\{\hat{m}_{\hat{\mathbf{x}}}^k\}_{\hat{\mathbf{x}} \in \hat{\Sigma}}$, i.e.:

$$(h_{\hat{\mathbf{x}}, \hat{\mathbf{y}}}^{u,k})_* \hat{m}_{\hat{\mathbf{x}}}^k = \hat{m}_{\hat{\mathbf{y}}}^k$$

for all $\hat{\mathbf{y}} \in W_{\mathrm{loc}}^u(\hat{\mathbf{x}})$ and $\hat{\mu}$ -almost every $\hat{\mathbf{x}} \in \hat{\Sigma}$. Let us consider for m_k the disintegration formed by the conditional measures as in Equation (4.5).

We want to prove that, given any continuous function $\varphi: \mathbb{P}\mathbb{R}^2 \rightarrow \mathbb{R}$ and any $\varepsilon > 0$, one can choose a $\delta > 0$ independent of k such that

$$\left| \int \varphi \, dm_{\hat{\mathbf{y}}}^k - \int \varphi \, dm_{\hat{\mathbf{x}}}^k \right| < \varepsilon$$

for every $\hat{\mathbf{x}}, \hat{\mathbf{y}} \in \Sigma$ with $\mathrm{dist}_{\rho}(\hat{\mathbf{x}}, \hat{\mathbf{y}}) < \delta$. As we have taken k randomly, and as the choice of δ does not depend on k , it would follow that the family of conditional measures $\{m_{\hat{\mathbf{x}}}^1, m_{\hat{\mathbf{x}}}^2, \dots\}_{\hat{\mathbf{x}} \in \Sigma}$ is equicontinuous in the weak-* topology.

Let us take an $m \in \mathbb{N}^*$ large enough so that the u -holonomy block $\mathcal{D}_m^u$ (see Definition 4) intersects each cylinder. Now, for all $i = 0, \dots, \kappa - 1$, fix any point $\hat{\mathbf{x}}_i \in [0; i]$ and define the following sets:

$$\mathcal{D}_{i,m} = \{\hat{\mathbf{y}} \in W_{\mathrm{loc}}^s(\hat{\mathbf{x}}_i) : W_{\mathrm{loc}}^u(\hat{\mathbf{y}}) \cap \mathcal{D}_m^u \neq \emptyset\} \text{ and } \mathcal{C}_{i,m} = \bigcup_{\hat{\mathbf{y}} \in \mathcal{D}_{i,m}} W_{\mathrm{loc}}^u(\hat{\mathbf{y}}).$$

It is worth mentioning that the sets $\{\mathcal{C}_{i,m}\}_{i=0, \dots, \kappa-1}$ are well defined: none of them depends on the choice of the points $\{\hat{\mathbf{x}}_i\}_{i=0, \dots, \kappa-1}$.

Recall the definition and the properties of the unstable holonomy map given at Lemma 4.3.1. Let us observe that, for any $\hat{\mathbf{x}}, \hat{\mathbf{y}} \in \Sigma$ in the same cylinder $[0; i]$, we have:

$$\int \varphi \, dm_{\hat{\mathbf{y}}}^k = \int_{P^{-1}(\hat{\mathbf{y}})} \int \varphi \, d\hat{m}_{\hat{\mathbf{y}}}^k \, d\hat{\mu}_{\hat{\mathbf{y}}}(\hat{\mathbf{y}})$$

$$\begin{aligned}
&= \int_{P^{-1}(\mathbf{x}) \cap \mathcal{C}_{i,m}} \left(\int \varphi \, d\hat{m}_{h_{\mathbf{x},\mathbf{y}}(\hat{\mathbf{x}})}^k \right) J_{\mathbf{x},\mathbf{y}}(\hat{\mathbf{x}}) \, d\hat{\mu}_{\mathbf{x}}(\hat{\mathbf{x}}) + \\
&\quad + \int_{P^{-1}(\mathbf{x}) \cap \mathcal{C}_{i,m}^c} \int \varphi \, d\hat{m}_{\hat{\mathbf{y}}}^k \, d\hat{\mu}_{\mathbf{y}}(\hat{\mathbf{y}}) \\
&= \int_{P^{-1}(\mathbf{x}) \cap \mathcal{C}_{i,m}} \left(\int \varphi \circ h_{\hat{\mathbf{x}},\hat{\mathbf{y}}}^{u,k} \, d\hat{m}_{\hat{\mathbf{x}}}^k \right) J_{\mathbf{x},\mathbf{y}}(\hat{\mathbf{x}}) \, d\hat{\mu}_{\mathbf{x}}(\hat{\mathbf{x}}) + \\
&\quad + \int_{P^{-1}(\mathbf{x}) \cap \mathcal{C}_{i,m}^c} \int \varphi \, d\hat{m}_{\hat{\mathbf{y}}}^k \, d\hat{\mu}_{\mathbf{y}}(\hat{\mathbf{y}}),
\end{aligned}$$

where we had used the definition of the conditional measure $m_{\mathbf{y}}^k$ and we had changed variables $\hat{\mathbf{y}} = h_{\mathbf{x},\mathbf{y}}(\hat{\mathbf{x}})$.

It is possible to show by local product structure that

$$\hat{\mu}_{\mathbf{y}}(P^{-1}(\mathbf{x}) \cap \mathcal{C}_{i,m}^c) = \hat{\mu}([0; i] \setminus \mathcal{C}_{i,m}) < \frac{1}{m}.$$

Hence

$$\begin{aligned}
\left| \int \varphi \, d\hat{m}_{\mathbf{y}}^k - \int \varphi \, d\hat{m}_{\mathbf{x}}^k \right| &= \left| \int_{P^{-1}(\mathbf{x}) \cap \mathcal{C}_{i,m}} \int J_{\mathbf{x},\mathbf{y}}(\hat{\mathbf{x}}) \varphi \circ h_{\hat{\mathbf{x}},\hat{\mathbf{y}}}^{u,k} \, d\hat{m}_{\hat{\mathbf{x}}}^k \, d\hat{\mu}_{\mathbf{x}}(\hat{\mathbf{x}}) + \right. \\
&\quad + \int_{P^{-1}(\mathbf{x}) \cap \mathcal{C}_{i,m}^c} \int \varphi \, d\hat{m}_{\hat{\mathbf{y}}}^k \, d\hat{\mu}_{\mathbf{y}}(\hat{\mathbf{y}}) - \\
&\quad - \int_{P^{-1}(\mathbf{x}) \cap \mathcal{C}_{i,m}} \int \varphi \, d\hat{m}_{\hat{\mathbf{x}}}^k \, d\hat{\mu}_{\mathbf{x}}(\hat{\mathbf{x}}) - \\
&\quad \left. - \int_{P^{-1}(\mathbf{x}) \cap \mathcal{C}_{i,m}^c} \int \varphi \, d\hat{m}_{\hat{\mathbf{x}}}^k \, d\hat{\mu}_{\mathbf{x}}(\hat{\mathbf{x}}) \right| \\
&\leq \int_{P^{-1}(\mathbf{x}) \cap \mathcal{C}_{i,m}} \int \|J_{\mathbf{x},\mathbf{y}} \varphi \circ h_{\hat{\mathbf{x}},\hat{\mathbf{y}}}^{u,k} - \varphi\| \, d\hat{m}_{\hat{\mathbf{x}}}^k \, d\hat{\mu}_{\mathbf{x}}(\hat{\mathbf{x}}) + \\
&\quad + \int_{P^{-1}(\mathbf{x}) \cap \mathcal{C}_{i,m}^c} \int |\varphi| \, d\hat{m}_{\hat{\mathbf{y}}}^k \, d\hat{\mu}_{\mathbf{y}}(\hat{\mathbf{y}}) + \\
&\quad + \int_{P^{-1}(\mathbf{x}) \cap \mathcal{C}_{i,m}^c} \int |\varphi| \, d\hat{m}_{\hat{\mathbf{x}}}^k \, d\hat{\mu}_{\mathbf{x}}(\hat{\mathbf{x}}) \\
&\leq \int_{P^{-1}(\mathbf{x}) \cap \mathcal{C}_{i,m}} \int \|J_{\mathbf{x},\mathbf{y}} \varphi \circ h_{\hat{\mathbf{x}},\hat{\mathbf{y}}}^{u,k} - \varphi\| \, d\hat{m}_{\hat{\mathbf{x}}}^k \, d\hat{\mu}_{\mathbf{x}}(\hat{\mathbf{x}}) + \\
&\quad + \frac{2C}{m},
\end{aligned}$$

where $C = \sup |\varphi|$ (recall that φ is bounded).

Increasing m if necessary, we can assume $\frac{2C}{m} < \frac{\varepsilon}{2}$. Thus it is sufficient to find a $\delta > 0$ such that $\text{dist}_{\rho}(\mathbf{x}, \mathbf{y}) < \delta$ implies

$$\|J_{\mathbf{x},\mathbf{y}}(\hat{\mathbf{x}}) \cdot \varphi \circ h_{\hat{\mathbf{x}},\hat{\mathbf{y}}}^{u,k} - \varphi\| < \frac{\varepsilon}{2} \quad (4.7)$$

for every $\hat{\mathbf{x}} \in P^{-1}(\mathbf{x}) \cap \mathcal{C}_{i,m}$ and $\hat{\mathbf{y}} = h_{\mathbf{x},\mathbf{y}}(\hat{\mathbf{x}})$.

In order to find such δ , observe that, since the Jacobian $J_{\mathbf{x},\mathbf{y}}$ depends continuously on $(\mathbf{x}, \mathbf{y})$, we can choose a $\delta' > 0$ satisfying

$$\|J_{\mathbf{x},\mathbf{y}} - 1\| < \frac{\varepsilon}{4C}$$

whenever $\text{dist}_\rho(\mathbf{x}, \mathbf{y}) < \delta'$. On the other hand, the continuity of φ implies the existence of an $\varepsilon' > 0$ such that

$$|\varphi[v] - \varphi[w]| < \frac{\varepsilon}{4}$$

for all $[v], [w] \in \mathbb{P}\mathbb{R}^2$ with $\text{dist}([v], [w]) < \varepsilon'$.

Now, as the sequence $\{H^{u,k}\}_{k \in \mathbb{N}^*}$ satisfies the equiproximity condition, we can choose another $\delta'_m > 0$ so that

$$\|H_{\hat{\mathbf{x}}, \hat{\mathbf{y}}}^{u,k} - \text{id}\| < \varepsilon'$$

for each $\hat{\mathbf{x}} \in \mathcal{C}_{i,m}$ and $\hat{\mathbf{y}} \in W_{\text{loc}}^u(\hat{\mathbf{x}})$ with $\text{dist}_\rho(\hat{\mathbf{x}}, \hat{\mathbf{y}}) < \delta'_m$. Observe that, given any $\mathbf{x}, \mathbf{y} \in \Sigma$ in the same cylinder $[0; i]$, we have $\text{dist}_\rho(\mathbf{x}, \mathbf{y}) = \text{dist}_\rho(\hat{\mathbf{x}}, \hat{\mathbf{y}})$ for every $\hat{\mathbf{x}} \in P^{-1}(\mathbf{x})$ and $\hat{\mathbf{y}} = h_{\mathbf{x},\mathbf{y}}(\hat{\mathbf{x}})$. Take, finally, $\delta = \min\{\delta', \delta'_m\}$.

Therefore if one takes $\mathbf{x}, \mathbf{y} \in \Sigma$ satisfying $\text{dist}_\rho(\mathbf{x}, \mathbf{y}) < \delta$, and also takes $\hat{\mathbf{y}} = h_{\mathbf{x},\mathbf{y}}(\hat{\mathbf{x}})$ for some $\hat{\mathbf{x}} \in P^{-1}(\mathbf{x}) \cap \mathcal{C}_{i,m}$, then

$$\begin{aligned} \|J_{\mathbf{x},\mathbf{y}} \cdot \varphi \circ h_{\hat{\mathbf{x}}, \hat{\mathbf{y}}}^{u,k} - \varphi\| &\leq \|J_{\mathbf{x},\mathbf{y}} \cdot \varphi \circ h_{\hat{\mathbf{x}}, \hat{\mathbf{y}}}^{u,k} - \varphi \circ h_{\hat{\mathbf{x}}, \hat{\mathbf{y}}}^{u,k}\| + \|\varphi \circ h_{\hat{\mathbf{x}}, \hat{\mathbf{y}}}^{u,k} - \varphi\| \\ &\leq \sup |\varphi \circ h_{\hat{\mathbf{x}}, \hat{\mathbf{y}}}^{u,k}| \|J_{\mathbf{x},\mathbf{y}} - 1\| + \|\varphi \circ h_{\hat{\mathbf{x}}, \hat{\mathbf{y}}}^{u,k} - \varphi\| \\ &< C \cdot \frac{\varepsilon}{4C} + \frac{\varepsilon}{4} = \frac{\varepsilon}{2} \end{aligned}$$

as we wanted in Equation (4.7). $\square$

Backes, Brown and Butler shown in [BBB18, Lemma 4.8] that the disintegrations stated in the previous proposition have the following property that will be useful in next section. We are not going to prove it here.

Proposition 4.3.3. *Considering the hypotheses of Proposition 4.3.2, the disintegrations found satisfy: $m_{\mathbf{x}}^k \rightarrow m_{\mathbf{x}}$ in the weak-* topology for every $\mathbf{x} \in \Sigma$. In fact, this convergence is uniform in $\mathbf{x}$.*

Before we finish that chapter and begin with the proof of Theorem B and Theorem C, let us show that the Lyapunov exponents remain unchanged when we project our problem from the two-sided to the one-sided shift setting.

Let us consider a linear cocycle $\hat{B}: \hat{\Sigma} \rightarrow \text{SL}(2, \mathbb{R})$ constant along local stable sets, and the linear cocycle $B: \Sigma \rightarrow \text{SL}(2, \mathbb{R})$ induced by it, that is, such that $\hat{B} = B \circ P$. Then, for $\hat{\mu}$ -almost every $\hat{\mathbf{x}} \in \hat{\Sigma}$, we have:

$$\begin{aligned} \lambda_+(\hat{B}) &= \lim_{n \in \mathbb{N}^*} \frac{1}{n} \log \|\hat{B}^n(\hat{\mathbf{x}})\| \\ &= \lim_{n \in \mathbb{N}^*} \frac{1}{n} \log \|(B \circ P)^n(\hat{\mathbf{x}})\| \end{aligned}$$

$$\begin{aligned}
&= \lim_{n \in \mathbb{N}^*} \frac{1}{n} \log \|B^n(P(\hat{\mathbf{x}}))\| \\
&= \lim_{n \in \mathbb{N}^*} \frac{1}{n} \log \|B^n(\mathbf{x})\|, \quad (\text{where } \mathbf{x} = P(\hat{\mathbf{x}}))
\end{aligned}$$

and, since $\hat{\mu}$ -almost every $\hat{\mathbf{x}}$ implies μ -almost every $\mathbf{x}$, where $\mu = P_*\hat{\mu}$ as in Equation (4.4), we conclude:

$$\lambda_+(\hat{B}) = \lambda_+(B). \quad (4.8)$$

Chapter 5

Proof of the theorems

We devote this chapter to the proofs of Theorem 2.1.1 and Theorem 2.1.2. This is the last step in order to conclude Theorem B and Theorem C.

5.1 Proof of Theorem 2.1.1

In this section we are in position to prove Theorem 2.1.1. Remember that such result implies Theorem B, and hence Theorem A. We are going to reproduce here the statement of the theorem to make easier to follow the proof.

Theorem 2.1.1. *Let $(\hat{\Sigma}, \hat{f})$ be a subshift of finite type and $\hat{\mu}$ be an $\hat{f}$ -ergodic probability measure on $\hat{\Sigma}$ that has local product structure and full support. Consider $(\hat{A}, H^s) \in \mathcal{H}_s^0(2, \hat{\Sigma})$ satisfying the 2-invariance condition. If one takes any sequence $\{(\hat{A}_k, H^{s,k}, H^{u,k})\}_{k \in \mathbb{N}^*}$ of linear cocycles with uniform s -holonomies and non-uniform u -holonomies such that*

$$(1) \quad (\hat{A}_k, H^{s,k}) \rightarrow (\hat{A}, H^s) \text{ in } \mathcal{H}_s^0(2, \hat{\Sigma}); \text{ and}$$

$$(2) \quad \{H^{u,k}\}_{k \in \mathbb{N}^*} \text{ satisfies the equiproximity condition,}$$

then $\{\lambda_+(\hat{A}_k)\}_{k \in \mathbb{N}^}$ converges to $\lambda_+(\hat{A})$.*

Proof. We begin by recalling that we can consider $(\hat{A}, H^s)$ a diagonal cocycle due to the 2-invariance condition.

Now, for each $k \in \mathbb{N}^*$ consider the $\mathbb{P}F_{\hat{A}_k}$ -ergodic probability measure $\hat{m}_k$ that projects down to $\hat{\mu}$, and satisfies the equation:

$$\lambda_+(\hat{A}_k) = \int \Phi_k d\hat{m}_k, \quad \text{where} \quad \Phi_k(\hat{\mathbf{x}}, [v]) = \log \frac{\|\hat{A}_k(\hat{\mathbf{x}}) \cdot v\|}{\|v\|}.$$

The existence of such measure was proved in Proposition 3.2.4.

Although the cited proposition also claims that each one of these measures $\hat{m}_k$ can be taken as a u -state for $(\hat{A}_k, H^{u,k})$, we may wonder whether or not it is also an s -state for $(\hat{A}_k, H^{s,k})$. As commented at Section 3.2, the key property of su -states is given by Proposition 3.2.1, and it motivates us to split our proof in two cases, namely: (i) when there are infinitely many su -states in the sequence $\{\hat{m}_k\}_{k \in \mathbb{N}^*}$; and (ii) when there is only a finite number of those measures. We will deal with these cases separately.

Case 1. There is a subsequence $\{\hat{m}_k\}_{k \in \mathbb{N}'}$ composed by su -states. To simplify the notation we can assume it to the whole sequence.

In order to show the convergence in this case, we are going to apply the sufficient condition established in Proposition 3.4.1. Let us then verify its hypotheses.

As we have mentioned, the desirable feature of su -states is that they admit a continuous and invariant disintegration (see Proposition 3.2.1). Thus, for each $k \in \mathbb{N}^*$, let $\{\hat{m}_k^{\hat{x}}\}_{\hat{x} \in \hat{\Sigma}}$ be a disintegration of $\hat{m}_k$ such that: it varies continuously with the point $\hat{x}$, and it is invariant by the projective cocycle $\mathbb{P}\hat{A}_k$ and both projective holonomies $h^{s,k} = \mathbb{P}H^{s,k}$ and $h^{u,k} = \mathbb{P}H^{u,k}$.

Recall that, for each $k \in \mathbb{N}^*$, the invariance by u -holonomies happens in a full-measure set $\Sigma_k^u \subset \hat{\Sigma}$, as we are dealing with the non-uniform case. Thus the intersection $\Sigma^u = \bigcap_{k \in \mathbb{N}^*} \Sigma_k^u$ still has full $\hat{\mu}$ -measure. Besides, the set Γ of points in $\hat{\Sigma}$ where we can apply Theorem 4.1.2 (that is, where the sequence of u -holonomies converges) also has full $\hat{\mu}$ -measure. Finally, observe that the set Θ of points in $\hat{\Sigma}$ having all symbols in their non-negative entries has positive $\hat{\mu}$ -measure. Thus we can take a point $\hat{\mathbf{q}}$ in the intersection $\Sigma^u \cap \Gamma \cap \Theta$, as its measure is positive.

Notice that, since $\hat{A}_k \rightarrow \hat{A}$ in the C^0 -topology, and $\{H^{u,k}\}_{k \in \mathbb{N}^*}$ satisfies the equiproximity condition, Theorem 4.1.2 states that exists a projective u -holonomy h^u for $\hat{A}$ restricted to the set $W_{\text{loc}}^u(\hat{\mathbf{q}})$. Besides, the referred theorem also states that one can take a subsequence $\{h^{u,k}\}_{k \in \mathbb{N}'}$ converging uniformly to h^u in $W_{\text{loc}}^u(\hat{\mathbf{q}})$.

Therefore we are in position to apply Proposition 3.4.1 to conclude that $\{\lambda_+(\hat{A}_k)\}_{k \in \mathbb{N}'}$ converges to $\lambda_+(\hat{A})$, as we wanted.

Case 2. There exists a $k_0 \in \mathbb{N}^*$ such that the measure $\hat{m}_k$ is not an su -state for each $k \geq k_0$. Again, to simplify the notation we can assume $k_0 = 1$.

The proof in this case is by contradiction. This will be done through the so called “energy method” introduced by Avila, Eskin and Viana (see [Via14, Chapter 10] for a precise explanation of this method). However, the calculations of that method was already done in [BBB18, § 7] for a setting similar to the one we are considering. Thus our goal is to prove that we have the same tools used in such calculations. In order to do this we need to project our hypotheses to the one-sided subshift of finite type. But first we must introduce some useful elements.

By compactness we can assume, without loss of generality, that the sequence $\{\hat{m}_k\}_{k \in \mathbb{N}^*}$ weakly-* converges to some probability measure $\hat{m}$. Using that $\hat{m}_k$ is a $\mathbb{P}F_{\hat{A}_k}$ -invariant measure projecting down to $\hat{\mu}$, and that the sequence $\{\hat{A}_k\}_{k \in \mathbb{N}^*}$ converges to $\hat{A}$ in the C^0 -topology, one can see that $\hat{m}$ is a $\mathbb{P}F_{\hat{A}}$ -invariant measure projecting down to $\hat{\mu}$.

Fix for now any $k \in \mathbb{N}^*$. Applying Lemma 4.2.1 to both $(\hat{A}, H^s)$ and $(\hat{A}_k, H^{s,k}, H^{u,k})$, we obtain linear cocycles $\tilde{A}$ and $\tilde{A}_k$ constant along local stable sets such that the latter admits a u -holonomy $\tilde{H}^{u,k}$. Once again, applying Lemma 4.2.3, we get measures $\tilde{m}$ and $\tilde{m}_k$ projecting down to $\hat{\mu}$ and such that the latter is a u -state for $(\tilde{A}_k, \tilde{H}^{u,k})$. We would like to observe that we still have the convergence $\tilde{m}_k \rightarrow \tilde{m}$ in the weak-* topology. In addition, the straightened-out

cocycle $\tilde{A}$ is in the diagonal form, as discussed in Remark 4.2.2.

Let us translate our problem to the non-invertible case of a one-sided subshift of finite type. First consider the projection $P: \hat{\Sigma} \rightarrow \Sigma$. We can then take the linear cocycles $A, A_k: \Sigma \rightarrow \text{SL}(2, \mathbb{R})$ induced from $\tilde{A}$ and $\tilde{A}_k$ respectively, that is, such that $\tilde{A} = A \circ P$ and $\tilde{A}_k = A_k \circ P$. We would like to observe again that we have the convergence $A_k \rightarrow A$ in the C^0 -topology. Lastly, let m be the measure induced by $\tilde{m}$ as in Equation (4.6), and m_k be the one induced by the u -state $\tilde{m}_k$ with respect to A_k .

According to Propositions 4.3.2 and 4.3.3, those measures admit disintegrations $\{m_{\mathbf{x}}\}_{\mathbf{x} \in \Sigma}$ and $\{m_{\mathbf{x}}^k\}_{\mathbf{x} \in \Sigma}$ forming together an equicontinuous family and such that $m_{\mathbf{x}}^k \rightarrow m_{\mathbf{x}}$ uniformly in $\mathbf{x} \in \Sigma$. Moreover, Freijo and Marin proved in [FM21, Proposition 12] that the conditional measure $m_{\mathbf{x}}^k$ is non-atomic for every $k \in \mathbb{N}^*$ and any $\mathbf{x} \in \Sigma$.

Since the linear cocycle $\tilde{A}$ preserves the coordinate axes, and the projection P is surjective, we can show that the induced cocycle A also preserves the coordinate axes, which are represented by the projective points $a, r \in \mathbb{P}\mathbb{R}^2$. In addition, combining Equation (4.5) and the proof of Lemma 3.3.5, there exists a constant $\beta \in [0, 1]$ such that $m_{\mathbf{x}} = \beta \delta_a + (1 - \beta) \delta_r$ for all $\mathbf{x} \in \Sigma$.

However, since we cannot have $\lambda_+(A_k) \rightarrow \lambda_-(A)$ (because this implies null Lyapunov exponents for A), we conclude that $\beta \neq 0$. On the other hand, if we suppose (for the sake of a contradiction) that $\lambda_+(A_k) \not\rightarrow \lambda_+(A)$, then we also get $\beta \neq 1$. Therefore $\beta \in (0, 1)$.

From this point we have all one needs to follow the calculations of the energy method in [BBB18, § 7] leading to a contradiction. This means that $\lambda_+(A_k) \rightarrow \lambda_+(A)$. Finally, we can combine Equations (4.3) and (4.8), that show the invariance of the Lyapunov exponents, to conclude the theorem. $\square$

5.2 Proof of Theorem 2.1.2

In this section we prove Theorem 2.1.2 as a final stage in concluding Theorem C. To make easier to follow the proof, we reproduce here the statement of such result.

Theorem 2.1.2. *Let $(\hat{\Sigma}, \hat{f})$ be a subshift of finite type and $\hat{\mu}$ be an $\hat{f}$ -ergodic probability measure on $\hat{\Sigma}$ that has local product structure and full support. Consider $(\hat{A}, H^s) \in \mathcal{H}_s^0(2, \hat{\Sigma})$ with non-zero Lyapunov exponents and satisfying the 2-invariance condition. If one takes a sequence $\{(\hat{A}_k, H^{s,k}, H^{u,k})\}_{k \in \mathbb{N}^*}$ of linear cocycles with uniform s -holonomies and non-uniform u -holonomies, and a set $\hat{\mathcal{D}} \subset \hat{\Sigma}$ with positive $\hat{\mu}$ -measure such that:*

- (1) $(\hat{A}_k, H^{s,k}) \rightarrow (\hat{A}, H^s)$ in $\mathcal{H}_s^0(2, \hat{\Sigma})$; and
- (2) *for each $\hat{\mathbf{q}} \in \hat{\mathcal{D}}$, exists a u -holonomy H^u for the cocycle $\hat{A}$ restricted to $W_{\text{loc}}^u(\hat{\mathbf{q}})$ so that the sequence $\{H^{u,k}\}_{k \in \mathbb{N}^*}$ converges uniformly to H^u in such set,*

then $\lambda_+(\hat{A}_k) > 0$ for every sufficiently large $k \in \mathbb{N}^$.*

Proof. Recall that we can consider $(\hat{A}, H^s)$ as a diagonal cocycle due to the 2-invariance condition.

We use Proposition 3.4.1 in order to conclude this result. Suppose by contradiction that exists a subsequence of $\{\hat{A}_k\}_{k \in \mathbb{N}^*}$ formed by linear cocycles with null Lyapunov exponents. Without loss of generality we can assume that the whole sequence satisfies this property from now on.

Given any $k \in \mathbb{N}^*$, let us take a measure $\hat{m}_k$ as in Proposition 3.2.4. Since we are assuming $\lambda_+(\hat{A}_k) = 0$, the Invariance Principle (Proposition 3.2.2) asserts that $\hat{m}_k$ is an su -state and hence, by Proposition 3.2.1, it admits a continuous disintegration that is invariant by $\mathbb{P}\hat{A}_k$ and both projective holonomies.

Since $H^{u,k}$ is a non-uniform u -holonomy, such invariance only holds in a full-measure subset of $\hat{\Sigma}$, for each $k \in \mathbb{N}^*$. Hence the intersection of these subsets still has full $\hat{\mu}$ -measure. Nevertheless, once $\hat{\mathcal{D}}$ has positive $\hat{\mu}$ -measure, it follows that the set $\hat{\mathcal{D}}_*$, defined as the intersection of all mentioned sets, also has positive $\hat{\mu}$ -measure. Thus, we conclude that the disintegration considered above of each measure $\hat{m}_k$ (with $k \in \mathbb{N}^*$) is also invariant by the projective u -holonomies $h^{u,k}|_{W_{\text{loc}}^u(\hat{\mathbf{q}})} = \mathbb{P}H^{u,k}|_{W_{\text{loc}}^u(\hat{\mathbf{q}})}$, where $\hat{\mathbf{q}} \in \hat{\mathcal{D}}_*$. Notice that our hypotheses assure that $h^{u,k}|_{W_{\text{loc}}^u(\hat{\mathbf{q}})} \rightarrow h^u|_{W_{\text{loc}}^u(\hat{\mathbf{q}})} = \mathbb{P}H^u|_{W_{\text{loc}}^u(\hat{\mathbf{q}})}$ uniformly, and this holds for each $\hat{\mathbf{q}} \in \hat{\mathcal{D}}_*$.

In order to make use of Proposition 3.4.1, it remains only to find a point in $\hat{\mathcal{D}}_*$ that has all symbols in its non-negative entries. Equivalently, we need to find a point in $\hat{\mathcal{D}}_*$ whose future orbit intersects every cylinder of rank 1. However, this is achieved by the mixing property of $(\hat{f}, \hat{\mu})$, since cylinders has positive $\hat{\mu}$ -measure. Therefore we can apply Proposition 3.4.1 and conclude that $\{\lambda_+(\hat{A}_k)\}_{k \in \mathbb{N}^*}$ converges to $\lambda_+(\hat{A})$, contradicting the fact that $\lambda_+(\hat{A}) > 0$. This finish the proof. $\square$

Bibliography

- [AV10] Artur Avila and Marcelo Viana. Extremal Lyapunov exponents: an invariance principle and applications. *Invent. Math.*, 181(1):115–178, 2010.
- [BBB18] Lucas Backes, Aaron Brown, and Clark Butler. Continuity of Lyapunov exponents for cocycles with invariant holonomies. *J. Mod. Dyn.*, 12:223–260, 2018.
- [BGV03] Christian Bonatti, Xavier Gómez-Mont, and Marcelo Viana. Généricité d’exposants de Lyapunov non-nuls pour des produits déterministes de matrices. *Ann. Inst. H. Poincaré Anal. Non Linéaire*, 20(4):579–624, 2003.
- [Boc02] Jairo Bochi. Genericity of zero Lyapunov exponents. *Ergodic Theory Dynam. Systems*, 22(6):1667–1696, 2002.
- [Bow70] Rufus Bowen. Markov partitions for Axiom A diffeomorphisms. *Amer. J. Math.*, 92:725–747, 1970.
- [BV04] Christian Bonatti and Marcelo Viana. Lyapunov exponents with multiplicity 1 for deterministic products of matrices. *Ergodic Theory Dynam. Systems*, 24(5):1295–1330, 2004.
- [BV17] Carlos Bocker-Neto and Marcelo Viana. Continuity of Lyapunov exponents for random two-dimensional matrices. *Ergodic Theory Dynam. Systems*, 37(5):1413–1442, 2017.
- [Dix84] Jacques Dixmier. *General topology*. Undergraduate Texts in Mathematics. Springer-Verlag, New York, 1984. Translated from the French by Sterling K. Berberian.
- [FK60] Hillel Furstenberg and Harry Kesten. Products of random matrices. *Ann. Math. Statist.*, 31:457–469, 1960.
- [FM21] Catalina Freijo and Karina Marin. Continuity of Lyapunov exponents for non-uniformly fiber-bunched cocycles. *Ergodic Theory Dynam. Systems*, 41(12):3740–3767, 2021.
- [Kal11] Boris Kalinin. Livšic theorem for matrix cocycles. *Ann. of Math. (2)*, 173(2):1025–1042, 2011.

- [Kin68] John F. C. Kingman. The ergodic theory of subadditive stochastic processes. *J. Roy. Statist. Soc. Ser. B*, 30:499–510, 1968.
- [KN11] Anatole Katok and Viorel Nițică. *Rigidity in higher rank abelian group actions. Volume I*, volume 185 of *Cambridge Tracts in Mathematics*. Cambridge University Press, Cambridge, 2011. Introduction and cocycle problem.
- [KS16] Boris Kalinin and Victoria Sadovskaya. Holonomies and cohomology for cocycles over partially hyperbolic diffeomorphisms. *Discrete Contin. Dyn. Syst.*, 36(1):245–259, 2016.
- [Led84] François Ledrappier. Quelques propriétés des exposants caractéristiques. École d’été de probabilités de Saint-Flour XII-1982, Lect. Notes Math. 1097, 305–396 (1984)., 1984.
- [Led86] François Ledrappier. Positivity of the exponent for stationary sequences of matrices. Lyapunov exponents, Proc. Workshop, Bremen/Ger. 1984, Lect. Notes Math. 1186, 56–73 (1986)., 1986.
- [Lya92] Aleksandr M. Lyapunov. The general problem of the stability of motion. *Internat. J. Control*, 55(3):521–790, 1992. Translated by A. T. Fuller from Édouard Davaux’s French translation (1907) of the 1892 Russian original, With an editorial (historical introduction) by Fuller, a biography of Lyapunov by V. I. Smirnov, and the bibliography of Lyapunov’s works collected by J. F. Barrett, Lyapunov centenary issue.
- [Ose68] Valery I. Oseledets. A multiplicative ergodic theorem. Characteristic Ljapunov, exponents of dynamical systems. *Trudy Moskov. Mat. Obšč.*, 19:179–210, 1968.
- [Rok49] Vladimir Abramovich Rokhlin. Über die Grundbegriffe der Maßtheorie. *Mat. Sb., Nov. Ser.*, 25:107–150, 1949.
- [Via08] Marcelo Viana. Almost all cocycles over any hyperbolic system have nonvanishing Lyapunov exponents. *Ann. of Math. (2)*, 167(2):643–680, 2008.
- [Via14] Marcelo Viana. *Lectures on Lyapunov exponents*, volume 145 of *Cambridge Studies in Advanced Mathematics*. Cambridge University Press, Cambridge, 2014.
- [Via20] Marcelo Viana. (Dis)continuity of Lyapunov exponents. *Ergodic Theory Dynam. Systems*, 40(3):577–611, 2020.
- [VO16] Marcelo Viana and Klerley Oliveira. *Foundations of ergodic theory*, volume 151 of *Camb. Stud. Adv. Math.* Cambridge: Cambridge University Press, 2016.

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- [VY19] Marcelo Viana and Jiagang Yang. Continuity of Lyapunov exponents in the C^0 topology. *Israel J. Math.*, 229(1):461–485, 2019.

Appendix A

Holonomies in the Bocker–Viana’s counterexample

In this chapter we explore the equiproximity condition (Definition 5) somewhat more. In fact, we use a classic counterexample to show that in the absence of the mentioned condition we cannot expect convergence of the Lyapunov exponents. We begin by exhibiting such example given by Bocker and Viana of a linear cocycle that is not a Hölder-continuity point.

A.1 Bocker–Viana’s counterexample

Consider a (two-sided) full shift $(\hat{\Sigma}, \hat{f})$ of 2 symbols, and a Bernoulli measure $\hat{\mu} = (p_0\delta_0 + p_1\delta_1)^{\mathbb{Z}}$, where the constants $p_0, p_1 \in (0, 1)$ are distinct. Finally, for any $\sigma > 4$, let $\hat{A}: \hat{\Sigma} \rightarrow \text{SL}(2, \mathbb{R})$ be given by:

$$\hat{A}(\hat{\mathbf{x}}) = \begin{pmatrix} \sigma & 0 \\ 0 & \sigma^{-1} \end{pmatrix} \text{ if } x_0 = 0 \quad \text{and} \quad \hat{A}(\hat{\mathbf{x}}) = \begin{pmatrix} \sigma^{-1} & 0 \\ 0 & \sigma \end{pmatrix} \text{ if } x_0 = 1.$$

Notice that the map $\hat{A}$ is α -Hölder continuous (for any $\alpha > 0$), and also that the cocycle generated by it is locally constant (*cf.* Example 1.1.1) and diagonal (hence it satisfies the 2-invariance condition). In addition, it is possible to show that its Lyapunov exponents are distinct from zero. Indeed,

$$\lambda_{\pm}(\hat{A}) = \pm |p_0 - p_1| \log \sigma.$$

Fix for now a $k \in \mathbb{N}^*$. Let us consider the number $n = 2k + 1$ and the cylinder $Z_n = [0; 1, \dots, 1, 0, \dots, 0, 0]$, where the symbol ‘1’ appears k times and the symbol ‘0’ appears $k + 1$ times. We would like to observe that the iterates $f^i(Z_n)$ are pairwise disjoint for all $i = 0, \dots, 2k$.

Now, let us take constants $\varepsilon_n = \sigma^{-k}$ and $\delta_n = \arctan \varepsilon_n$. Then define the map $\hat{A}_n: \hat{\Sigma} \rightarrow \text{SL}(2, \mathbb{R})$ by

$$\hat{A}_n(\hat{\mathbf{x}}) = \begin{cases} \hat{A}(\hat{\mathbf{x}}) \cdot R_{\delta_n} & \text{if } \hat{\mathbf{x}} \in f^k(Z_n) \\ \hat{A}(\hat{\mathbf{x}}) \cdot \begin{pmatrix} 1 & 0 \\ \varepsilon_n & 1 \end{pmatrix} & \text{if } \hat{\mathbf{x}} \in Z_n \cup f^{2k}(Z_n) \\ \hat{A}(\hat{\mathbf{x}}) & \text{otherwise,} \end{cases} \quad (\text{A.1})$$

where R_{δ_n} is the rotation map with angle δ_n . Notice again that the map $\hat{A}_n$ is also α -Hölder continuous.

Bocker and Viana have shown in [BV17] that $\hat{A}_n \rightarrow \hat{A}$ in the Hölder topology, and furthermore that the Lyapunov exponents of every $\hat{A}_n$ vanish. Therefore it is a counterexample for the continuity problem of Lyapunov exponents of Hölder-continuous cocycles.

A.2 Convergence of holonomies

We are interested in this section in the following result.

Proposition A.2.1. *In the setting of Bocker–Viana’s counterexample, the non-uniform invariant holonomies for the linear cocycles A_n converge to the invariant holonomy for the locally constant linear cocycle A in a full-measure set.*

Proof. We would like to recall that the local stable set of a sequence $\hat{\mathbf{x}} \in M$ is given by:

$$W_{\text{loc}}^s(\hat{\mathbf{x}}) = \{\hat{\mathbf{y}} \in M : x_i = y_i \text{ for all } i \in \mathbb{N}\}.$$

Let us consider sequences $\hat{\mathbf{x}} \in M$ and $\hat{\mathbf{y}}, \hat{\mathbf{z}} \in W_{\text{loc}}^s(\hat{\mathbf{x}})$, and fix for now a number $n = 2k + 1$. Recall that, since $\lambda_+(A_n) = 0$, the linear cocycle A_n is non-uniformly fibre-bunched. Hence, exist non-uniform invariant holonomies for A_n over f .

Our goal at the moment is to calculate the s -holonomy relative to A_n at the points $\hat{\mathbf{y}}$ and $\hat{\mathbf{z}}$, that is, the linear isomorphism

$$H_{\hat{\mathbf{y}}, \hat{\mathbf{z}}}^{s,n} = \lim_{m \in \mathbb{N}^*} A_n^m(\hat{\mathbf{z}})^{-1} \circ A_n^m(\hat{\mathbf{y}}).$$

Observe that we need to know whether $\hat{\mathbf{y}}, \hat{\mathbf{z}} \in \bigcup_{i=0}^{2k} f^i(Z_n)$ to determine the matrices $A_n^m(\hat{\mathbf{y}})$ and $A_n^m(\hat{\mathbf{z}})^{-1}$, for any $m \in \mathbb{N}^*$. In order to do this, it is necessary to know the values y_j and z_j for $j = -2k, \dots, 0, \dots, 2k$.

Now, since $\hat{\mathbf{y}}$ and $\hat{\mathbf{z}}$ have the same non-negative entries, we can conclude that $A_n(f^m(\hat{\mathbf{y}})) = A_n(f^m(\hat{\mathbf{z}}))$ for every $m \geq n$. Therefore,

$$\begin{aligned} H_{\hat{\mathbf{y}}, \hat{\mathbf{z}}}^{s,n} &= \lim_{m \in \mathbb{N}^*} A_n^m(\hat{\mathbf{z}})^{-1} \circ A_n^m(\hat{\mathbf{y}}) \\ &= \lim_{m \geq n} (A_n^{m-n+1}(f^{n-1}(\hat{\mathbf{z}})) \circ A_n^{n-1}(\hat{\mathbf{z}}))^{-1} \circ (A_n^{m-n+1}(f^{n-1}(\hat{\mathbf{y}})) \circ A_n^{n-1}(\hat{\mathbf{y}})) \\ &= \lim_{m \geq n} A_n^{n-1}(\hat{\mathbf{z}})^{-1} \circ (A_n^{m-n+1}(f^{n-1}(\hat{\mathbf{z}}))^{-1} \circ A_n^{m-n+1}(f^{n-1}(\hat{\mathbf{y}}))) \circ A_n^{n-1}(\hat{\mathbf{y}}) \\ &= \lim_{m \geq n} A_n^{n-1}(\hat{\mathbf{z}})^{-1} \circ A_n^{n-1}(\hat{\mathbf{y}}) \\ &= A_n^{n-1}(\hat{\mathbf{z}})^{-1} \circ A_n^{n-1}(\hat{\mathbf{y}}). \end{aligned} \tag{A.2}$$

With this in mind, we know that $H_{\hat{\mathbf{y}}, \hat{\mathbf{z}}}^{s,n} = \text{id}$ whenever none of the n firsts iterates of $\hat{\mathbf{y}}$ and $\hat{\mathbf{z}}$ lie in $Z_n \cup f^k(Z_n) \cup f^{2k}(Z_n)$; that is, whenever

$$\hat{\mathbf{y}}, \hat{\mathbf{z}} \notin \bigcup_{i=0}^{n-1} f^{-i}(Z_n) \cup f^{k-i}(Z_n) \cup f^{2k-i}(Z_n) = \bigcup_{i=-k}^{3k} f^{k-i}(Z_n).$$

Our next goal is to determine whether or not the sequence $\{H_{\hat{\mathbf{y}}, \hat{\mathbf{z}}}^{s,n}\}_{n=2k+1}^{k \in \mathbb{N}}$ of s -holonomies is convergent. According to our previous calculation, we need to guarantee that $\hat{\mathbf{y}}$ and $\hat{\mathbf{z}}$ are not in cylinders of the type $f^{k-i}(Z_n)$ for some $i \in \mathbb{Z}$ and for infinite values of $n = 2k + 1$, with $k \in \mathbb{N}$. Let us define then the set

$$\mathcal{C} = \{\hat{\mathbf{x}} \in M : \exists i < j \in \mathbb{N} \text{ such that } x_i = 0 \text{ and } x_j = 1\}.$$

We would like to observe first that if $\hat{\mathbf{x}} \notin \mathcal{C}$, then exists $j \in \mathbb{N}$ such that $x_i = 1$ for each $0 \leq i < j$, and $x_i = 0$ for each $i \geq j$. In this case, $\hat{\mathbf{x}} \in Z_{2j+1}$ and can be in infinitely many cylinders $f^{k-j}(Z_n)$ with $k \in \mathbb{N}$. Notice also that $W_{\text{loc}}^s(\hat{\mathbf{x}}) \subset \mathcal{C}$ for every $\hat{\mathbf{x}} \in \mathcal{C}$.

Summarising, if one consider a sequence $\hat{\mathbf{x}} \in \mathcal{C}$ and take another sequences $\hat{\mathbf{y}}, \hat{\mathbf{z}} \in W_{\text{loc}}^s(\hat{\mathbf{x}})$, then $H_{\hat{\mathbf{y}}, \hat{\mathbf{z}}}^{s,n} \neq \text{id}$ for only a finite number of values of $n \in \mathbb{N}^*$. Hence, we can conclude that $H_{\hat{\mathbf{y}}, \hat{\mathbf{z}}}^{s,n} \rightarrow \text{id}$ in this case. Since A is a locally constant linear cocycle, it admits the identity map as an s -holonomy. We conclude then that the non-uniform s -holonomies for A_n converge to the s -holonomy for A in the set $\mathcal{C}$.

It remains to show that $\mathcal{C}$ has full μ -measure. To achieve this, we are going to approximate it by sets of the form

$$\mathcal{C}_m = \{\hat{\mathbf{x}} \in M : \exists 0 \leq i < j < m \text{ such that } x_i = 0 \text{ and } x_j = 1\}$$

with $m \in \mathbb{N}$. Define the cylinder $Z_m^j = [0; 1, \dots, 1, 0, \dots, 0]$ where the symbol '1' appears j times and the symbol '0' appears $m - j$ times, for any $m \in \mathbb{N}$ and $0 \leq j < m$. By definition, each of these cylinders satisfies $\mu(Z_m^j) = p_1^j p_0^{m-j}$.

Given any $m \in \mathbb{N}$, we would like to point that the set $\mathcal{C}_m$ can be rewritten as $\mathcal{C}_m = \{\hat{\mathbf{x}} \in M : \hat{\mathbf{x}} \notin \bigcup_{j=0}^{m-1} Z_m^j\} = \left(\bigcup_{j=0}^{m-1} Z_m^j\right)^c$. Now, since the cylinders $Z_m^0, \dots, Z_m^{m-1}$ are pairwise disjoint, we can calculate

$$\begin{aligned} \mu(\mathcal{C}_m) &= 1 - \mu\left(\bigcup_{j=0}^{m-1} Z_m^j\right) = 1 - \sum_{j=0}^{m-1} \mu(Z_m^j) \\ &= 1 - \sum_{j=0}^{m-1} p_1^j p_0^{m-j} = 1 - p_0^m \sum_{j=0}^{m-1} \left(\frac{p_1}{p_0}\right)^j \end{aligned}$$

to conclude that

$$\begin{aligned} \lim_{m \in \mathbb{N}} \mu(\mathcal{C}_m) &= 1 - \lim_{m \in \mathbb{N}} p_0^m \sum_{j=0}^{m-1} \left(\frac{p_1}{p_0}\right)^j \geq 1 - \lim_{m \in \mathbb{N}} p_0^m m \left(\frac{p_1}{p_0}\right)^{m-1} \\ &= 1 - p_0 \lim_{m \in \mathbb{N}} m p_1^{m-1} = 1. \end{aligned}$$

Finally, since the sets $\mathcal{C}_m \rightarrow \mathcal{C}$ in the Hausdorff topology, we obtain that $\mu(\mathcal{C}) \geq \lim_{m \in \mathbb{N}} \mu(\mathcal{C}_m) = 1$, as we have claimed. $\square$

A.3 Non-equiproximity

We begin this section by showing that, although the set $\mathcal{C}$ (of points where the holonomies converge) has full μ -measure, its complement is not empty.

Proposition A.3.1. *In the setting of Bocker–Viana’s counterexample, there are points $\hat{\mathbf{y}}, \hat{\mathbf{z}} \in M$ in the same local stable set such that the sequence $\{H_{\hat{\mathbf{y}}, \hat{\mathbf{z}}}^{s,n}\}_n$ of linear isomorphisms does not converge.*

Proof. We would like to start remembering that $H_{\hat{\mathbf{y}}, \hat{\mathbf{z}}}^{s,n} = A_n^{n-1}(\hat{\mathbf{z}})^{-1} \circ A_n^{n-1}(\hat{\mathbf{y}})$ for every n odd and every points $\hat{\mathbf{y}}, \hat{\mathbf{z}} \in M$ in the same local stable set, as we have seen in Equation (A.2).

Let us define now the sequences $\hat{\mathbf{y}}$ and $\hat{\mathbf{z}}$ by the respectively elements:

$$y_j = \begin{cases} 1 & \text{if } j < 0 \\ 0 & \text{otherwise} \end{cases} \quad \text{and} \quad z_j = \begin{cases} 1 & \text{if } -N \leq j < 0 \\ 0 & \text{otherwise,} \end{cases} \quad (\text{A.3})$$

for some $N \in \mathbb{N}^*$ fixed. It is easy to see that these points are in the local stable set of the zero sequence, for instance. Besides, they satisfy the following properties for every $n = 2k + 1$:

- $\hat{\mathbf{y}} \in f^k(Z_n)$, $f^k(\hat{\mathbf{y}}) \in f^{2k}(Z_n)$ but $f^i(\hat{\mathbf{y}}) \notin Z_n \cup f^k(Z_n) \cup f^{2k}(Z_n)$ for all $0 < i \leq 2k$ with $i \neq k$;
- $f^i(\hat{\mathbf{z}}) \notin Z_n \cup f^k(Z_n) \cup f^{2k}(Z_n)$ for all $0 \leq i \leq 2k$, as long as $k > N$.

With this and the definition of the cocycle A_n in Equation (A.1) in mind, we can compute the expression of $A_n^{n-1}(\hat{\mathbf{y}})$ and $A_n^{n-1}(\hat{\mathbf{z}})^{-1}$ for each $n = 2k + 1$ with $k > N$ as follows:

$$\begin{aligned} A_n^{n-1}(\hat{\mathbf{y}}) &= A_n(f^{2k-1}(\hat{\mathbf{y}})) \cdots A_n(f^k(\hat{\mathbf{y}})) \cdots A_n(f(\hat{\mathbf{y}})) A_n(\hat{\mathbf{y}}) \\ &= \underbrace{A(\hat{\mathbf{y}}) \cdots A(\hat{\mathbf{y}})}_{k-1 \text{ times}} \left[A(\hat{\mathbf{y}}) \begin{pmatrix} 1 & 0 \\ \varepsilon_n & 1 \end{pmatrix} \right] \underbrace{A(\hat{\mathbf{y}}) \cdots A(\hat{\mathbf{y}})}_{k-1 \text{ times}} [A(\hat{\mathbf{y}}) R_{\delta_n}] \\ &= \begin{pmatrix} \sigma & 0 \\ 0 & \sigma^{-1} \end{pmatrix}^k \begin{pmatrix} 1 & 0 \\ \varepsilon_n & 1 \end{pmatrix} \begin{pmatrix} \sigma & 0 \\ 0 & \sigma^{-1} \end{pmatrix}^k \begin{pmatrix} \cos \delta_n & -\sin \delta_n \\ \sin \delta_n & \cos \delta_n \end{pmatrix} \\ &= \begin{pmatrix} \sigma^{2k} \cos \delta_n & -\sigma^{2k} \sin \delta_n \\ \sigma^{-k} \cos \delta_n + \sigma^{-2k} \sin \delta_n & \sigma^{-2k} \cos \delta_n - \sigma^{-k} \sin \delta_n \end{pmatrix} \end{aligned}$$

and

$$\begin{aligned} A_n^{n-1}(\hat{\mathbf{z}})^{-1} &= [A_n(f^{2k-1}(\hat{\mathbf{z}})) \cdots A_n(f(\hat{\mathbf{z}})) A_n(\hat{\mathbf{z}})]^{-1} \\ &= A_n(\hat{\mathbf{z}})^{-1} A_n(f(\hat{\mathbf{z}}))^{-1} \cdots A_n(f^{2k-1}(\hat{\mathbf{z}}))^{-1} \\ &= \underbrace{A(\hat{\mathbf{z}})^{-1} \cdots A(\hat{\mathbf{z}})^{-1}}_{2k \text{ times}} \\ &= \begin{pmatrix} \sigma^{-1} & 0 \\ 0 & \sigma \end{pmatrix}^{2k} = \begin{pmatrix} \sigma^{-2k} & 0 \\ 0 & \sigma^{2k} \end{pmatrix}. \end{aligned}$$

Therefore, for any $n = 2k + 1$ with $k > N$, the linear isomorphism $H_{\hat{\mathbf{y}}, \hat{\mathbf{z}}}^{s,n}$ is given by

$$H_{\hat{\mathbf{y}}, \hat{\mathbf{z}}}^{s,n} = A_n^{n-1}(\hat{\mathbf{z}})^{-1} \circ A_n^{n-1}(\hat{\mathbf{y}})$$

$$= \begin{pmatrix} \cos \delta_n & -\sin \delta_n \\ \sigma^k \cos \delta_n + \sin \delta_n & \cos \delta_n - \sigma^k \sin \delta_n \end{pmatrix}. \quad (\text{A.4})$$

This implies that $H_{\hat{\mathbf{y}}, \hat{\mathbf{z}}}^{s,n} \not\rightarrow \text{id}$. Indeed, using the max norm in the space $\text{SL}(2, \mathbb{R})$, we obtain:

$$\begin{aligned} \|H_{\hat{\mathbf{y}}, \hat{\mathbf{z}}}^{s,n} - \text{id}\|_{\max} &= \left\| \begin{pmatrix} \cos \delta_n - 1 & -\sin \delta_n \\ \sigma^k \cos \delta_n + \sin \delta_n & \cos \delta_n - \sigma^k \sin \delta_n - 1 \end{pmatrix} \right\|_{\max} \\ &\geq |\sigma^k \cos \delta_n + \sin \delta_n| \rightarrow \infty. \end{aligned}$$

Since the s -holonomy of the locally constant linear cocycle A is the identity map, the proof is completed. $\square$

One important feature that can occur to some linear cocycles is the equiproximity condition in the sequence of holonomies. Recall Definition 5 for this discussion. Our next result shows that, in the Bocker–Viana’s counterexample, that sequence *does not* satisfy the equiproximity condition.

Proposition A.3.2. *In the setting of Bocker–Viana’s counterexample, for all $\varepsilon > 0$ and $\delta > 0$, one can choose δ -close points $\hat{\mathbf{y}}, \hat{\mathbf{z}} \in M$ in the same local stable set such that $\|H_{\hat{\mathbf{y}}, \hat{\mathbf{z}}}^{s,n} - \text{id}\| > \varepsilon$ for some odd n .*

Proof. First let us choose an $N \in \mathbb{N}^*$ such that the points $\hat{\mathbf{y}}, \hat{\mathbf{z}} \in M$ defined by Equation (A.3) satisfy $\text{dist}_\rho(\hat{\mathbf{y}}, \hat{\mathbf{z}}) < \delta$.

Accordingly to the proof of the previous proposition, given any $n = 2k + 1$, the linear isomorphism $H_{\hat{\mathbf{y}}, \hat{\mathbf{z}}}^{s,n}$ is defined by Equation (A.4) and satisfy

$$\|H_{\hat{\mathbf{y}}, \hat{\mathbf{z}}}^{s,n} - \text{id}\|_{\max} > |\sigma^k \cos \delta_n + \sin \delta_n|,$$

provided that $k > N$.

Therefore, if one takes the number $n = 2k + 1$ with $k > N$ such that $|\sigma^k \cos \delta_n + \sin \delta_n| > \varepsilon$, then we are done. $\square$